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EXAMINING DATA QUALITY MANAGEMENT PRACTICES IN LIGHT OF THE ADOPTION OF LLMS AND GENAI: CLOSING THE GAP BETWEEN IQ THEORY AND IQ PRACTICE

VENANT HABAMUNGU

May 2026

A dissertation submitted to the Graduate School
University of Arkansas at Little Rock

in partial fulfillment of the requirements for the degree of
DOCTOR OF PHILOSOPHY
in Computer and Information Sciences: Information Quality

in the Department of Computer and Information Sciences
in the Donaghey College of Science, Technology, Engineering, and Mathematics

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Abstract

The rapid adoption of Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) has intensified longstanding data quality concerns while exposing important gaps in current Data Quality Management (DQM) practice. This dissertation examines whether widely used DQM frameworks sufficiently address semantic data quality dimensions that are critical for trustworthy GenAI and LLM applications. Guided by Information Quality (IQ) theory, particularly the Wang and Strong conceptualization of data quality, the study evaluates selected DQM frameworks used in practice to determine the extent to which they emphasize semantic dimensions such as accuracy, value-added, relevancy, objectivity, reputation, believability, security, and appropriate amount of data. The study also investigates the persistent misconception that data validation alone is equivalent to data accuracy, especially when validation is not performed against an authoritative data source or system of record.

A qualitative, multi-phase research design was used. First, a comparative document analysis was conducted on nine widely referenced DQM frameworks to identify the degree to which theoretical data quality dimensions are represented in practice. Second, a two-round Delphi study was conducted to validate the framework analysis and obtain expert consensus on the implications of the findings. In Round 1, fifteen expert panelists responded to twenty-three structured statements and seven open-ended questions concerning the adequacy of current DQM practice in the context of GenAI and LLM adoption. In Round 2, fourteen experts ranked the relative importance of semantic dimensions and evaluated practical measures and approaches for incorporating them into DQM activities. Descriptive statistics were used to summarize

Round 2 rankings, while thematic analysis was used to synthesize qualitative feedback from both rounds.

The findings indicate that current DQM practice continues to place heavier emphasis on syntactic dimensions such as validity, completeness, consistency, timeliness, and uniqueness than on semantic dimensions required for context-sensitive, trustworthy AI outputs. Delphi panelists strongly agreed that DQM practices must evolve in response to GenAI and LLM adoption, and they consistently identified accuracy, relevance, trustworthiness, and context-sensitive evaluation as high priorities. The findings further suggest that human review, cross-source validation, and authoritative reference data remain indispensable, even where automation is used. This study contributes by documenting the gap between IQ theory and IQ practice, validating that gap through expert consensus, and offering recommendations for strengthening DQM and data governance practices without proposing a wholly new framework. The dissertation concludes that organizations can improve trust, transparency, and risk management in GenAI and LLM implementations by increasing the priority of semantic data quality dimensions within existing DQM programs.

Dedication

I would like to dedicate this achievement to those who are most precious, have inspired, encouraged, and sacrificed for me throughout this journey.

TO,

... my father, **Felix Rwizhirwa**, who dedicated himself for his children and worked really hard to give me the best of everything in life! You are my role model in everything that I do in this life.

... my mother, **Leonie Ngezahumuremyi**, who not just brought me into this beautiful world, but also gave me wings to fly. You are the reason I could embark on this journey!

... my wife, **Edith Umulisa Habamungu**, whose love, and faith sustained me. I could not have done this without you being by my side. Especially, Thanks for being the best Mother to my children, this kept me going!

... my children, **Mael Izihirwa Habamungu**, **Mae Igabe Habamungu**, who motivated me to do better, and to be better. No matter what I achieve in life, you are the BEST and most precious. I certainly gave birth to you, but in real you gave me life!

... my sister, **Bernadette Mukashema**, whom we grew up together, who worked hard and sacrificed a lot to support my various successes and achievements in this life, especially my education from childhood to this PhD milestone. Thank you for being such a great big sister.

...my cousin and brother, **Bahati Philbert Kayihura**, thank you for your encouragement, guidance, and unwavering support, which inspired me to pursue higher education in the United States. Your influence laid the foundation for this higher degree. I am deeply grateful for your exceptional love and friendship.

... my entire family, colleagues, teachers, professors, and friends for their patience, encouragement, guidance and unwavering presence, all of which were essential to the completion of this work.

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Thank you ALL!

Protocol or IRB Approval Statement



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TO: Venant Habamungu / Dr. John R. Talburt, Advisor
FROM: UA Little Rock Institutional Review Board
CC: Office of Research Compliance
DATE: October 16, 2025

PROTOCOL: 25-076-R1 "EXAMINING DATA QUALITY MANAGEMENT PRACTICES IN LIGHT OF THE ADOPTION OF GENERATIVE ARTIFICIAL INTELLIGENCE(GenAI) AND LARGE LANGUAGE MODELS (LLMs): CLOSING THE GAP BETWEEN IQ THEORY AND IQ PRACTICE"

Thank you for your recent submission to the University of Arkansas at Little Rock Institutional Review Board (IRB). The IRB (FWA #00002205) has reviewed **Protocol 25-076-R1 "EXAMINING DATA QUALITY MANAGEMENT PRACTICES IN LIGHT OF THE ADOPTION OF GENERATIVE ARTIFICIAL INTELLIGENCE(GenAI) AND LARGE LANGUAGE MODELS (LLMs): CLOSING THE GAP BETWEEN IQ THEORY AND IQ PRACTICE"**. The review was limited to activities described in the protocol narrative. On the basis of the review of this protocol, we find that it meets the IRB's criteria for protection of human participants. Your protocol has been **approved** as Human Participant Research. You may now begin to collect data.

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Best of luck with your study.

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Chapter 1: Introduction

1.1 Background

As Data Quality and Data Governance professionals, we must first consider whether Data Quality Management (DQM) practices need to be examined in light of the adoption of LLM and GenAI. The answer is a resounding yes. Examining DQM practices to support GenAI and Large Language Models (LLMs) reveals emerging challenges such as bias mitigation, model interpretability, synthetic data validation, and ethical compliance. Key areas for enhancement include expanding Data Quality Dimensions for LLMs and GenAI, strengthening Data Governance and Policies, implementing AI-driven automated Data Quality Monitoring, establishing a Synthetic Data Validation Framework, ensuring comprehensive Model and Data Documentation, and prioritizing Semantics Data Quality Dimensions, which have long been overlooked in many Data Quality and Governance practices.

A robust Data Quality Management (DQM) and Data Governance (DG) framework is crucial for Generative AI and LLMs, as data quality directly influences the accuracy, reliability, and ethical integrity of AI-generated outputs. Generative AI models, such as LLMs, GANs, and diffusion models, depend on large datasets to generate realistic content. A well-designed DQM framework upholds data integrity, fairness, and accuracy, enhancing AI performance and real-world applicability. To fully leverage AI while minimizing risks, organizations must prioritize strong data governance. Moreover, the issue of "digital inbreeding," where AI deteriorates due to poor-quality or AI-generated training data, underscores the importance of maintaining high data quality. Implementing a comprehensive DQM framework helps address these risks and ensures the development of trustworthy AI systems. [41], [42]

In the book *Beyond Accuracy: What Data Quality Means to Data Consumers* by Richard Y. Wang and Diane M. Strong, the authors highlight the significant social and economic

consequences of poor data quality (DQ). While organizations are making strides in enhancing data quality through practical methods and tools, their efforts often concentrate too narrowly on accuracy. To achieve meaningful improvements, it is essential to understand how data quality is perceived by data consumers, the individuals who rely on and utilize the data. [2]

Data Quality Management (DQM) is a discipline that encompasses methods for measuring and enhancing the quality and integrity of an organization's data. It serves as a fundamental component for compliance with data governance frameworks such as BCBS239 (Basel Committee on Banking Supervision 239) and DFAST (Dodd-Frank Act Stress Testing). In Chapter 7, titled "Data Quality," of the book Data Governance Guide for BCBS239 and DFAST Compliance, the author acknowledges the importance of Data Quality Dimensions within the Data Governance Program. They emphasize that a bank's board and senior management should actively support the identification, assessment, and management of data quality risks as an integral part of the institution's overall risk management framework.[10]

Generative AI (GenAI) systems encounter major challenges concerning data quality and governance. Inadequate data quality can produce misleading or unreliable AI outputs, diminishing the technology's effectiveness and credibility. Problems such as data inconsistencies, inaccuracies, and biases can become magnified in GenAI applications, leading to results that fail to accurately reflect the intended information. [12]

The swift advancement of GenAI has surpassed the creation of strong data governance frameworks. This absence of clear guidelines creates challenges in maintaining data security, privacy, and regulatory compliance. Organizations frequently face difficulties with data integration, cleaning, and storage, all of which are essential for successful GenAI implementation. In the absence of adequate governance, there is a risk of security breaches, compliance issues, and a decline in model output quality. [13]

To tackle these challenges, companies are concentrating on long-term strategies that prioritize data accuracy, consistency, and diversity. Establishing continuous feedback loops and

periodically reviewing and examining governance frameworks are crucial steps to avoid security breaches and maintain high-quality AI outputs. Furthermore, organizations are forming strategic partnerships with data providers to effectively manage the complexities of the AI data lifecycle.[14]

However, with all the efforts taking place and adopting LLMs and GenAI, there have been less discussions, research and recommendations for practices to prioritize semantic Data Quality dimensions that can support risk monitoring and mitigations due to the issues introduced by the adoption of these models.

1.2 Problem Statement

The widespread adoption of Large Language Models (LLMs) and Generative Artificial Intelligence (GenAI) had introduced significant challenges for existing Data Quality Management (DQM) practices. While these technologies enabled advanced automation and data-driven decision-making, they also amplified risks related to bias, hallucination, misinformation, lack of transparency, inconsistent accuracy driven by training context, deskilling of employees, filter bubbles, inadvertent leakage of sensitive information, and declining trust in model outputs. These issues exposed fundamental limitations in how data quality had traditionally been defined, measured, and governed within organizations.

Current DQM frameworks and practices predominantly emphasized syntactic data quality dimensions, including validity, completeness, consistency, timeliness, uniqueness, and structural conformity. Although these dimensions remained necessary for ensuring operational data reliability, they proved insufficient for addressing the semantic challenges introduced by GenAI and LLMs. Semantic data quality dimensions—those concerned with meaning, context, relevance, bias, and interpretability—were not consistently prioritized or operationalized in widely adopted DQM frameworks, despite their critical importance to AI-driven systems.

The limited focus on semantic data quality created a substantial gap between Information Quality (IQ) theory and IQ practice. Established IQ theory frameworks, such as Beyond Accuracy and Total Data Quality Management, had long emphasized a broader set of dimensions that included accuracy of meaning, value added, relevance, objectivity, believability, reputation, security, and appropriate volume of data. However, the analysis of prevailing DQM practices demonstrated that many of these dimensions were either weakly addressed or omitted entirely in operational frameworks, particularly those most relevant to the performance and trustworthiness of GenAI and LLM outputs.

This misalignment posed heightened risks for organizations adopting GenAI and LLM technologies. Failure to adequately address semantic data quality dimensions increased exposure to regulatory non-compliance, financial loss, reputational damage, erosion of stakeholder trust, and unreliable AI-generated outputs. The research further identified a persistent misconception within data quality practice that equated data validation with data accuracy. While validation confirmed adherence to predefined rules or formats, it did not guarantee semantic correctness or contextual accuracy unless validation was performed against authoritative data sources. This misunderstanding further undermined data integrity in AI-enabled environments.

Consequently, a critical problem existed in the inability of current DQM practices to ensure data quality integrity suitable for GenAI and LLM adoption. Data quality and data governance functions had not sufficiently evolved to account for the semantic demands of language-based AI systems or to address the interdisciplinary risks they introduced. Addressing this problem required a systematic re-examination of DQM frameworks, increased prioritization of semantic data quality dimensions, and stronger collaboration across data governance, data science, data

risk management, security, and data product disciplines. Without such changes, organizations risked deploying GenAI and LLM solutions that lacked transparency, reliability, and trust, thereby undermining the intended value of these technologies.

1.3 Purpose of the Study

The purpose of this study was to examine contemporary Data Quality Management (DQM) practices in light of the widespread adoption of Large Language Models (LLMs) and Generative Artificial Intelligence (GenAI), with the intent of closing the persistent gap between Information Quality (IQ) theory and IQ practice. While advances in AI technologies have significantly expanded organizational capabilities, they have also amplified risks related to data meaning, context, bias, explainability, and trustworthiness. This study evaluated whether widely adopted DQM frameworks adequately addressed the semantic data quality dimensions required to support reliable, transparent, and trustworthy AI-driven outcomes.

Information Quality theory, including foundational frameworks such as Beyond Accuracy and Total Data Quality Management (TDQM), has long emphasized a comprehensive set of data quality dimensions that extend beyond syntactic correctness to include semantic attributes such as meaning, relevance, interpretability, and contextual integrity. In contrast, many DQM frameworks commonly used in practice were found to prioritize syntactic dimensions of data quality, including structure, format, completeness, and conformance. Although sufficient for traditional reporting and transactional systems, this syntactic emphasis proved increasingly inadequate in environments where GenAI and LLMs relied extensively on semantic coherence to generate valid and trustworthy outputs.

Accordingly, this research systematically analyzed a set of widely implemented DQM frameworks to assess their alignment with IQ theory in their treatment of data quality

dimensions. The frameworks examined included the Data Management Body of Knowledge (DAMA-DMBOK), Conformed Dimensions of Data Quality (CDDQ), Data Management Capability Assessment Model (DCAM), Six Sigma Data Quality (SSDQ), Data Quality Assessment Framework (DQAF), ISO 8000 Part 61, Three-Dimensional Analysis, Ten Steps to Data Quality by McGilvray, and BCBS 239. These frameworks represented commonly adopted industry standards, regulatory guidance, and operational methodologies shaping data quality practice across multiple sectors.

The central objective of the study was to identify and expose gaps in current DQM practices, particularly the underprioritization or omission of semantic data quality dimensions that were essential for managing the risks introduced by GenAI and LLM adoption. The analysis examined how frequently, and explicitly semantic dimensions were acknowledged within the selected frameworks and compared those findings with the full set of dimensions advanced by IQ theory. This comparison revealed a consistent misalignment between theory and practice, with semantic dimensions related to meaning, context, interpretability, and trust receiving limited attention in operational frameworks.

To validate and strengthen these findings, the study incorporated a Delphi methodology involving a panel of subject-matter experts in data quality, data governance, and AI-enabled systems. The Delphi study was used to achieve expert consensus regarding the existence and significance of the identified gaps, as well as to solicit feedback on the prioritization of semantic data quality dimensions and practical recommendations for enhancing existing DQM frameworks. This consensus-based approach ensured that the study's conclusions were both theoretically sound and grounded in professional practice.

The study also addressed prevalent misconceptions within data quality practice, particularly the

assumption that data validation alone ensured data accuracy. While validation processes confirmed compliance with predefined rules and formats, they did not guarantee semantic correctness or contextual accuracy—attributes critical for GenAI and LLM performance. The findings emphasized the importance of authoritative data sources, semantic consistency, and governance mechanisms in establishing data integrity within AI-driven environments.

Overall, the purpose of this research was to contribute to both academic scholarship and professional practice by clarifying how DQM frameworks must evolve to meet the semantic demands of AI-driven technologies. By explicitly examining the gap between IQ theory and IQ practice, the study provided actionable insights to support organizations in mitigating AI-related risks, improving confidence in AI-generated outputs, and strengthening trust in data-driven decision-making. The outcomes of this research support the advancement of more transparent, resilient, and semantically grounded data quality and governance practices suitable for the era of GenAI and LLMs.

1.4 Research Questions and Hypotheses

1.4.1 Quantitative Component of the Study

RQ1: To what extent do current DQM practices incorporate semantic data quality dimensions for GenAI and LLM applications?

H0: Current DQM practices sufficiently incorporate semantic DQ dimensions.

H1: Current DQM practices do not sufficiently incorporate semantic DQ dimensions.

1.4.2 Qualitative Component of the Study

RQ2: What semantic DQ dimensions are perceived as most critical for GenAI/LLMs?

RQ3: What actions should organizations take to prioritize semantic DQ?

1.5 Significance and Limitations of the Study

This study is significant as it provides a thorough analysis and documentation of existing gaps in DQM practices, raising awareness and encouraging organizations to critically assess their current approaches and the data quality dimensions they prioritize. With the adoption of GenAI and LLMs pushing the boundaries of data quality practices, this research is both timely and essential. Examining Data Quality Management for data quality dimension gaps is crucial in today's landscape, given the challenges introduced by LLMs and GenAI applications. The study aims to improve the performance and trustworthiness of these technologies while contributing to Data Quality Management and Data Governance programs. Additionally, it helps bridge the gap between data quality theory and practice in response to the growing application of generative AI and LLM models.

This research is crucial for ensuring the integrity, reliability, and ethical compliance of data used in Generative AI systems while also fostering trust in model outcomes by adopting the recommendations from this study. By raising the priority of semantic data quality dimensions, organizations can minimize risks, enhance transparency, and strengthen confidence in GenAI and LLMs processes and products.

This study will not develop a new DQM framework or introduce new Data Quality dimensions nor propose calculation methods for proposed semantic dimensions. However, future research could explore this opportunity by proposing more robust frameworks which prioritizes semantic dimensions essential for measuring, monitoring, and mitigating risks associated with LLMs and GenAI. Instead, this research establishes a strong foundation, and its recommendations can be utilized to enhance current practices, ensuring organizations maximize the value of adopting LLMs and GenAI models.

1.6 Future Work

As this study does not aim to modify existing DQM frameworks or introduce new Data Quality dimensions for GenAI and LLMs, future research could expand on its recommendations to develop new robust DQM frameworks or introduce additional DQ dimensions to account for issues introduced by the adoption of LLMs and GenAI. Also, the study was limited to exposing the gaps on the overlooked dimensions, future researchers based on the outcome of this study can develop calculations formula or methodology for measuring those semantic dimensions and measures experts highly rated or recommended.

Rather than directly addressing Data Quality issues, the primary goal of this study was to raise awareness, highlight existing gaps, and ensure that current practices prioritize the often-overlooked semantic Data Quality dimensions.

Additionally, the adaptability and effectiveness of the recommended approach can be further enhanced by expanding its scope of application.

1.7 Summary

The first chapter sets the stage for the dissertation by stressing the need to examine Data Quality Management (DQM) practices in the era of Generative AI (GenAI) and Large Language Models (LLMs). Existing DQM frameworks and data governance programs focus primarily on syntactic data quality dimensions such as validity, completeness and consistency. While necessary, these dimensions are insufficient to address challenges posed by GenAI and LLMs—bias mitigation, hallucinations, misinformation, transparency and ethical compliance. The chapter argues for greater emphasis on semantic data quality dimensions—accuracy, value added, relevancy, objectivity, reputation, believability, security and appropriate amount of data—because they influence the meaning, context and trustworthiness of data used by AI systems.

The problem statement highlights that most widely used DQM frameworks—including the DAMA Data Management Body of Knowledge, Conformed Dimensions of Data Quality

(CDDQ), Data Management Capability Assessment Model (DCAM), Six Sigma Data Quality framework, Data Quality Assessment Framework (DQAF), Three-Dimensional Analysis, ISO 8000, Ten Steps of Data Quality and the BCBS239 standard—pay insufficient attention to these semantic dimensions. This gap exposes organizations adopting GenAI and LLMs to regulatory penalties, financial losses, erosion of stakeholder trust and unreliable model outputs. To explore this issue, the dissertation compared the coverage of semantic dimensions in these practice-oriented frameworks with information quality theory frameworks such as Beyond Accuracy and Total Data Quality Management (TDQM).

The purpose of the study was to examine and document the gap in current DQM practices and seek expert consensus, via a Delphi study, on the need to prioritize semantic data quality dimensions. By identifying overlooked dimensions and soliciting feedback from data quality specialists, the research aims to develop recommendations for improving DQM frameworks without introducing new dimensions or calculation methods. Future studies could extend this work by proposing formulas or developing new frameworks tailored to GenAI/LLM environments.

This research is significant because it raises awareness of the disconnect between data quality theory and practice and provides a roadmap for enhancing DQM and data governance. By advocating for stronger attention to semantic data quality, the study aims to bolster the integrity, reliability and ethical compliance of AI-driven systems, thereby supporting organizational transparency, mitigating risk and fostering confidence in GenAI and LLM outputs. Although the study acknowledges that its impact depends on practitioner adoption and further methodological development, it underscores the urgency of revisiting data quality priorities as AI technologies continue to evolve.

Chapter 2: Literature Review

2.1 Introduction

This literature review examined and synthesized scholarly and professional research on Data Quality Management (DQM) frameworks to assess their effectiveness in addressing the challenges introduced by the adoption of Large Language Models (LLMs) and Generative Artificial Intelligence (GenAI). The review focused on identifying gaps between Information Quality (IQ) theory and prevailing DQM practices, with particular attention to the limited emphasis placed on semantic data quality dimensions that are essential for ensuring reliable, trustworthy, and contextually accurate AI-generated outputs.

The review analyzed a selection of widely adopted DQM frameworks, standards, and methodologies that continue to shape data quality practices across industries. These included, but were not limited to, the Data Management Body of Knowledge (DAMA-DMBOK), Conformed Dimensions of Data Quality (CDDQ), Data Management Capability Assessment Model (DCAM), Six Sigma Data Quality (SSDQ), Data Quality Assessment Framework (DQAF), Three-Dimensional Analysis, ISO 8000 Part 61, Ten Steps to Data Quality by McGilvray, and BCBS 239 issued by the Basel Committee on Banking Supervision. Together, these frameworks represented commonly implemented approaches to managing and assessing data quality in enterprise and regulated environments.

The literature revealed that most DQM frameworks in practice emphasized data quality outcomes without clearly defining what should be measured or providing guidance on prioritizing data quality dimensions. In particular, the analysis demonstrated that these frameworks predominantly focused on syntactic data quality attributes, such as structure, completeness, consistency, and conformity, while providing limited or implicit treatment of semantic

dimensions. This imbalance was found to be increasingly problematic in the context of GenAI and LLMs, which depend heavily on semantic accuracy, contextual relevance, and meaning preservation to generate dependable results.

To assess the extent of this misalignment, the literature review evaluated how frequently and explicitly the reviewed frameworks addressed the fifteen data quality dimensions proposed in the Wang–Strong Information Quality framework. The findings indicated that many of the semantic dimensions advocated by IQ theory—particularly those related to meaning, relevance, believability, and contextual accuracy—were either underrepresented or omitted in operational DQM frameworks. These omissions are especially consequential for addressing issues such as bias, hallucination, and inconsistency in GenAI and LLM outputs.

Building on these findings, the literature review established the foundation for a Delphi study designed to validate the identified gaps through expert consensus. The Delphi process was used to confirm the relevance and severity of the underprioritization of semantic data quality dimensions and to solicit expert input on the prioritization and integration of these dimensions into DQM practice. By synthesizing theoretical perspectives and applied frameworks, the literature review framed the central problem addressed by this dissertation and positioned the study to contribute toward closing the gap between IQ theory and IQ practice in the era of GenAI and LLM adoption.

2.2 Literature Search Strategy

This dissertation employed a systematic, iterative, and transparent literature search strategy to identify, evaluate, and synthesize relevant academic and professional sources addressing data quality, data governance, and the implications of generative artificial

intelligence (GenAI) and large language models (LLMs). Given the interdisciplinary scope of the study and the rapid pace of technological evolution, the search strategy was designed to balance scholarly rigor, methodological depth, practical relevance, and contemporaneity.

The literature search was guided by core research themes derived directly from the dissertation study's topic and questions. These themes included data quality dimensions, data quality and governance management frameworks, semantic and syntactic data quality, data validation and trust, GenAI and LLM architectures, AI governance, and the impact of data quality on the reliability, bias, and trustworthiness of AI-generated outputs. Keywords and phrases were iteratively developed and refined, incorporating combinations such as 'data quality frameworks,' 'data governance standards,' 'semantic data quality dimensions,' 'generative artificial intelligence,' 'large language models,' 'AI governance,' and 'data quality impact on GenAI.' Boolean operators were systematically applied to expand and refine search results.

Primary searches for peer-reviewed academic literature were conducted using established scholarly databases, including IEEE Xplore, ACM Digital Library, Scopus, Web of Science, ScienceDirect, and Google Scholar. These databases were selected for their strong coverage of information systems, data management, and artificial intelligence research. Peer-reviewed journal articles and high-quality conference proceedings were prioritized to ensure theoretical robustness and empirical validity.

To establish foundational concepts and enduring theoretical frameworks, authoritative academic and professional books published by reputable scholarly presses were reviewed. These sources provided historical grounding, methodological clarity, and widely accepted definitions central to data quality and governance research. In addition, internationally recognized standards, institutional frameworks, and public policy documents were incorporated to ensure alignment

with prevailing professional practices and regulatory expectations, particularly within enterprise and regulated environments.

Recognizing that research on GenAI and LLMs is evolving faster than the traditional academic publication cycle, carefully curated industry reports, institutional research outputs, and expert-authored organizational publications were selectively included. These sources were used to contextualize emerging trends, practical challenges, and real-world adoption issues, while core theoretical arguments remained grounded in peer-reviewed and authoritative scholarly literature.

All identified sources underwent a critical appraisal process based on explicit inclusion and exclusion criteria. Inclusion criteria required demonstrated topical relevance, credible authorship, reputable publication venues, methodological rigor, and alignment with the study's objectives. Exclusion criteria eliminated non-authoritative opinion pieces, unsupported claims, and sources lacking clear provenance. Tertiary sources such as Wikipedia were used solely for preliminary topic familiarization and were replaced with authoritative secondary scholarly references in the final citation set.

The literature review process was iterative, incorporating backward and forward citation tracking to identify seminal works as well as recent contributions. This approach ensured comprehensive coverage of both foundational theories and current developments, enabling a rigorous examination of the gap between information quality theory and its practical application in the context of GenAI and LLM adoption. Through this structured strategy, the resulting literature base provides a credible, defensible, and academically sound foundation for the analyses and conclusions presented in this dissertation.

2.3 Nine Top List DQM Framework Selection Process

This study deliberately selected nine Data Quality Management (DQM) frameworks to examine how contemporary data quality practices align with, or diverge from, established Information Quality (IQ) theory in the context of Large Language Models (LLMs) and Generative Artificial Intelligence (GenAI). The selection process was grounded in a systematic, iterative, and transparent literature search strategy designed to balance theoretical rigor with practical relevance across industries where data quality failures carry material operational, regulatory, and ethical consequences. Given the interdisciplinary and rapidly evolving nature of GenAI and LLM adoption, it was essential to focus on frameworks that not only possess strong academic and institutional foundations but are also actively used by practitioners in sectors such as banking, finance, healthcare, government, academia, and regulated industries.

The nine frameworks selected—DAMA-DMBOK, Conformed Dimensions of Data Quality (CDDQ), Data Management Capability Assessment Model (DCAM), Six Sigma Data Quality (SSDQ), Data Quality Assessment Framework (DQAF), Three-Dimensional Analysis, ISO 8000 Part 61, Ten Steps to Data Quality by McGilvray, and BCBS 239—emerged consistently during the literature review as the most frequently cited, implemented, and relied upon references for defining, operationalizing, measuring, and governing data quality in practice. Unlike many niche or proprietary models, these frameworks represent a convergence of academic research, professional standards, regulatory guidance, and real-world operational deployment. Their persistence across time and industries made them particularly suitable for evaluating how data quality dimensions are interpreted and prioritized in practice, especially when contrasted with the conceptual IQ framework articulated by Wang and Strong in *Beyond Accuracy*.

An important criterion for selection was the degree to which each framework explicitly or implicitly proposed data quality dimensions that could be examined, compared, and cross-referenced against the fifteen dimensions articulated in IQ theory. While numerous other frameworks and models were identified during the literature search, many were excluded

because they either lacked a clear articulation of data quality dimensions, focused narrowly on technical implementation without conceptual grounding, or were confined to limited industry-specific use cases. In contrast, the selected nine frameworks offered sufficiently articulated perspectives on data quality attributes—whether through dimensions, principles, capabilities, or assessment criteria—to enable systematic comparison with Wang and Strong’s consumer-oriented conceptual model.

Another factor influencing the selection was the practical authority of these frameworks within organizations responsible for managing high-risk, high-impact data environments. For example, DAMA-DMBOK serves as a globally recognized reference for data management professionals; DCAM is widely adopted by financial institutions to assess and mature enterprise data capabilities; BCBS 239 carries regulatory authority for systemically important banks; and ISO 8000-61 provides an internationally recognized process reference model. Similarly, McGilvray’s Ten Steps framework and SSDQ are frequently used by practitioners to design and execute data quality initiatives, while DQAF and Three-Dimensional Analysis offer structured assessment and profiling approaches. Together, these frameworks collectively reflect how data quality is operationalized across enterprise, regulatory, and analytical contexts.

Crucially, these frameworks were not selected because they are deficient; rather, they were chosen precisely because they represent best practice within contemporary DQM as it is commonly implemented. Each framework excels in areas such as governance enablement, process discipline, measurement consistency, compliance, and operational scalability. As such, they provide a fair and non-biased basis for evaluating whether even the most authoritative and widely adopted DQM approaches sufficiently address semantic data quality dimensions—dimensions that have become central to the reliability, trustworthiness, and ethical performance of GenAI and LLM systems.

The comparative analysis revealed that, despite their strengths, these frameworks collectively prioritize syntactic data quality dimensions—such as validity, completeness,

consistency, timeliness, uniqueness, and conformity—because these attributes are easier to define, automate, measure, and audit. While necessary, this syntactic emphasis reflects legacy data environments primarily designed for transactional processing, reporting, and rule-based validation. In contrast, semantic data quality dimensions emphasized in IQ theory—such as meaning accuracy, relevance, value added, objectivity, believability, reputation, contextual fitness, security, and appropriate amount of data—were inconsistently addressed, loosely defined, or omitted altogether in practical frameworks.

This imbalance became particularly consequential in the context of GenAI and LLM adoption, where models are trained to infer meaning, context, and intent rather than merely conform to structural rules. The selected frameworks, by virtue of their widespread adoption, provided compelling evidence that prevailing data quality practices have not evolved at the same pace as AI technologies. As a result, organizations relying on these frameworks risk deploying AI systems that are syntactically correct yet semantically flawed, thereby increasing exposure to bias, hallucination, misinformation, regulatory non-compliance, loss of stakeholder trust, and reputational damage.

Finally, selecting these nine frameworks enabled a comprehensive and defensible assessment of the theory–practice gap central to this dissertation. By comparing widely trusted, authoritative DQM frameworks against the Beyond Accuracy conceptual model, the study avoided the bias of evaluating marginal or lesser-used approaches and instead focused on the very practices organizations depend upon most. This selection strategy ensured that any identified gaps in semantic data quality were not artifacts of weak framework choice but indicative of systemic issues within mainstream Data Quality Management practice. Consequently, the findings offer high external validity and provide a credible foundation for the subsequent Delphi study, which sought expert consensus on prioritizing semantic data quality dimensions essential for trustworthy and reliable GenAI and LLM deployments.

2.3.1 Summary Rationale and Cross-Industry Relevance

The identification of the nine frameworks followed a systematic, iterative, and transparent literature review strategy designed to balance scholarly rigor with practical relevance. Searches were guided by the dissertation topic and research questions and were organized around core themes including data quality dimensions, data quality management frameworks, data governance standards, semantic and syntactic data quality, data validation and trust, and the impact of GenAI and LLMs on reliability, bias, and trustworthiness. Keywords were iteratively refined using combinations such as data quality frameworks, data governance standards, semantic data quality dimensions, generative artificial intelligence, large language models, AI governance, and data quality impact on GenAI, with Boolean operators used to broaden and narrow the search as needed. The review drew on peer-reviewed databases such as IEEE Xplore, ACM Digital Library, Scopus, Web of Science, ScienceDirect, and Google Scholar, while also incorporating authoritative books, institutional standards, regulatory guidance, and carefully curated industry sources in order to capture both foundational theory and current professional practice.

Frameworks were retained only when they met clear inclusion criteria: topical relevance to data quality management, credible authorship or institutional authority, recognized use in professional practice, and sufficiently explicit treatment of data quality dimensions, principles, capabilities, or assessment criteria to allow comparison with the fifteen dimensions in Wang and Strong's IQ framework. Models were excluded when they were narrowly technical, proprietary, weakly documented, or limited to specialized use cases without broader cross-industry relevance. The review process used backward and forward citation tracking and repeated appraisal until no additional widely used, dimension-oriented frameworks with comparable authority and applicability were identified. At that point, the search was deliberately bounded at nine frameworks because these nine consistently appeared as the most cited, most operationally relevant, and most defensible set for evaluating how mainstream DQM practice

aligns with IQ theory across banking, finance, healthcare, government, technology, retail, academia, and regulated environments. Their breadth of adoption, combined with their role in governance, assessment, compliance, and DQ measurement, made them an appropriate preliminary discovery set to take forward into the Delphi study for expert validation.

The review stopped at nine when no additional frameworks were found that were both widely used across industry and sufficiently explicit about data quality dimensions to support a credible IQ-versus-practice comparison.

Table 1

Summary of the Nine Selected DQM Frameworks

Framework	Brief description	Why retained / relevance	Popularity / authority	Common industry usage
DAMA-DMBOK	Global body of knowledge for enterprise data management.	Represents mainstream practitioner guidance and commonly referenced DQ dimensions in organizations.	Widely recognized professional reference; strong enterprise authority.	Banking, financial services, healthcare, government, retail, technology, academia
CDDQ	Framework organizing conformed data quality dimensions.	Useful for mapping operational DQ dimensions to conceptual IQ dimensions.	Frequently cited in DQ dimension discussions and comparative work.	Banking, enterprise data programs, analytics teams, academia
DCAM	Capability assessment model for enterprise data management and control.	Shows how DQ is operationalized in high-risk, regulated environments.	Strong institutional authority through the EDM Council; widely adopted in finance.	Banking, financial services, risk and compliance, regulators
SSDQ	Six Sigma-oriented framework for executing DQ improvement initiatives.	Captures process discipline, measurement, and continuous improvement in practice.	Well known in practitioner and quality-improvement contexts.	Manufacturing, operations, finance, healthcare, technology
DQAF	Structured framework for assessing and profiling data quality.	Provides practical assessment criteria that can be compared against IQ theory.	Commonly referenced where formal DQ assessment is needed.	Government, statistical agencies, finance, healthcare, academia
Three-Dimensional Analysis	Practical profiling approach that evaluates data quality from multiple analytical angles.	Adds an applied, assessment-based perspective to the framework set.	Useful as a practitioner-oriented profiling method.	Analytics, data profiling, technology, consulting, academia
ISO 8000-61	International process reference model for data quality management.	Brings formal standardization and globally recognized institutional grounding.	High authority because it is an international standard.	Global enterprises, regulated industries, government, manufacturing, technology
Ten Steps to Data Quality	Practitioner-focused roadmap for designing and running DQ programs.	Important because it is widely used in real-world DQ initiative execution.	Popular practitioner reference with strong operational usability.	Banking, healthcare, retail, technology, consulting, academia
BCBS 239	Regulatory framework for risk data aggregation and reporting.	Demonstrates how DQ is governed where reliability, traceability, and compliance are critical.	High regulatory authority for systemically important banks.	Banking, financial services, regulators, risk and compliance

Note - The table summarizes why these frameworks were suitable for a preliminary, practice-oriented comparison of DQM against IQ theory before Delphi validation. Industry usage categories reflect the sectors in which the framework is commonly referenced or applied in the dissertation literature review.

2.4 The Conceptual Data Quality Frameworks

2.4.1 Beyond Accuracy: What Data Quality Means to Data Consumer

Wang and Strong's conceptual Data Quality framework served as a foundational theoretical baseline for this study by defining data quality from the perspective of data consumers rather than system designers. In their seminal work *Beyond Accuracy: What Data Quality Means to Data Consumers*, Wang and Strong argued that data quality could not be adequately assessed by accuracy alone but instead should be understood as "fitness for use," reflecting how data supports users' decision-making needs [2]. Drawing on prior literature, quality theory, and empirical insights from data consumers, they proposed a hierarchical framework comprising four interrelated categories of data quality dimensions: intrinsic, contextual, representational, and accessibility. Intrinsic data quality encompassed attributes such as accuracy, objectivity, believability, and reputation, reflecting whether data were perceived as correct and trustworthy. Contextual data quality captured value-added, relevance, timeliness, completeness, and the appropriate amount of data, emphasizing the situational nature of data use. Representational data quality addressed interpretability, ease of understanding, representational consistency, and concise representation, focusing on how meaning was conveyed to users. Accessibility data quality incorporated both accessibility and access security, acknowledging that users must be able to retrieve data safely and effectively. This framework established that data must be accessible, interpretable, relevant, and credible to be considered high quality by consumers, and it provided a comprehensive theoretical lens through which this dissertation evaluated the extent to which contemporary Data Quality Management practices aligned with Information Quality theory, particularly in the context of LLMs and Generative AI [2].

Table 2

Description of Data Quality Dimension (Beyond Accuracy)

Description of Dimensions (Beyond Accuracy by Richard Wang and Diane M. Strong)		
Category	Dimension	Attribute list
Intrinsic Data Quality	Believability	Believable
	Accuracy	data are certified error-free, accurate, correct, flawless, reliable, errors can be easily identified, the integrity of the data, precise
	Objectivity	unbiased, objective
	Reputation	reputation of the data source, reputation of the data
Contextual Data Quality	Value-added	data give you a competitive edge, data add value to your operations
	Relevancy	applicable, relevant, interesting, usable
	Timeliness	age of data
	Completeness	breadth, depth, and scope of information contained in the data
	Appropriate amount of data	the amount of data
Representational Data Quality	Interpretability	interpretable
	Ease of understanding	easily understood, clear, readable
	Representational consistency	data are continuously presented in the same format, consistently represented, consistently formatted, data are compatible with previous data
	Concise representation	well-presented, concise, compactly represented, well-organized, aesthetically pleasing, form of presentation, well-formatted, format of the data
	Accessibility	accessible, retrievable, speed of access, available, up to date

Accessibility Data Quality	Access security	data cannot be accessed by competitors, data are of a proprietary nature, access to data can be restricted, secure
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Wang and Strong's consumer-oriented framework demonstrated that high data quality extended beyond syntactic correctness to include semantic attributes such as meaning, relevance, believability, and contextual appropriateness, which collectively defined data fitness for use [2]. While traditional Data Quality Management practices largely operationalized only syntactic dimensions, the framework highlighted that intrinsic, contextual, and representational quality were essential for ensuring that data were interpretable, trustworthy, and task-relevant. This theoretical emphasis became particularly salient in the context of Large Language Models and Generative Artificial Intelligence, where model outputs depended heavily on semantic coherence rather than structural validity alone. The limited incorporation of these semantic dimensions in contemporary DQM practices therefore exposed a critical gap between Information Quality theory and practice, contributing to risks such as hallucination, bias, and reduced trust in GenAI and LLM outputs [2].

2.4.2 Total Data Quality Management (TDQM)

The Total Data Quality Management (TDQM) framework, developed by Richard Y. Wang at the Massachusetts Institute of Technology, adopted a holistic and process-oriented approach to managing information quality by treating data as an information product that moved through an information manufacturing system [20], [45]. Drawing inspiration from the Deming cycle in Total Quality Management, TDQM conceptualized data quality improvement as a continuous cycle of defining, measuring, analyzing, and improving information quality. Rather than prescribing a fixed set of data quality dimensions, the framework emphasized that organizations should define and prioritize quality dimensions based on consumer needs and usage contexts, thereby reinforcing the principle of information quality as fitness for use. This

adaptability made TDQM applicable across diverse organizational settings while maintaining strong theoretical grounding.

Within the TDQM methodology, information quality was explicitly linked to the multidimensional framework proposed in *Beyond Accuracy*, encompassing intrinsic, contextual, representational, and accessibility dimensions [45]. Accuracy was positioned as only one component of intrinsic quality, while contextual relevance, interpretability, and accessibility were recognized as equally critical to effective data use. TDQM research demonstrated that information quality problems were often most visible to consumers, particularly in contextual dimensions; however, responsibility for improvement rested with information producers and governance structures rather than end users. By emphasizing proactive, continuous improvement supported by cross-functional collaboration and executive commitment, TDQM established a theoretical and methodological foundation for evaluating how Data Quality Management practices should evolve—an insight that became increasingly pertinent in the context of LLMs and Generative AI, where semantic understanding and contextual fitness were central to trustworthy system outputs [20], [45].

2.5 Data Management Body of Knowledge (DAMA - DMBOK)

The Data Management Body of Knowledge (DAMA-DMBOK) framed data and information as critical enterprise assets whose value depended on deliberate, coordinated, and cross-functional management across the data lifecycle, encompassing governance, quality, security, and use [1]. The framework emphasized that organizations could not derive strategic value from data by chance; instead, effective data management required strong leadership, shared responsibility between business and information technology roles, and sustained collaboration to align data with business needs. High-quality data was positioned as foundational to organizational performance, decision-making, and risk mitigation, with data quality defined as “fit for purpose” and operationalized through measurable dimensions intended

to support objective assessment and transparency. However, while DAMA-DMBOK provided a comprehensive management and governance foundation, its practical guidance largely emphasized standards-driven, measurable dimensions and processes, leaving limited explicit treatment of how data meaning, interpretation, and contextual understanding should be prioritized. This orientation reflected broader industry practice and contributed to the ongoing gap between Information Quality theory and practice that this dissertation examined in the context of LLM and GenAI adoption, where trustworthy outcomes depended increasingly on semantic coherence rather than solely on structural or procedural correctness [1].

Table 3

Common Dimensions of Data Quality (DAMA-DMBOK framework)

Dimension of Quality	Description
Validity	Validity refers to whether data values are consistent with a defined domain of values. A domain of values may consist of: 1. A data type (e.g., text with or without special characters, a number, or a date) 2. A data range (e.g., a numeric range, date range, or a set of valid text values in a reference table) 3. A format (e.g., a phone number must contain country, area codes, and the correct number of digits or a currency must have only two decimal places) 4. The precision of expected values. For example, a date-stamp must be recorded to a millisecond or a number recorded to no more than two decimal places. 5. The time frame (certain values may also only be valid for a specific length of time, for example, data that is generated from RFID (radio frequency ID) or some scientific data sets.)
Completeness	Completeness refers to whether all the required data is present. For example, are all the mandatory components of an address populated, including the street number, street name, city, and country? For example, if the customer account is not marked as a debt risk, there must not be a debt manager appointed. Completeness can be measured at the column, record, or data set level. 1. Are columns/data elements populated to the level expected? (Some columns are mandatory. Optional columns are populated only under specific conditions.)

	2. Are records populated correctly? (Records with different statuses may have different expectations for completeness.) 3. Does the data set contain all the records expected?
Consistency	Consistency is ensuring that data values are coded using the same approach, assessment and valuation criteria. Consistency is between two different data items, potentially within a data set and between data sets, and across time. Consistency may be defined between one set of data element values and another data element set within the same record (record-level consistency), between one set of data element values and another data element values and the same data element set within the same records at different points in time (temporal consistency). Consistency can also be used to refer to consistency of format. <u>Take care not to confuse consistency with accuracy or correctness</u> Consistency can be measured using the following ways: 1. Cross record consistency is between data values in the same column – are all the customer addresses recording the head-office details or do some record service delivery centers? 2. Between associated data sets, where data sets are expected to have values that are linked. For example, when moving data from data capture systems to data warehousing systems, a consistency quality measure will maintain confidence in the values in the data warehouse. 3. Across time can be applied to any of the previous cases. For example, has the grading of students been the same over time? If a subject is completed previously, would the same learning performance yield the same grade today?
Integrity	Integrity refers to the lack of incoherent values and broken relationships in data. Integrity can be measured using the following ways: 1. Coherence – where one data value implies a limited range in another data value, and they are matching. For example, a country has a defined set of states or provinces. If a customer address is in a certain country, they must use a state or province name for the associated set. For example, Alberta is a province of Canada, Pisa is a province of Italy, and Tasmania is a state of Australia. 2. Parent-child (referential integrity) – where every child must have a parent data value. For example, all the customer address country names must exist on the valid list of countries where the organization is authorized to sell products to. Data sets without integrity are seen as corrupted or have data loss. Data sets without referential integrity have 'orphans' – invalid reference keys, or duplicates' – identical rows which may negatively affect aggregation functions

Timeliness	Data Timeliness refers to the time that data needs to become accessible to a user after its capture or update. Timeliness is the expectation and actual delay before the data becomes available. For example: An electricity utility needs to maintain its power grid and must have demand data measured and available to a system operator within a few seconds. Otherwise, electricity generation may get out of alignment with demand, causing an energy excess or shortage. As another example, a statistical agency of government produces the Gross Domestic Product (GDP) report two months after the end of the quarter. In order to meet this timeliness requirements, all the data collections that make up the GDP will have their own specific timeliness requirement.
Currency	Currency is the date that the data was last updated relative to now and likelihood that it is still correct. Different data sets will have different currency expectations from relatively static data to highly volatile data. Static data remains current for a long period. Volatile data remains current for a long period. A business' bank account is relatively volatile as the balance is constantly changing. Volatile data, like stock prices on financial web pages, will often be shown with an "as-of-time", so that data consumers understand the risk that the data has changed since it was recorded. During the day, while the markets are open, such data will be updated frequently – it is volatile. Once market closes, the data will remain unchanged but will still be current since the market itself is inactive – it is static. Key terms used in currency include: 1. Update time – the time stamp of the last update 2. Volatility measures the rate of change of the date. It may measure across all the data values or measure within a segment. For example, the rate that domestic customers change their address may be different from the rate that corporate customers change their address. 3. Latency measures the time between when the data was created and when it was made available for use. For example, overnight population of a data warehouse may end up with different data latencies. Data can have a latency of 1 day for data entered into the system early on the prior day, but only a few minutes for data generated just before the load. (See chapter 8)
Reasonableness	Reasonableness asks whether a data pattern meets expectations. For example, whether a distribution of sales in a geographic area makes sense based on what is known about the customers in that area. For example, based on previous customer logins at 5pm, are today's customer logins to our systems out of the ordinary? Key terms used in reasonableness include: 1. Fixed values – this is where the measure is fixed, not dependent on previous behavior. The maximum currency value of a transaction may be limited in size by the technical limits of

	a bank's systems and other values bigger than this value is "unreasonable". 2. Benchmark value – it is possible to measure the average, deviation or any other statistical function and measure the latest set of values conforms to expected boundaries. For example, there have been 5, 000 and 10, 000 customer address changes per day, but yesterday there were 2,000,000. On a government data collection, the average size company has 20 employees, but the latest returns had an unusual number with an average of 1,500 employees reported.
Uniqueness/Deduplication	Uniqueness states that no real-world entity exists more than once within the data set. Asserting uniqueness of the entities within a data set implies that each row relates to each unique entity in the real world, and only that specific entity. Uniqueness can be identified using the following techniques: 1. Key Structure – where there are duplicate keys within the data set (see chapter 5). For example, in a data set containing customers, many of which share the same customer number. 2. Related data – where other data in the data may point to a duplicate. For example, two customers with different customer numbers, but having the same name, date of birth, and office address. 3. Inquiries or complaints that users cannot access certain data. For example, a customer with multiple services can only access some of their services as the others are recorded under a different customer number (which is duplicative).
Accuracy	Accuracy refers to the degree that data correctly represents 'real-life' entities. For example, is the person's name in our database actually the person's name in real life? Does the customer actually use that email address? Accuracy is difficult to measure unless an organization can reproduce data collection or manually confirm accuracy of records. Most measures of accuracy use the other dimensions to imply accuracy. Common techniques to improve accuracy include: 1. Checking consistency with a data source that has been verified as accurate; a government company register (system of record) or a commercial company Information Broker (system of reference). 2. Ongoing calibration of devices with reality. For example, over time, weather gauges may drift from providing accurate measurements. An annual program of field inspections visits 10% of the devices and recalibrates them to a standard to ensure they are delivering accurate measurements. 3. Sampling data values with reality. For example, small number of emails are sent to customers (mainly) to drive sales, but also

	have the benefit of checking the accuracy of the email address (does it exist, is it used)?
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2.6 Conformed Dimensions of Data Quality (CDDQ)

The Conformed Dimensions of Data Quality (CDDQ) framework, developed by Dan Myers, sought to address longstanding fragmentation in Data Quality Management practice by proposing a standardized, cross-industry set of data quality dimensions designed to improve clarity, consistency, and measurement [47], [48]. The framework was motivated by the recognition that conflicting definitions and overlapping terminology across data quality methodologies undermined effective communication, comparability, and repeatability, particularly in enterprise and analytical environments. CDDQ emphasized standardization as a prerequisite for objective measurement, continuous improvement, and meaningful comparison across departments, organizations, and industries, aligning with principles established in Six Sigma and Lean methodologies. Structurally, the framework decomposed data quality into hierarchical elements—dimensions, underlying concepts, and metrics—thereby providing a practical mechanism for translating abstract quality characteristics into measurable indicators. While CDDQ contributed significantly to harmonizing data quality language and measurement in practice, its adoption also illustrated a broader tendency within DQM frameworks to privilege operational consistency over deeper semantic considerations, a limitation that became increasingly salient with the adoption of Large Language Models and Generative Artificial Intelligence, where shared definitions, contextual meaning, and interpretability were central to bridging the gap between Information Quality theory and practice [47], [48].

Table 4

Conformed Dimensions of Data Quality

Conformed Dimension	Definition	Underlying Concepts	Non-Standard Terminology
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			for Dimension
Completeness	Completeness measures the degree of population of data values in a data set.	Record Population, Attribute Population, Truncation, Existence	Fill Rate, Coverage, Usability, Scope
Accuracy	Accuracy measures the degree to which data factually represents its associated real-world object, event,	Agree with Real-world, Match to Agreed Source	Consistency
Consistency	Consistency measures whether or not data is equivalent across systems or location of storage.	Equivalence of Redundant or Distributed Data, Format Consistency, Logical Consistency, Temporal Consistency	Integrity, Concurrence, Coherence
Validity	Validity measures whether a value conforms to a preset standard.	Values in Specified Range, Values Conform to Business Rule, Domain of Predefined Values, Values Conform to Data Type, Values Conform to	Conformity, Accuracy, Integrity, Reasonableness, Compliance
Timeliness	Timeliness is a measure of time between when data is expected versus made available.	Time Expectation for Availability, Manual Float, Electronic Float	Currency, Lag Time, Latency, Information Float
Currency	Currency measures how quickly data reflects the real-world concept that it represents.	Current with World it Models	Timeliness
Integrity	Integrity measures the structural or relational quality of data sets.	Referential Integrity, Uniqueness, Cardinality	Validity, Duplication
Accessibility	Accessibility measures how easy it is to acquire data when needed, how long it is retained, and how access is controlled.	Ease of Obtaining Data, Access Control, Retention	Availability, Security
Precision	Precision is the measurement or classification detail used in specifying an attribute's domain.	Precision of Data Value, Granularity, Domain Precision	Coverage, Detail
Lineage	Lineage measures whether factual documentation exists about where data came from, how it was transformed, where it went and end-to-end graphical illustration.	Source Documentation, Segment Documentation, Target Documentation, End-to-End Graphical Documentation	
Representation	Representation measures ease of understanding data, consistency of presentation, appropriate media choice	Easy to Read and Interpret, Presentation Language, Media Appropriate, Metadata Availability	Presentation

	ce, and availability of documentation (metadata).	ty, Includes Measurement Units	
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Conformed Dimensions of Data Quality (CDDQ) by Dan Myers 2019 Release 4.3

2.7 Data Management Capability Assessment Model (DCAM)

The Data Management Capability Assessment Model (DCAM) defined a comprehensive set of organizational capabilities required to establish and sustain mature data management practices, including a structured and operationally focused approach to Data Quality Management (DQM) grounded in business value and execution realism [3]. Within DCAM, data quality was framed as an attribute of fitness for purpose rather than a standalone process, emphasizing capabilities such as data profiling, rule definition, measurement, defect management, root-cause analysis, and remediation across the data supply chain. The framework operationalized data quality through a set of largely syntactic dimensions—including accuracy, completeness, coverage, conformity, consistency, timeliness, and uniqueness—reinforced by iterative profiling, grading, and rule-based validation tied to business requirements. While this capability-driven approach strengthened consistency, governance integration, and operational control, it largely emphasized measurable structural attributes of data, providing limited explicit guidance on semantic dimensions related to meaning, contextual interpretation, and believability. This limitation became increasingly consequential with the adoption of Large Language Models and Generative Artificial Intelligence, where reliable outputs depended not only on syntactic correctness but also on semantic coherence and contextual integrity, thereby exposing a gap between DCAM-aligned DQM practice and Information Quality theory that this dissertation examined [3].

2.8 Six Sigma Data Quality (SSDQ) Management

The Six Sigma Data Quality (SSDQ) framework applied the principles of Six Sigma and the DMAIC cycle—define, measure, analyze, improve, and control—to systematically reduce data defects and improve decision-making reliability through statistically driven process

improvement [50], [51], [53]. SSDQ operationalized data quality through measurable dimensions such as accuracy, completeness, consistency, timeliness, validity, uniqueness, and integrity, emphasizing defect reduction, standardization, and continuous monitoring across business processes. This approach proved effective in environments where structural correctness and process efficiency were paramount, including finance, healthcare, and manufacturing. However, while SSDQ incorporated relevance as a dimension, its primary focus remained on syntactic and rule-based attributes that could be statistically measured and controlled. As a result, the framework offered limited guidance for addressing semantic data quality dimensions related to meaning, contextual interpretation, and truthfulness—capabilities that became critical with the adoption of Large Language Models and Generative Artificial Intelligence. This emphasis on defect minimization over semantic fitness for use illustrated a broader gap between Information Quality theory and data quality practice, particularly in AI-driven contexts where reliable outputs depended on more than structural accuracy alone [50], [51], [53].

2.9 Data Quality Assessment Framework (DQAF)

The Data Quality Assessment Framework (DQAF), developed by the International Monetary Fund and grounded in the United Nations' Fundamental Principles of Official Statistics, provided a structured approach for evaluating the quality of statistical data through five dimensions: assurances of integrity, methodological soundness, accuracy and reliability, serviceability, and accessibility [20], [46]. Designed primarily for official statistics such as economic indicators and census data, the framework emphasized objectivity, adherence to internationally accepted methodologies, and consistency in data dissemination, offering clear criteria for assessing governance, processes, and statistical products. While the DQAF proved effective for evaluating the reliability and comparability of structured statistical datasets and was widely adopted by central banks and international institutions, its focus remained largely on methodological rigor and procedural transparency. Consequently, the framework offered limited

guidance for addressing semantic data quality dimensions related to meaning, contextual interpretation, and inferential trust—capabilities that became increasingly critical with the adoption of Large Language Models and Generative Artificial Intelligence. This limitation underscored a broader gap between Information Quality theory and applied data quality assessment practices, particularly in AI-driven environments where semantic coherence and contextual fitness were central to trustworthy outcomes [20], [46].

2.10 Three-Dimensional Analysis

Lindsey's Three-Dimensional Data Profiling framework defined data profiling as a systematic process for examining existing datasets to derive empirical statistics that functioned as accurate, evidence-based metadata describing the true structure and content of data [5]. Through column, dependency, and redundancy profiling, the approach enabled organizations to assess data conformance, uncover anomalies, evaluate integration risks, and monitor data quality over time, thereby reducing project uncertainty and preventing late-stage failures. Profiling techniques generated insights into domains, patterns, frequency distributions, and interdependencies within and across datasets, offering strong support for structural validation and metadata accuracy. However, while this approach effectively strengthened syntactic understanding of data and facilitated early risk detection, it largely relied on observable characteristics and statistical regularities, providing limited support for semantic interpretation, contextual meaning, and inferential trust. As such, Lindsey's profiling framework exemplified a broader orientation within Data Quality Management practice that privileged structural insight over semantic fitness for use—an imbalance that became increasingly consequential with the adoption of Large Language Models and Generative Artificial Intelligence, where trustworthy outputs depended on meaning and context rather than structural correctness alone [5].

2.11 ISO 8000 Part 61 Standard: Data Quality Management: Process Reference Model

ISO 8000-61 established a process-oriented reference model for Data Quality Management, focusing on defining, implementing, measuring, and continuously improving the processes that create, manage, and use data across its lifecycle [6]. Rather than prescribing specific data quality dimensions, the standard emphasized organizational maturity, repeatable processes, and governance integration through principles such as the process approach, continuous improvement, and structured accountability across organizational roles. Central to the framework was the application of the Plan-Do-Check-Act cycle, supported by data-related services and resource provisioning, to ensure sustainable data quality improvement aligned with business requirements. While ISO 8000-61 strengthened operational discipline, governance alignment, and consistency in data quality management practices, its deliberate avoidance of explicit data quality dimensions limited its guidance on evaluating meaning, context, and interpretability. This process-centric orientation reflected a broader trend in Data Quality Management practice and contributed to the gap between Information Quality theory and practice that became increasingly apparent with the adoption of Large Language Models and Generative Artificial Intelligence, where semantic understanding and contextual fidelity were essential for trustworthy outcomes [6].

2.12 Ten Steps to Quality Data and Trusted Information by Danette McGilvray

McGilvray's Ten Steps to Quality Data and Trusted Information framework addressed the persistent problem of data quality failures undermining enterprise initiatives such as data integration, data warehousing, and system migrations, where distrust in data often reduced the realized value of major investments [43]. The methodology provided a structured, end-to-end approach that linked technical data quality activities to business outcomes through steps that defined business needs, assessed the information environment, evaluated data quality dimensions, quantified business impact, identified root causes, and implemented sustainable

improvements and controls. Central to the framework was the premise that data quality must be measurable and aligned to business priorities, with selected data quality dimensions applied pragmatically rather than through a universal standard. While the approach proved effective in operational environments by emphasizing measurement, root-cause remediation, governance, and communication, its dimension selection was driven primarily by feasibility within existing constraints, often privileging easily measurable attributes over deeper semantic considerations. This pragmatic orientation reflected broader Data Quality Management practice and contributed to the gap between Information Quality theory and practice that became increasingly visible with the adoption of Large Language Models and Generative Artificial Intelligence, where trust, meaning, and contextual correctness were as critical as defect reduction and process control [43].

Table 5

Data Quality Dimensions - Ten Steps to Quality Data framework (First Edition)

Dimension of Quality	Description
Data Specifications	A measure of the existence, completeness, quality, and documentation of data standards, data models, business rules, metadata, and reference data.
Data Integrity Fundamentals	A measure of the existence, validity, structure, content and other basic characteristics of the data.
Duplication	A measure of unwanted duplication existing within or across systems for a particular field, record, or data set.
Accuracy	A measure of the correctness of the content of the data (which requires an authoritative source of reference to be identified and accessible).
Consistency and Synchronization	A measure of the equivalence of information stored or used in various data stores, applications, and systems, and the processes for making data equivalent.
Timeliness and Availability	A measure of the degree to which data are current and available for use as specified and in the timeframe in which they are expected.
Ease-of-Use and Maintainability	A measure of the degree to which data can be accessed and used and the degree to which data can be updated, maintained, and managed.
Data Coverage	A measure of the availability and comprehensiveness of data compared to the total data universe or population of interest.
Presentation Quality	A measure of how information is presented to and collected from those who utilize it. Format and appearance support appropriate use of information.

Perception, Relevance, and Trust	A measure of the perception of and confidence in the quality of the data; the importance, value, and relevance of the data to business needs.
Data Decay	A measure of the rate of negative change to the data.
Transactability	A measure of the degree to which data will produce the desired business transaction or outcome.

McGilvray's second edition of *Executing Data Quality Projects* framed Data Quality Management as a pragmatic, business-driven discipline that relied on selectively defined data quality dimensions to plan, measure, and sustain improvement initiatives in increasingly complex data environments, including AI, machine learning, and analytics-driven systems [49]. The Ten Steps methodology emphasized the flexible selection of dimensions based on business relevance, feasibility, and resource constraints, enabling organizations to establish baselines, prioritize remediation, and operationalize monitoring over time. While this adaptable approach strengthened implementation discipline and governance integration, it also favored dimensions that were operationally convenient and easily measurable, such as transactability and structural correctness, over semantic attributes related to meaning, interpretability, and contextual trust. This emphasis illustrated a persistent gap between Information Quality theory and practice that became more pronounced with the adoption of Large Language Models and Generative Artificial Intelligence, where reliable outcomes depended on semantic fitness for use rather than transactional success alone [49].

Table 6

Data Quality Dimensions -Ten Steps to Data Quality (Second Edition)

Data Quality Dimensions in the Ten Steps Process 2nd edition – Names, Definitions, and Notes	
Substep. Dimension Name	Definition, and Notes
3.1. Perception of Relevance and Trust	The subjective opinion of those who use information and/or those who create, maintain, and dispose of data regarding: 1) Relevance - which data if of most value and importance to them, and 2) Trust-their confidence in the quality of data to meet their needs. Notes: It is often said that perception is reality, and people act in accordance with their perceptions. Users' feelings about relevance

	and trust influence whether they readily accept and use the organization's data and information (or not). If users believe the data quality is poor, it is less likely they will use the organization's data sources or they will create their own spreadsheets or databases to manage their data. This leads to the proliferation of "spreadmarts" with duplicate and inconsistent data, often without adequate access and security controls in place. This dimension gathers the opinions of those who use and/or manage the data and information through a formalized survey (individual interviews, group workshops, online surveys, etc.). While in contact with users, it makes sense to ask questions about both the value/business impact viewpoint, it is included as both a Data Quality Dimension in Step 3.1 and as a Business Impact Technique in Step 4.7. Opinions about the quality of the data can be compared to other data quality assessment results showing the actual data quality. This allows you to uncover and address any gaps between perception and reality. An assessment of the Perception of Relevance and Trust could be conducted on an annual or biennial basis much as employee satisfaction surveys are conducted. Results are compared for trends over time and ascertain: 1) If the data being managed is still relevant to the users, and 2) If actions to prevent data quality issues are leading to expected increases in trust of data.
3.2. Data Specification	Data Specifications include any information and documentation that provide context, structure, and meaning to data. Data Specifications provide information needed to make, build, produce, assess, use, and manage data and information. Examples include metadata, data standards, reference data, data models, and business rules. Without the existence, completeness, and quality of data specifications it is difficult to produce high-quality data and harder to measure, understand, and manage the quality of data content. Notes: Data specifications provide the standard against which to compare data quality assessment results. They also provide instructions for manually entering data, designing data load programs, updating information, and developing applications.
3.3. Data Integrity Fundamentals	The existence (completeness/fill rate), validity, structure, content, and other basic characteristics of data. Notes: Most other dimensions of quality build on what is learned in Data Integrity Fundamentals. If you don't know anything else about your data, you need to know what is learned in this assessment. Data Integrity Fundamentals uses the technique of data profiling to assess basic characteristics of data such as completeness/fill rate, validity, lists of values and frequency distributions, patterns, ranges, maximum and minimum values, precision, and referential integrity. What other lists may call out as separate dimensions of data quality (e.g., completeness and validity) are part of this overarching dimension because all can be assessed using the same technique of data profiling.

	Note that this dimension assesses the quality, which is different than how you may choose to report results on a report or dashboard. You may want to report completeness and validity separately, for example, or you may combine results and report quality by a particular data field or dataset. Don't confuse the reporting categorization and presentation with detail you need to know and learn from this data quality dimension.
3.4. Accuracy	The correctness of the content of data as compared to an agreed-upon and accessible authoritative source of reference. Notes: The word accuracy is often used by some as synonym for data quality in general. Data professionals know they are not the same. Data accuracy requires comparing the data to the real-world object it represents (the authoritative source of reference). Sometimes it is not possible to access the real-world object represented by the data, and in those cases a carefully chosen substitute for it may be used as the authoritative source of reference. The assessment of accuracy can be a manual and time-consuming process. Following are examples of what is learned through assessing Data Integrity Fundamentals using the technique of data profiling compared to what is learned in Accuracy assessment: 1. Data profiling reveals if an item number contains a valid code indicating a make or buy part, but only an accuracy assessment using someone familiar with item #123 can determine if that particular item actually is a make item or buy item. 2. Data profiling shows if a customer record has a valid pattern for a postal code, but only an accuracy assessment that contacts the customer or uses a secondary authoritative source such as a postal service list can tell if it is his or her postal code. 3. Data profiling displays the value for the number of products on hand in an inventory database and confirms it is the correct data type. But only by completing an accuracy assessment with someone manually counting or scanning products on the shelf and comparing that number to the record in the inventory system can you know if the inventory count in the database is an accurate reflection of the inventory on hand.
3.5. Uniqueness and Deduplication	The uniqueness (positive) or unwanted duplication (negative) of data (fields, records, or datasets) existing within or across systems or datastores. Notes: There are many hidden costs associated with duplicate records. For example, duplicative vendor records with the same name and different addresses make it difficult to ensure that payment is sent to the correct address. When purchases by one company are associated with duplicate master records, the credit limit for that company can unknowingly be exceeded. This can expose the business to unnecessary credit risks.

	Identifying duplicates requires different processes and tools than Data Integrity Fundamentals, Accuracy, or the other data quality dimensions.
3.6. Consistency and Synchronization	Equivalence of data stored or used in various datastores, applications, and systems. Notes: Equivalence is the degree to which data stored in multiple places are conceptually equal. The same technique of data profiling used in Data Integrity Fundamentals can be used on the same data as it appears in multiple datastores or datasets. Results are compared for consistency. Consistency and Synchronization is called out as a separate data quality dimension because each datastore that is compared adds time and work to your project. If it is necessary, it should be done, but knowing that you are assessing consistency and synchronization, in addition to Data Integrity Fundamentals will help you better plan for the additional resources and time required.
3.7. Timeliness	Data and information are current and ready for use as specified and in the time frame in which they are expected. Notes: This data quality dimension is easily added onto Consistency and Synchronization if the factor of how long it takes for the data to move from one data store to another is a concern. It is called out separately here because you may or may not want to look at Timeliness. It is easy to add in or take out as needed. As the world changes, there will always be a gap between when the real-world object changes and when data representing it is updated in database and ready to be utilized. This assessment can also examine the timing of data throughout its life cycle and determine if data is being updated in time to meet business needs.
3.8. Access	The ability to control how authorized users can view, modify, use, or otherwise process data and information. Notes: Access is often summed up as enabling the right individuals to access the right resources, at the right times, under the right circumstances. Defining the right individuals, right resources, right times, and right circumstances are business decisions, which need to balance enabling access against protecting sensitive data and information. Other functions such as Information Security or a separate Access Management team often typically manage access. Your data quality team should work closely with them if your project addresses this data quality dimension of Access
3.9. Security and Privacy	Security is the ability to protect data and information assets from unauthorized access, use, disclosure, disruption, modification, or destruction (US Department of Commerce, n.d.). Privacy for an individual is the ability to have some control over how data about them as a person is collected and used. For an organization, it is the ability to comply with how people want their data to be collected, shared, and used. Notes: Securing and protecting data require specialized expertise. This dimension does NOT replace that. However, data professionals should find and work with information security teams and others responsible for privacy in their organizations to ensure:

	1) Appropriate security and privacy safeguards are in place for the data with which they are concerned, 2) They understand the security and privacy requirements with which the data they manage must comply. The data quality dimension of Access (step 3.8) is focused on access for authorized users. Security also includes access, but it is focused on unauthorized access. The Information Security world (InfoSec) often talks about protecting through the CIA Triad: Confidentiality, Integrity, and Availability.
3.10. Presentational Quality	The format, appearance, and display of data and information support their collection and uses. Notes: Presentation quality applies to user interfaces, reports, surveys, dashboards, etc. Presentation quality can affect the quality of the data when it is collected. For example, does a dropdown list contain valid options or codes? Are questions framed in such a way that users understand what is being asked and can provide correct responses? Are the various methods for collecting the same data presented in a consistent manner? Presenting information in a way that is understandable from the user's perspective increases the quality of data from the time of collection. In this respect, good presentation quality is an excellent way to prevent data quality errors. Presentation quality also applies when using the data. For example, is the user interpreting a report correctly? Does it have descriptive column headings and include dates? Do the graphics, reports, or user interfaces correctly reflect the data behind them? If the data in the datastore is of high quality itself but is misunderstood when shown to user, you still do not have data quality.
3.11. Data Coverage	The comprehensiveness of data available compared to the total data universe or population of interest. Notes: This dimension looks at how the datastore reflects the total population of interest to the business. For example, a database should contain all customers in North and South America, but there are concerns that the database really reflects only a portion of the company's customers. Coverage in this example is the percentage of customers actually captured in the datastore compared to the population of all customers that should be in it. This dimension is more in-depth than general decisions made about population, coverage, and selection criteria when developing the project data capture and assessment plans.
3.12. Data Decay	Rate of negative change to the data. Notes: Knowing data decay rates helps determine whether mechanisms should be put into place to maintain the data and what the frequency of those updates should be. This dimension is an example where just the idea of the dimension can spur you to action, often without the need for an n-depth assessment. Volatile data requiring a high reliability level requires more frequent updates than data with a lower decay rate or data where the quality level may not need to be so high. If essential data is already known to change quickly, then efforts should also quickly move to finding solutions. Determine how to become aware of the change in the

	real world. Develop processes where data can be updated within the organization as soon and as close to the real-world change as possible.
3.13. Usability and Transactability	Data produces the desired business transaction, outcome, or use for which it was intended. Notes: This dimension is a way to determine if the data is fit for purpose and is a final checkpoint of data quality. Even if the right people have defined business requirements and prepared data to meet them, the data must produce the expected outcome or use – or we still do not have high-quality data. Can the data be used to complete a transaction? Can the invoice be created? Can a sales order be completed? Can a lab order be produced? Can an insurance claim be initiated? Can an item master record be used properly when building a Bill of Materials? Can a report be correctly generated? If the answer is no, we still do not have high-quality data.
3.14. Other Relevant Data Quality Dimensions	Other characteristics, aspects, or features of data and information that are deemed to be important for your organization to define, measure, improve, monitor, and manage. Notes: There may be characteristics of data that are specific to an organization, uses of the data, or aspects of data quality issues that are not already covered in the list of data quality dimensions. For example, within one corporate finance group, the main concern was reconciliation between systems. Some could call this completeness, but the word “reconciliation” is clearer to them and specific to their context. If another dimension is chosen, it can still be used within the context of the full Ten Steps Process, just like any of the others

McGilvray’s *Ten Steps to Data Quality* (Second Edition) articulated a structured yet flexible methodology for assessing, improving, and sustaining data quality by explicitly linking technical remediation activities to business objectives and governance practices [49]. The framework emphasized that trusted data was essential for effective decision-making, operational performance, and regulatory compliance, and it operationalized improvement through a sequence of steps that identified business needs, analyzed the data environment, assessed quality issues, and addressed root causes. Central to the methodology was the use of data quality dimensions—such as accuracy, completeness, consistency, timeliness, uniqueness, validity, integrity, and presentational quality—as practical criteria for measurement and prioritization. While this dimension-driven, repeatable approach strengthened organizational discipline and enabled scalable implementation, it primarily focused on dimensions that could be

readily operationalized within existing processes. As a result, deeper semantic concerns related to meaning, contextual interpretation, and trustworthiness were less explicitly addressed, illustrating a continuing gap between Information Quality theory and applied Data Quality Management practice that became increasingly consequential with the adoption of Large Language Models and Generative Artificial Intelligence [49].

2.13 BCBS239 (Basel Committee on Banking Supervision)

Regulatory guidance issued by the Basel Committee on Banking Supervision underscored the central role of Data Quality Management within enterprise risk and governance frameworks, explicitly asserting that data quality risks must be actively identified, assessed, and managed by senior leadership as part of an institution's overall risk management responsibilities [10]. Under BCBS 239, effective risk data aggregation and reporting depended fundamentally on the integrity, accuracy, and reliability of underlying data, positioning Data Quality Management as a foundational capability for regulatory compliance and sound decision-making. Consistent with this regulatory perspective, data quality dimensions were recognized as measurable characteristics used to assess data against defined standards, enabling organizations to formalize data quality rules within governance programs [10]. Although no universally accepted industry standard for data quality dimensions existed, these dimensions provided a practical mechanism for operationalizing data quality expectations in regulated environments. However, regulatory implementations frequently emphasized control, accuracy, and consistency to support compliance objectives, often prioritizing syntactic assurance over semantic understanding. This regulatory orientation further illustrated the gap between Information Quality theory and practice examined in this dissertation, a gap that became increasingly significant with the adoption of Large Language Models and Generative Artificial Intelligence, where risks extended beyond regulatory correctness to include meaning, context, and trust in AI-generated outputs [10].

Table 7

Data Quality Dimensions (BCBS239)

DQ Dimension	Definition	Sample DQ Rules
1.Completeness	The degree to which data elements are populated	General Ledger Account ID cannot be NULL
2.Conformity	The degree to which data elements correspond to expected formats or valid values or ranges of values	State should be from the agreed-upon list of code values for state
3.Consistency	The degree of relational integrity between data elements and other data elements	The insurance policy expiration date should be greater than or equal to the policy effective date
4.Synchronization	The degree to which data elements are consistent from one data store to the next	Customer ID in the master data hub must be the same as the one in the legacy customer file
5.Uniqueness	The degree to which data elements are unique within a data store	Customer IDs should be unique in the customer reference table
6.Timeliness	The degree to which data is available on a timely basis	Global value at risk must be calculated by 10 a.m. Eastern time Monday to Friday
7.Accuracy	The degree to which data elements are accurate	Customer NAICS code must be accurate

2.14 Other Related Materials

Recent scholarship highlighted a growing shift in Data Quality Management from a predominantly syntactic orientation toward greater emphasis on semantic data curation, particularly in response to the adoption of Generative AI and Large Language Models [8], [11]. Talburt argued that data quality must be grounded in explicitly defined requirements expressed through data quality dimensions, yet noted that despite the existence of numerous frameworks—such as those proposed by Wang and Strong, McGilvray, and ISO standards—organizations commonly operationalized only a limited subset of fundamental dimensions.

These dimensions, including accuracy, completeness, consistency, timeliness, validity, and relational integrity, were favored because they could be detected and corrected through rule-based processes, whereas semantic dimensions required deeper contextual understanding. Talburt further demonstrated that prevalent practices often conflated data validation with true data accuracy, resulting in a false sense of data correctness. This distinction became increasingly consequential in GenAI-enabled environments, where unverified but “valid” data could propagate semantic errors, hallucinations, and bias, reinforcing the disconnect between theoretical definitions of information quality and their practical implementation in modern DQM programs [8], [11].

Several applied frameworks were developed to help organizations assess and improve their data quality capabilities in a more systematic manner, including maturity and scorecard-based approaches [20], [21]. The Data Quality Maturity Model provided a staged roadmap for progressing from basic awareness toward optimized governance and continuous improvement, enabling organizations to evaluate their current capabilities and plan targeted enhancements. Complementarily, the Data Quality Scorecard operationalized data quality through measurable indicators aligned with predefined criteria, supporting ongoing monitoring and accountability. While these tools strengthened organizational discipline and measurement practices, their effectiveness depended on the dimensions selected for assessment, which in practice often mirrored the same syntactic bias identified in other DQM frameworks. As a result, although maturity models and scorecards improved governance transparency and operational consistency, they did not, by themselves, resolve deficiencies related to semantic interpretation and contextual fitness that became critical in AI-driven decision environments [20], [21].

The rapid advancement of Generative AI and LLM technologies further intensified the need to reassess traditional data quality assumptions, as these models relied heavily on semantic coherence rather than structural correctness alone [28], [29]. Comprehensive reviews of Generative AI research documented significant progress in architectures such as GANs,

GPT-based transformers, diffusion models, and multimodal learning systems, demonstrating their capacity to generate complex, human-like outputs across diverse domains. At the same time, these studies identified persistent challenges related to bias, reliability, explainability, and trustworthiness, many of which were rooted in deficiencies in training data quality and governance. Surveys of emerging research landscapes underscored how innovations such as mixture-of-experts models and multimodal systems amplified dependency on high-quality semantic information, thereby exposing limitations in existing Data Quality Management practices. Collectively, this body of literature reinforced the central premise of this dissertation: that the adoption of LLMs and Generative AI had widened the gap between Information Quality theory and practice, necessitating a reprioritization of semantic data quality dimensions within contemporary DQM frameworks [28], [29].

2.15 A Glance View of the Referenced DQ Dimensions Highlighting the Gaps

Summary Overview of the DQ Dimensions referenced in the below selected reviewed DQM Frameworks compared to the preliminary conceptual DQ frameworks by Richard Y. Wang and Diane M. Strong: Data Management Body of Knowledge (DAMA-DMBOK), Conformed Dimensions of Data Quality (CDDQ), Data Management Capability Assessment Model (DCAM), Six Sigma Data Quality (SSDQ) Framework, Data Quality Assessment Framework (DQAF), Three-Dimensional Analysis(3DA), ISO 8000 Part 61 Standard, Ten Steps of DQ by McGilvray(10SDQ), BCBS239 (Basel Committee on Banking Supervision).

Table 8

Data Quality Dimensions Cross-reference IQ Theory vs IQ Practice Frameworks

Conceptual Frameworks Dimensions	DMBOK	CDDQ	DCAM	SSDQ	DQAF	3DA	ISO8000	10SDQ	BCBS239
Believability									
Accuracy	X	X	X	X	X(i)			X(m)	X
Objectivity					X(j)				

Reputation									
Value-added									
Relevancy				X				X(n)	
Timeliness	X(a)	X(c)	X	X	X(k)			X(o)	X
Completeness	X	X	X	X				X(p)	X
Appropriate amount of data									
Interpretability		X(w)			X(v)				
Ease of understanding		X(d)						X(u)	
Representational consistency	X(b)	X(e)	X(f)	X(h)		X(l)		X(q)	X(t)
Concise representation									
Accessibility		X	X(g)		X			X(r)	
Access security								X(s)	

DQ dimensions referenced in reviewed frameworks versus the conceptual DQ Frameworks.

DQ Dimensions highlighted in red color in the table above means “DQ dimensions that are related to semantic features of LLMs and GenAI that need an increase of priority in the DQM practices. When a letter X is checked in the table above it means that the respective framework recommends that respective dimensions. Some DQ definitions in these frameworks don't match word to word with the conceptual models' definitions, analysis consisted of careful and thorough reviewing and understanding of the meaning associated with each dimension in these selected frameworks in order to appropriately know which conceptual dimension each dimension is referencing. Some conceptual dimensions are being referenced multiple times in these selected frameworks but only choosing different DQ names where the assigned definitions meaning wise match the conceptual frameworks, and these instances are presented

in Table 8 below. To see the definition associated with those specific dimensions that are referenced more than once or given a different name, see in the literature review for that specific framework.

Analysis discovered a lack of consistency in the naming convention and defining data quality dimensions regardless of the set of dimensions recommended in Information Quality (IQ) theory. Data Quality dimensions are being represented more than once in some of the analyzed frameworks and in some cases using different naming conventions or definitions not exactly matching verbiage as used in the conceptual theoretical information quality frameworks. Analysis consisted of careful and thorough revision of each data quality dimension as defined in reviewed frameworks in order to know which dimension in IQ theory is being covered.

Table 9

Conceptual Data Quality Dimensions Frequency in the reviewed frameworks

This table shows conceptual dimensions that are being represented more than once in these frameworks and or using different names

DAMA-DMBOK

(a):

DAMA-DMBOK recommends Timeliness

DAMA-DMBOK recommends Currency for Timeliness

(b):

DAMA-DMBOK recommends Consistency for Representational consistency

DAMA-DMBOK recommends Uniqueness/Deduplication for Representational consistency

DAMA-DMBOK recommends Validity for Representational consistency

DAMA-DMBOK recommends Integrity for Representational consistency

DAMA-DMBOK recommends Reasonableness for Representational consistency

CDDQ

(c):

CDDQ recommends Timeliness

CDDQ recommend Currency for Timeliness

(d):

CDDQ recommends Representation for Ease of understanding

(e):

CDDQ recommends Consistency for Representational consistency

CDDQ recommends Validity for Representational consistency

CDDQ recommends Integrity for Representational consistency

(w):

CDDQ recommends Precision for Interpretability

CDDQ recommends Lineage for Interpretability

DCAM

(f):

DCAM recommends Consistency for Representational consistency

DCAM recommends Conformity for Representational consistency

DCAM recommends Uniqueness for Representational consistency

(g):

DCAM recommends Coverage for Accessibility

SSDQ

(h):

SSQD recommends Consistency for Representational consistency
 SSQD recommends Validity for Representational consistency
 SSQD recommends Integrity for Representational consistency
 SSQD recommends Uniqueness for Representational consistency

DQAF

(i):

DQAF recommends "Accuracy and reliability" for Accuracy

(j):

DQAF recommends "Assurances of integrity" for Objectivity

(k):

DQAF recommends Serviceability for Timeliness

(v):

DQAF recommend "Methodological soundness" for Interpretability

3DA

(l):

3DA recommends "Redundancy profiling" for Representational consistency

10SDQ

(m):

10SDQ recommends Accuracy

10SDQ recommends Data Decay for Accuracy,

(n)

10SDQ recommends "Perception of Relevance and Trust" for Relevancy,

10SDQ recommends "Other Relevant Data Quality Dimensions" for Relevancy

(o):

10SDQ (2nd edition) recommends "Timeliness"

10SDQ (1st edition) recommends "Timeliness and Availability" for Timeliness

(p):

10SDQ recommends "Data Specifications" for Completeness

10SDQ recommends "Data Integrity Fundamentals" for Completeness

10SDQ recommends "Data Coverage" for Completeness

(q):

10SDQ recommends "Uniqueness and Deduplication" for Representational consistency

10SDQ recommends "Consistency and Synchronization" for Representational consistency

10SDQ recommends "Presentation Quality" for Representational consistency

(r):

10SDQ (2nd edition) recommends Access 10SDQ for Accessibility

10SDQ (1st edition) recommends "Ease of-Use and Maintainability" for Accessibility

(s):

10SDQ recommends "Security and Privacy" for Access Security

(u):

10SDQ recommends "Usability and Transactability" for Ease of understanding

BCBS239

(t):

BCBS239 recommends Consistency for Representational consistency

BCBS239 recommends Conformity for Representational consistency

BCBS239 recommends Synchronization for Representational consistency

BCBS239 recommends Uniqueness for Representational consistency

Overall, the literature review demonstrated a persistent and consequential gap between Information Quality theory and Data Quality Management practice in the context of Large Language Models and Generative Artificial Intelligence. While theoretical frameworks emphasized multidimensional, consumer-oriented definitions of data quality that included semantic attributes such as meaning, relevance, objectivity, believability, and contextual fitness, the majority of widely adopted DQM frameworks reviewed prioritized syntactic dimensions that

were easier to define, measure, and automate. The comparative analysis of nine prominent frameworks revealed limited and inconsistent treatment of semantic data quality dimensions that were critical for addressing GenAI-related issues, including bias, hallucination, trust degradation, and unreliable outputs. As organizations increasingly relied on LLMs and GenAI systems trained to infer meaning rather than validate structure, these omissions would expose material risks spanning regulatory compliance, financial performance, and stakeholder trust. Consequently, the literature collectively underscored the need to re-examine and realign Data Quality Management practices to elevate semantic data quality dimensions, thereby closing the gap between Information Quality theory and practice and supporting the development of trustworthy, reliable, and ethically sound GenAI-enabled systems.

2.16 Literature Review Summary

This literature review examined foundational Information Quality theories, widely adopted Data Quality Management (DQM) frameworks, and emerging research on Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) to assess how well current practices addressed the quality challenges introduced by AI-driven systems. Conceptual models such as *Beyond Accuracy* and Total Data Quality Management (TDQM) framed data quality as multidimensional and consumer-oriented, emphasizing that fitness for use depended on semantic attributes including meaning, relevance, objectivity, believability, and context. In contrast, the review of nine prominent DQM frameworks—including DAMA-DMBOK, DCAM, CDDQ, SSDQ, DQAF, ISO 8000-61, Three-Dimensional Analysis, Ten Steps of Data Quality, and BCBS 239—revealed a strong and consistent emphasis on syntactic dimensions such as validity, completeness, consistency, timeliness, and uniqueness. Although these frameworks provided valuable guidance for governance, regulatory compliance, and operational control, they rarely articulated what semantic aspects should be measured or how such dimensions should be prioritized, measured, or governed in practice.

When cross-referenced against Wang and Strong's Information Quality framework, the analysis demonstrated that semantic data quality dimensions critical to GenAI and LLM performance—including accuracy of meaning, value added, relevancy, objectivity, reputation, believability, security, and appropriate amount of data—received limited and inconsistent coverage across practical frameworks. Compounding this issue, the literature identified a lack of standardization in terminology and definitions, with similar dimensions labeled differently across frameworks, hindering comparison, benchmarking, and adoption. At the same time, GenAI and LLM research highlighted escalating risks related to bias, hallucination, misinformation, explainability, privacy, and trust, all of which were closely tied to deficiencies in semantic data quality rather than structural correctness alone. Collectively, the literature established that although existing DQM frameworks were effective for managing traditional data environments, they were not sufficiently aligned with the semantic demands of AI-driven models. This misalignment widened the gap between Information Quality theory and practice, underscoring the need for explicit prioritization, standardization, and governance of semantic data quality dimensions to support trustworthy, reliable, and ethically sound GenAI and LLM deployments.

Chapter 3: Methodology

3.1 Research Design

This study employed a qualitative, multi-phase design anchored in document analysis and the Delphi technique. The first phase examined selected Data Quality Management frameworks widely used in organizational practice and compared their coverage of data quality dimensions against Information Quality theory. The second phase used a Delphi study to validate the initial framework analysis, obtain expert judgment, and develop practice-oriented recommendations for strengthening DQM in light of the adoption of GenAI and LLMs. This design was appropriate because the research problem concerns both conceptual gaps in current practice and the need for expert consensus on how those gaps should be interpreted and addressed.

3.2 Methodological Rationale

The methodology was selected to answer a problem that is partly documentary and partly judgment based. Document analysis enabled a systematic review of formal frameworks used in practice, while the Delphi method provided a structured process for eliciting informed consensus from subject-matter experts. The Delphi approach was especially suitable because the study addressed an emerging area where standards are still evolving, empirical benchmarks are limited, and expert interpretation is central to identifying practical priorities. The combination of framework analysis and Delphi validation therefore strengthened the study by linking theoretical review, practice analysis, and expert evaluation.

3.3 Research Purpose and Questions

The purpose of this study was to examine whether widely used DQM frameworks sufficiently address semantic data quality dimensions required for trustworthy GenAI and LLM

applications, and to validate those findings through expert consensus. The study was guided by the following research questions:

RQ1: To what extent do current DQM practices incorporate semantic data quality dimensions for GenAI and LLM applications?

RQ2: What semantic DQ dimensions are perceived as most critical for GenAI/LLMs?

RQ3: What actions should organizations take to prioritize semantic DQ?

3.4 Phase One: Comparative Framework Analysis

Phase One consisted of a comparative qualitative review of major DQM frameworks referenced in academic and professional practice. The frameworks selected in the dissertation included DAMA-DMBOK, CDDQ, DCAM, SSDQ, DQAF, Three-Dimensional Analysis, ISO 8000-61, Ten Steps of Data Quality by McGilvray, and BCBS 239. These frameworks were reviewed against the fifteen data quality dimensions identified in IQ theory, particularly the Wang and Strong framework and related Total Data Quality Management concepts. The analysis focused on identifying which dimensions were explicitly defined, emphasized, or operationalized in each framework, and whether semantic dimensions central to meaning, relevance, context, credibility, and trust were underrepresented relative to syntactic dimensions. The framework comparison generated the preliminary findings that were later presented to Delphi participants for validation and critique.

3.5 Phase Two: Delphi Study Design

The second phase of the study used a two-round Delphi design. The Delphi technique is a structured method for obtaining informed judgments from a panel of experts through iterative rounds of questioning and controlled feedback. This study used two rounds because the first round established agreement with the core problem statements and collected open-ended recommendations, while the second round refined those recommendations through rating and ranking exercises. The Delphi design supported anonymity, reduced the influence of dominant

personalities, and allowed participants to reconsider their views in light of summarized group responses.

3.6 Panel Selection and Participants

A purposive expert sampling strategy was used. Participants were selected because of their professional or scholarly expertise in Data Quality Management, Information Quality, Data Governance, data architecture, data risk, or closely related disciplines. In Round 1, fifteen anonymized panelists participated. In Round 2, fourteen experts participated, although one question received thirteen usable responses because one participant indicated insufficient expertise for that item. Panelists were de-identified in the study to preserve confidentiality and were referred to by anonymous identifiers. The decision to use an expert panel rather than a large general sample was consistent with Delphi methodology, which prioritizes depth of expertise over population representativeness.

3.6.1 Delphi Panel Selection Process

The credibility and reliability of a Delphi study are fundamentally dependent on the expertise and diversity of the panel of experts who participate in the consensus-building process. In recognition of this requirement, this study implemented a rigorous, transparent, and ethically grounded Delphi panel selection process to ensure that panelists possessed the depth of knowledge, professional experience, and disciplinary breadth necessary to evaluate Data Quality Management (DQM) practices in the context of Large Language Models (LLMs) and Generative Artificial Intelligence (GenAI). The selection strategy was explicitly designed to support the study's objective of examining the gap between Information Quality (IQ) theory and contemporary DQM practice while ensuring the validity and defensibility of the resulting expert consensus.

Panel recruitment followed a purposive expert sampling strategy, which is considered appropriate and methodologically sound for Delphi research when the objective is to obtain

informed judgment rather than statistical representation. Qualified participants were identified and recruited through direct email invitations issued by the Principal Investigator (PI) and the Research Advisor. Potential panelists were selected based on clearly defined inclusion criteria, including advanced academic credentials, demonstrated research contributions, and substantial professional experience relevant to data quality, data governance, information quality, and AI-enabled systems. Recruitment channels included academic affiliations, professional networks, peer-reviewed publications, industry forums, professional referrals, and LinkedIn-based professional visibility.

All prospective participants received a standardized recruitment email detailing the purpose of the study, scope of participation, anticipated time commitment, voluntary nature of involvement, and contact information for both the PI and the Research Advisor. The recruitment materials also included the Institutional Review Board (IRB)-approved informed consent document. Participation was entirely voluntary, and panelists were informed of their right to withdraw at any time without penalty. Eligibility was limited to individuals aged 18 years or older who satisfied the study's professional expertise requirements. Throughout the recruitment and participation phases, confidentiality was strictly maintained in accordance with ethical standards and IRB requirements.

To ensure a robust and credible panel, the minimum acceptable panel size was established at ten experts up front, and with an upper limit of fifteen experts who participated, consistent with methodological recommendations for Delphi studies. Fewer than eight participants may limit diversity of thought, while panels exceeding fifteen members often yield diminishing returns in terms of novel insights. An initial pool of approximately fifty qualified candidates was identified, all of whom met the inclusion criteria. From this pool, twenty three experts consented to participate, among of which only fifteen experts were able to participate and completed the study, resulting in a panel size that balanced disciplinary diversity with manageable consensus iteration.

All Delphi panelists held higher education degrees background including master's and doctoral degrees in data-related or closely aligned disciplines, including Information Systems, Data Science, Information Quality, Computer Science, Data Governance, and Analytics. A significant proportion of panelists were active academic researchers with peer-reviewed publications in data management, data quality, information quality, AI governance, or related domains. Many panelists also held tenured or senior faculty positions at universities, contributing to curriculum development, research supervision, and scholarly dissemination in the areas most relevant to this study.

In addition to their academic credentials, a majority of panelists possessed extensive industry experience spanning multiple decades in sectors such as financial services, healthcare, technology, government, and regulated industries. This dual academic–practitioner background was a critical selection criterion, as it ensured that expert judgments reflected both theoretical rigor and operational realities. Panelists had direct experience designing, implementing, auditing, or governing data quality and data governance programs, including exposure to regulatory frameworks, enterprise-scale data platforms, and AI-enabled analytics environments.

The panel was intentionally composed to reflect multidisciplinary and cross-sector diversity. Expertise represented across the panel included Data Quality Management, Information Quality theory, Data Governance, enterprise data architecture, risk and compliance, AI and machine learning systems, GenAI and LLM technologies, and information technology leadership. This diversity strengthened the study by ensuring that the evaluation of semantic data quality dimensions was informed by multiple professional perspectives, reducing the likelihood of disciplinary bias and enhancing the comprehensiveness of the consensus outcomes.

A formal expertise matrix was developed to document panel qualifications across multiple dimensions, including academic attainment, research activity, industry practice experience, domain specialization, and AI-related technical exposure. Although individual

identities are withheld to preserve confidentiality, the matrix confirmed that each panelist met or exceeded the predefined expertise thresholds. Collectively, the panel demonstrated substantial depth across all areas directly relevant to assessing data quality dimensions in the context of GenAI and LLM adoption.

The composition and credentials of the Delphi panel provide strong evidence of methodological rigor and content validity. By engaging experts who are both thought leaders and practitioners in data quality and AI-related disciplines, the study ensured that the resulting consensus was not only theoretically informed but also grounded in real-world implementation challenges. The diversity, experience, and scholarly standing of the panelists support the credibility of the findings and affirm that the identified gaps in semantic data quality dimensions reflect systemic issues in contemporary DQM practice rather than isolated viewpoints.

In summary, the Delphi panel selection process was intentionally designed to maximize expertise, diversity, and ethical integrity. The resulting panel was well positioned to evaluate the adequacy of existing Data Quality Management frameworks, to assess their alignment with Information Quality theory, and to provide informed guidance on prioritizing semantic data quality dimensions necessary for trustworthy and reliable GenAI and LLM deployments. This rigorous selection process underpins the validity of the Delphi findings and strengthens the overall contribution of the dissertation.

3.6.2 Delphi Expert Panel – Anonymized Expertise Matrix

Table 9 presents an anonymized Delphi Expertise Matrix developed to document and demonstrate the qualifications, disciplinary breadth, and professional experience of the Delphi panel participants. Consistent with Delphi methodology and IRB confidentiality requirements, individual identities are not disclosed. The matrix summarizes key expertise dimensions relevant to this study, including academic credentials, research activity, industry experience, and subject-matter specialization. Collectively, the matrix provides evidence that all panelists met or

exceeded the predefined expertise criteria necessary to support a credible and defensible consensus on semantic data quality dimensions in the context of Generative AI (GenAI) and Large Language Models (LLMs). Moreover, all of these panelists have exposure and knowledge to the nine frameworks analyzed in this study to include the conceptual models.

Table 10

Delphi Expert Panel Expertise Matrix

Panelist (PN)ID	Highest Degree (PhD and/or MS) ?	Academic Role	Industry Experience (Years)	Primary Domain Expertise	Data Quality / IQ Expertise	Data Governance Experience	GenAI / LLM Technical Exposure	Research Publications
PN01	YES	Professor / Researcher	20+	Information Quality, DQM	High	High	Moderate	Peer-reviewed
PN02	YES	Professor	15+	Data Governance, Risk	High	High	Moderate	Peer-reviewed
PN03	YES	Professor	10+	Data Science, Analytics	High	Moderate	High	Peer-reviewed
PN04	YES	Professor / Practitioner	25+	Enterprise Data Architecture	High	High	Moderate	Peer-reviewed
PN05	YES	Research Scientist	12+	GenAI, ML Systems	Moderate	Moderate	High	Peer-reviewed
PN06	YES	Professor	18+	Statistics, Data Quality	High	Moderate	Low–Moderate	Peer-reviewed
PN07	YES	Industry Fellow	30+	Financial Data Governance	High	High	Moderate	Industry & Academic
PN08	YES	Professor	15+	Information Systems	High	Moderate	Low–Moderate	Peer-reviewed
PN09	YES	Professor / Advisor	20+	Healthcare Data Quality	High	High	Moderate	Peer-reviewed
PN10	YES	Researcher	10+	AI Ethics, Trust	Moderate	Moderate	High	Peer-reviewed
PN11	YES	Professor	22+	Enterprise Analytics	High	Moderate	Moderate	Peer-reviewed
PN12	YES	Industry Leader	28+	Regulated Data Systems	High	High	Moderate	Industry & Academic
PN13	YES	Professor	14+	Semantic Data Modeling	High	Moderate	High	Peer-reviewed

PN14	YES	Professor	19+	Information Governance	High	High	Moderate	Peer-reviewed
PN15	YES	Practitioner / Researcher	25+	AI-enabled Data Platforms	High	Moderate	High	Industry & Academic

The Delphi Expertise Matrix demonstrates that the panel collectively possessed deep, multidimensional expertise across data quality theory, enterprise data governance, regulatory environments, and GenAI/LLM technologies. The combination of advanced academic credentials, sustained research activity, and extensive industry practice provided a robust foundation for informed judgment. This breadth and depth of expertise strengthens the methodological rigor of the Delphi study and supports the credibility of the consensus findings regarding gaps in semantic data quality dimensions within contemporary Data Quality Management practice.

3.7 Instrumentation

Round 1 included twenty-three structured statements and seven open-ended questions. The structured statements asked panelists to indicate agreement or disagreement with propositions derived from the preliminary framework analysis, such as whether current DQM practice should be reexamined in light of GenAI and LLM adoption, whether semantic dimensions are under-prioritized in practice, and whether accuracy of meaning requires comparison against an authoritative data source or system of record. The open-ended questions invited participants to identify the most important semantic dimensions, recommend measures for assessing them, and propose efficient methods for gathering semantic DQ evidence.

Round 2 focused primarily on ranking and rating the synthesized responses generated from the open-ended first-round data. Participants used Likert-type scales to evaluate the importance of semantic dimensions and the perceived effectiveness or efficiency of proposed

measures and approaches. This second-round design allowed the study to move from exploratory input to more structured prioritization.

3.8 Data Collection Procedures

The study was conducted sequentially. First, the comparative framework analysis was completed and summarized into a research briefing for panelists. Second, round 1 Delphi materials were distributed to consenting participants. Panelists reviewed the study summary, responded to twenty-three statements, and provided narrative comments for seven open-ended items. Responses were collected, anonymized, and synthesized. Third, a second-round instrument was prepared using the main response patterns and recommendations identified in Round 1. Round 2 asked panelists to rank the importance of semantic dimensions and to evaluate candidate measures and practices for operationalizing semantic DQ. Throughout the process, participation remained voluntary, and the study design followed the dissertation draft's stated IRB and informed-consent requirements.

3.9 Data Analysis

Two forms of analysis were used. For the framework review and the open-ended Delphi responses, qualitative thematic analysis was conducted to identify recurring concepts, recommendations, and concerns. Similar comments were grouped into higher-order themes, such as the need to prioritize semantic dimensions, the importance of authoritative reference data, the role of automation, the continuing need for human review, and the limitations of current syntactic DQ practices.

For Round 2 quantitative items, descriptive statistics were used to summarize expert judgment. The analysis included measures of central tendency, such as mean, median, and mode, as well as measures of dispersion, including standard deviation and interquartile range (IQR). These measures were appropriate because Delphi studies commonly report both central tendency and dispersion to demonstrate consensus and variability. In the dissertation draft, high

consensus was interpreted using an 80 percent threshold for agreement in Round 1, while Round 2 distributions helped identify dimensions and actions receiving the strongest support.

3.10 Trustworthiness and Rigor

Several features of the study design strengthened methodological rigor. First, the study used triangulation across two evidence sources: documentary framework analysis and expert Delphi feedback. Second, anonymity reduced conformity pressure and encouraged candid expert responses. Third, the multi-round design allowed the researcher to move from initial validation to more refined prioritization. Fourth, the use of both qualitative synthesis and descriptive statistics provided a more comprehensive interpretation of the results. Finally, the study maintained conceptual alignment with IQ theory, ensuring that the comparison between theory and practice remained grounded in established data quality literature.

3.11 Ethical Considerations

This dissertation study operated under institutional review procedures and required informed consent before participation. Participation was voluntary, and panelists were permitted to withdraw at any time. Data were de-identified through panelist codes, and reporting emphasized aggregated results rather than individual attribution. These steps supported confidentiality and minimized risk to participants.

3.12 Delimitations and Limitations of the Methodology

The methodology was intentionally bounded. The study did not attempt to build a new DQM framework, define new data quality dimensions, or produce mathematical formulas for measuring semantic dimensions. Instead, it focused on identifying under-prioritized semantic dimensions within existing practice and validating that assessment through expert judgment. As with most Delphi studies, the results reflect informed expert consensus rather than statistical generalization to all industries or all practitioners. In addition, framework analysis depended on

published content and therefore captured what frameworks explicitly state more readily than how individual organizations may adapt those frameworks in practice.

3.13 Chapter Summary

This chapter described the methodology used to examine the gap between IQ theory and DQM practice in the era of GenAI and LLMs. A comparative framework analysis identified the preliminary gap, and a two-round Delphi study validated that gap and generated prioritized recommendations from experts. The next chapter presents the detailed findings from both phases of the study.

Chapter 4: Delphi Study Data Collection and Analysis Results

4.1 Introduction

This chapter presents the data collection procedures and analysis results of a two-round Delphi study examining data quality management (DQM) practices in the era of Generative AI (GenAI) and large language models (LLMs). The purpose of Chapter 4 is to explain how expert feedback was collected, describe the structure of the survey instrument, and summarize the findings across two iterative rounds. The study seeks to validate preliminary research observations that widely used DQM frameworks emphasize syntactic data quality dimensions while overlooking semantic dimensions essential for AI-driven applications. Through structured statements and open-ended questions, a panel of data quality experts were asked to evaluate the importance of semantic dimensions, rate improvement proposals for DQM frameworks, and offer recommendations.

This chapter is organized as follows: first, the data collection methodology is detailed, including participant recruitment, survey design, and data gathering procedures. Next, the analysis and results from the first round of the Delphi study are reported, highlighting the level of consensus on statements and synthesizing themes from open-ended responses. Finally, the second round analysis is presented, summarizing the experts' rankings of semantic data quality dimensions and evaluating proposed enhancements to DQM frameworks. The chapter concludes with a discussion of key insights and implications for practice.

4.2 Data Collection Methodology

The Delphi study employed a sequential, mixed-methods design involving two rounds of data collection. Participants were recruited based on their expertise in data quality management and familiarity with GenAI and LLM technologies. In the first round, fifteen anonymous experts representing academia, industry and data governance roles were invited to complete an online

survey. The survey comprised 23 Likert-type statements and 7 open-ended questions. The statements asked panelists to indicate their level of agreement (Agree, Disagree, or Neutral/None) with propositions related to DQM practices, semantic data quality dimensions, and challenges posed by GenAI/LLMs. The open-ended questions solicited qualitative feedback on priority semantic dimensions, suitable measures, data collection methods, DQ challenges, areas for enhancement of DQM practice, improvements to frameworks, and recommendations for prioritizing semantic DQ dimensions.

Panelists were instructed to read a preliminary findings summary that compared conceptual DQM frameworks (such as Beyond Accuracy and Total Data Quality Management) with practical frameworks (DAMA-DMBOK, CDDQ, DCAM, SSDQ, DQAF, Three-Dimensional Analysis, ISO 8000 Part 61, Ten Steps to Quality Data, and BCBS 239). The summary highlighted the lack of semantic DQ dimensions in practice. Using their expertise, panelists evaluated the statements and provided narrative responses. Responses were aggregated and analyzed for consensus, and common themes were extracted from qualitative feedback.

The second round of the Delphi study focused primarily on ranking the semantic data quality dimensions for the open-ended questions (Questions 24–30), as there was a broad consensus on questions 1-23 in the first round. Fourteen panelists from the first round participated. Ratings were analyzed using descriptive statistics (mean, median, mode, standard deviation and interquartile range) to assess central tendencies and agreement levels.

Throughout the study, anonymity was maintained to minimize bias and encourage candid responses. Data were collected using secure online survey tools, and descriptive and thematic analyses were conducted to interpret quantitative and qualitative findings. This systematic approach ensured robust data collection and facilitated expert consensus on neglected semantic data quality dimensions and practical recommendations for improving DQM frameworks in the GenAI/LLM context.

4.3 First Round Delphi Study Data Collection and Analysis

4.3.1 Participants' Instructions

- Below are the instructions which were given to the panelist in order to respond to the statements/questions for the first round.
- Read each statement and respond with Agree or Disagree based on your experience, expertise and the above research summary findings.
- Provide any additional comments, feedback or any other level of consensus to the statement in the "Additional Feedback" section beneath each statement.
- There are a total of 30 statements/questions in this Delphi study.
- Your insights are highly valued.

4.3.2 Analysis Introduction

The first round aimed to gauge whether existing data quality management frameworks adequately address semantic dimensions -aspects of data quality that relate to meaning and context and are often neglected. Fifteen anonymized panelists participated, rating their agreement with 23 Likert-type statements and providing qualitative feedback to seven open-ended questions. The analysis organized responses by question, grouped similar answers, and assessed the level of consensus achieved.

4.3.3 First Round Analysis Questions 1 - 23

Question 1 First Round Summary Analysis

Statement: *"Data Quality Management professionals in both academia and enterprise need to examine current Data Quality Management practices given the rise of GenAI and LLMs."*

Responses & Consensus: All 15 panelists agreed with this statement (100% agreement), indicating unanimous consensus that the advent of GenAI/LLMs warrants a reevaluation of current DQM practices. Several experts emphasized that GenAI has "exacerbated and accelerated the data quality problem" – in other words, existing data issues are amplified under AI.

Common Feedback Themes:

- **Amplification of Existing Issues:** Many noted that GenAI/LLMs amplify longstanding data quality problems (e.g., bias, drift, privacy breaches) and thus demand renewed focus on DQM (five panelists). For instance, one panelist warned that GenAI trained on biased data may amplify problems in ways that are hard to detect.
- **Different Data & New Criteria:** Panelists highlighted that LLMs rely on unstructured, probabilistic data with different quality criteria than traditional structured databases (two panelists). GenAI introduces new data types and risks, so most current frameworks (built for structured data) are insufficient.
- **Lack of DQ Awareness:** It was observed that many organizations still lack basic DQM awareness. One expert described colleagues “unsure of what the definition of data quality is” and no clear ownership for defining DQ requirements. This lack of clarity, coupled with GenAI’s complexity, makes reexamination critical.
- **GenAI as a Tool for DQM:** A few panelists suggested GenAI/LLMs themselves could help improve or automate parts of DQM processes (one panelist). They advocated exploring GenAI as a tool within DQM frameworks (for instance, to aid data cleaning or monitoring).

Question 2 First Round Summary Analysis

Statement: *“GenAI and LLMs introduce complex data challenges in practice.”*

Responses & Consensus: All 15 panelists agreed (100%) that GenAI/LLM adoption brings new, complex data quality challenges. There was a strong consensus that these technologies present novel issues beyond those handled by traditional DQ methods.

Common Feedback Themes:

- **New Types of DQ Problems:** Panelists listed specific challenges unique to GenAI/LLMs, including AI “hallucinations,” prompt sensitivity, context-window limitations, and stochastic/unpredictable outputs. These issues can degrade data quality in ways not seen with traditional IT systems.

- **“Garbage In, Garbage Out” at Scale:** Many experts echoed that feeding low-quality data into LLMs produces low-quality (and potentially harmful) outputs at much greater scale and speed. One noted that organizations rushing to implement GenAI without assessing data quality will get “garbage out very fast and in massive volumes.”
- **Governance and Oversight Gaps:** Panelists highlighted difficulties in governing AI systems and measuring success. Traditional data governance struggles when “the input data is unknown or not easily governed” emphasized by panelists. Ensuring oversight of what data LLMs consume – and how to validate it – is a complex new frontier.
- **Volume and Discovery:** The enormous volume of data used to train and run LLMs makes it hard to discover data quality issues. Traditional DQ tools may not scale to identify errors or inconsistencies across such vast, diverse datasets.
- **Confirmation of Existing Issues:** Panelists noted that these challenges reinforce long-known truths: ultimately, “AI succeeds or fails based on the quality of inputs provided”, so data quality remains fundamental. GenAI’s emergence simply shines a brighter light on data issues that may have been ignored before.

Question 3 First Round Summary Analysis

Statement: *“Traditional Data Quality Management practices may fall short due to the adoption of GenAI and LLMs for companies who are not enhancing their DQ practices.”*

Responses & Consensus: A strong majority agreed, 11 of 15 panelists (73.33%), supporting the notion that without updates, traditional DQM will be insufficient for GenAI. However, 2 panelists disagreed and 2 were neutral/uncertain, indicating some debate. Notably, one panelist explicitly “disagreed” and argued current practices can still work, while another could neither agree nor disagree at this stage.

Common Feedback Themes:

- **Need for Evolution:** Experts agreeing stressed that many traditional DQM frameworks focus on structured data and static rules, which cannot keep pace with GenAI’s dynamic, unstructured

inputs. GenAI uses heterogeneous data (text, images, etc.), so “unstructured data has a lot of variability... making it extremely difficult for traditional DQ methods to be successful”.

- Frameworks vs. Agility: A recurring observation was that organizations often treat DQM frameworks rigidly (“a framework, not a straitjacket” as one put it). Panelists felt companies must be more agile and continuously improve DQ processes in response to GenAI. Stagnant “checkbox” approaches to DQ were already struggling even “before GenAI came along.”

- Not a Blanket Dismissal: Some experts cautioned that saying traditional DQM “falls short” should not imply abandoning existing best practices entirely. For example, one agreed there could be gaps, “NOT to say that traditional DQM practices should be dismissed... assume many of them do apply and/or can be refined.” This nuanced view suggests we must build on, rather than replace, traditional DQ fundamentals when integrating GenAI.

- Dissenting Views: The two dissenting panelists believed current DQ practices can still suffice if properly applied. One argued “current practices will continue to work,” even if GenAI tools can add efficiency. Another panelist outright disagreed and later clarified that both traditional and enhanced practices are needed together. Essentially, they felt the statement’s phrasing risked implying an either/or choice between old and new frameworks, which they rejected.

Question 4 First Round Summary Analysis

Statement: *“Robust Data Quality Management frameworks are critical for Artificial Intelligence performance.”*

Responses & Consensus: The panel showed near-unanimous agreement on this point: all 15 panelists agreed (100%). Panelists strongly affirmed that having a solid DQM framework (and methodology) is crucial to reliable AI outcomes, although a few noted caveats in practice.

Common Feedback Themes:

- Foundational Importance: Many emphasized that “AI models trained on poor quality data will likely produce unreliable outputs,” so robust DQM is indeed critical to AI success. Especially in

high-stakes domains (finance, healthcare), comprehensive DQ frameworks underpin AI reliability and trust.

- Framework + Methodology: Panelists not only agreed on frameworks' importance, but some highlighted that a DQM framework must be paired with effective processes/methodologies. One noted the framework provides the "what and why," but "a methodology is equally important to provide the how and when." This suggests organizations need both a strong DQ model and an operational plan to implement it.

- Preventing "Silver Bullet" Thinking: Several comments warned against neglecting data fundamentals amid AI hype. Too often organizations chase AI as a "silver bullet" and ignore data quality, only to be reminded that "garbage in, garbage out" still applies. The consensus is that robust DQM must not be skipped if one expects AI to perform.

- Real-World Use of Frameworks: One panelist agreed in principle but remarked that while frameworks are conceptually critical, in practice they are "rarely used... in my organization." This implies a gap between DQ theory and practice – even if everyone knows DQ is critical, many practitioners may not actively use formal frameworks day-to-day. This insight supports the need to bridge IQ theory and practice (a core objective of this research).

Question 5 First Round Summary Analysis

Statement: *"Effective Data Quality Management practice would tremendously help industries minimize risk and maximize value through strong and robust data governance and data quality management programs in light of GenAI and LLMs adoption."*

Responses & Consensus: All panelists agreed or partially agreed with this broad statement. 93.33% (14/15) agreed outright, while one panelist expressed "partial" agreement, noting reservations about current governance efficacy. Overall, consensus was that strong DQM/governance is indeed essential to reap GenAI benefits while mitigating its risks.

Common Feedback Themes:

- DQM as Risk Mitigator: Panelists concurred that high-quality, well-governed data ensures

GenAI/LLM outputs are accurate, trustworthy, and compliant, “thereby reducing risk while maximizing value.” In other words, effective DQM acts as a safety net against the pitfalls of GenAI (like bad decisions from bad data).

- Fundamental Principle: Several noted this is true “Generative AI or not” – quality data is always critical for sound decisions. GenAI’s arrival simply raises the stakes. One participant remarked this statement is almost “a given,” implying it is intuitively true.

- Defining “High Quality” for AI: A few experts pointed out that in practice we need to define what “high-quality information” means for LLMs. Traditional data quality metrics may not directly translate to AI outputs. This suggests a future focus on establishing concrete DQ criteria/metrics for AI contexts (e.g., factual accuracy of outputs, absence of bias, etc.).

- Organizational Understanding: Panelists stressed the importance of a shared understanding of data quality within organizations. One wrote that it would help if everyone agreed on “what the definition of data quality is, and what issues a sound DQM practice ameliorates.” This hints that part of maximizing GenAI’s value is educational – ensuring stakeholders know why DQM matters for AI projects.

- Governance Challenges: One expert commented that “as currently practiced, data governance is failing” in many organizations. This sober note means that while effective DQM/governance would help, many current programs are not up to the task. It underscores the need to improve governance practices themselves to truly realize GenAI’s value safely.

Question 6 First Round Summary Analysis

Statement: *“Data Quality (good or poor) directly impacts GenAI and LLMs outputs such as ‘Accuracy, Reliability and Ethical integrity’.”*

Responses & Consensus: 14 out of 15 (93.33%) agreed that the quality of data directly determines the quality of AI outputs. No one disagreed; one panelist did not select agree/disagree, instead asking for clarification on “ethical integrity.” The overwhelming consensus was that “garbage in, garbage out” applies good data yields better AI results, bad

data yields poor results.

Common Feedback Themes:

- **Universal Principle:** Many noted this is true for all systems, not just AI – if anything, GenAI just makes the link more apparent. As one panelist succinctly put it, “Ultimately AI succeeds or fails based on the quality of answers it provides, which in turn depends on the quality of inputs.” Quality data is the fuel for quality AI output.
- **Reiterating GIGO:** Multiple experts literally echoed the phrase “garbage in, garbage out,” emphasizing that LLMs are no exception. If anything, GenAI can mask garbage with fluent language, but the underlying accuracy/reliability will still be poor if the data is poor.
- **Strong Programs Needed:** Panelists argued that organizations need very strong DQ programs to monitor and understand the data fed into AI models. Since GenAI can ingest massive, diverse data, only a robust DQ framework can ensure that “good” data (accurate, relevant, up to date) is being used to train or prompt the models.
- **Question Terminology:** One panelist didn’t fully understand the inclusion of “ethical integrity” as an output attribute. They commented that they do not usually consider “ethical integrity” an output quality per se, but rather a property of the process or system. This suggests the phrasing might mix concepts; accuracy and reliability are clear output metrics, but “ethical integrity” might need definition or could be interpreted as model behavior (e.g., not producing harmful content).

Question 7 First Round Summary Analysis

Statement: *“If DQM frameworks and practices are not examined and enhanced, mitigation of biases in GenAI/LLM outputs is among the challenges industries will be facing.”*

Responses & Consensus: 13 panelists (86.67%) agreed that poor/unrevised DQM will make bias in AI output harder to tackle. One panelist disagreed and one was uncertain. The majority view is that stronger DQM is needed to address AI bias, though a couple of experts felt the logical connection in the statement was not clearly expressed.

Common Feedback Themes:

- **Weak DQM => Bias Risk:** The consensus was that without robust DQM, organizations will struggle to detect and mitigate biases in AI. Bias in training data or outputs can lead to “unfair and unethical decisions that undermine public trust,” and weak DQ practices leave these issues unchecked. In short, enhancing DQM is seen as part of the solution to AI bias.

- **Lack of Awareness:** Panelists noted many organizations don’t even realize DQ frameworks exist, let alone consider bias in that context. This lack of DQM maturity exacerbates the bias problem - companies may deploy AI with biased data without any DQ oversight until something goes wrong.

- **Not Just Bias:** A few comments broadened the issue: it’s not only bias that’s a risk when DQM is neglected, but any number of output quality problems (e.g., misinformation, lack of context). One expert wrote “the problem is not only about bias... it goes to the core of how we use language itself,” implying that semantic nuances can introduce subtler biases or misinterpretations beyond obvious demographic bias. This underscores that semantic data quality (meaning, context) is deeply intertwined with bias mitigation.

Question 8 First Round Summary Analysis

Statement: *“If DQM frameworks and practices are not examined and enhanced, mitigation of hallucination in GenAI/LLM output is among the challenges industries will be facing.”*

Responses & Consensus: 11 panelists (73.33%) agreed that poor DQM will make it hard to address AI “hallucinations” (fabricated outputs), while 3 panelists disagreed. One panelist was uncertain (7%). Agreement was lower here than prior questions, reflecting some divergence on whether data quality practices can directly mitigate model hallucination, which some see as more of a model issue than a data issue.

Common Feedback Themes:

- **Data Quality Can Influence Hallucination:** Those in agreement argued that low-quality or insufficient training data contributes to hallucinations, so better data management can reduce

this risk. One noted “low quality data is one of the leading causes of hallucination in your models,” implying that improving data quality (e.g., ensuring factual sources) should mitigate hallucinated answers. Another said models might present false info (hallucinate) if not grounded in high-quality data, which impacts trust.

- Limits of DQ’s Role: The dissenting panelists felt hallucination is primarily a model architecture/training issue, not purely a data quality issue. One disagreed in principle, stating “hallucination and drift are specifically model issues, not data issues.” Another simply said, “hallucination is about the model, not about the data.” These experts suggest that even if you feed an LLM perfect data, the model might still hallucinate due to how it generates responses. In their view, enhanced DQM alone won’t solve hallucinations; other measures (like better model training techniques or prompt designs) are needed.

- Partial Agreement – DQ Helps Indirectly: Some who agreed still provided nuance. For example, one panelist fully agreed hallucination is a risk and that more reliable data can “partially mitigate” it, but they were “not sure how much a data quality framework can mitigate this risk” because factors like prompting and training objectives also play a role. This acknowledges DQM is necessary but maybe not sufficient to tackle hallucinations.

Question 9 First Round Summary Analysis

Statement: *“If DQM frameworks and practices are not examined and enhanced, mitigation of misinformation in GenAI/LLM output is among the challenges industries will be facing.”*

Responses & Consensus: The results for this question 13 of 15 (86.67%) panelists agreed, 1 panelist disagreed, and 1 panelist was uncertain. The panel mostly concurred that lacking strong DQM will make it hard to prevent or correct misinformation from AI. One dissent and one uncertainty again pointed to phrasing or nuance rather than rejecting the concept outright.

Common Feedback Themes:

- **Acknowledgment of the Misinformation Problem:** Many panelists agreed that AI systems can produce outputs that are believable but absolutely WRONG, and that improving data quality processes is part of the solution. Robust DQM helps ensure sources are verified and context is correct, reducing AI's tendency to generate misinformation. One expert wrote that "true accuracy... demands explicit alignment with authoritative semantic standards or records", reinforcing that authoritative data comparison (a DQ practice) is needed to catch misinformation.
- **GenAI Heightens Scrutiny:** A few noted that society is now far more likely to question the quality of inputs and outputs because of GenAI's high visibility. Thus, if DQM isn't improved, misinformation could proliferate and be quickly noticed, damaging trust. This underscores the urgency of having better DQ checks in place.
- **Definitional Clarity:** One panelist who agreed provided helpful definitions distinguishing misinformation (unintentional falsehood) vs disinformation (deliberately false), noting robust DQM can help address both. This suggests that enhanced DQM would include verifying facts and context to combat misinformation output, and maybe also tracking provenance to detect deliberate manipulation.

Question 10 First Round Summary Analysis

Statement: *"If DQM frameworks and practices are not examined and enhanced, there will be a lack of transparency of the GenAI and LLMs output."*

Responses & Consensus: 11 panelists (73.33%) agreed, 2 panelists disagreed, and 2 panelists were unsure. A couple of panelists weren't certain what was meant by "transparency of output" or how DQM ties to it, leading to the mixed responses.

Common Feedback Themes:

- **DQM and Transparency Links:** Those who agreed interpreted it as meaning that improved DQM (especially tracking data lineage, documentation, etc.) aids transparency of AI outputs. Effective DQM lets organizations "track data provenance, audit data usage, and understand model reasoning," thereby making AI decisions more transparent. In this sense, if we don't

enhance DQ practices (like metadata management, lineage tracking), AI outputs will remain a “black box” – i.e., lacking transparency.

- Transparency Beyond Data Quality: Panelists noted that transparency is also about the model and process, not just data. One panelist agreed but pointed out that even with good DQM, “transparency will always be an issue since companies will be reluctant to share their GenAI input–procedure–output.” Another expert agreed with the statement but added that the “black-box nature” of models themselves causes a lack of transparency beyond the data. In essence, poor data quality can further obscure output reasoning, but even great data practices won’t fully open the black box of complex models. This nuance suggests transparency is a multi-faceted challenge (data transparency, model interpretability, corporate openness, etc.).

- Disagreement Reasons: The two panelists who disagreed felt DQM has limited impact on transparency. One expert said they “do not believe DQ management will impact the transparency of the LLM output unless this refers to DQ practices on the training data inputs.” This panelist saw transparency as more related to explainable AI techniques or the model itself rather than data quality processes.

Question 11 First Round Summary Analysis

Statement: *“Inadvertent leakage of sensitive information is an emerging challenge industries will be facing due to the adoption of GenAI and LLMs, hence the need to examine traditional data quality frameworks and practices.”*

Responses & Consensus: 11 panelists (73.33%) agreed, 2 panelists disagreed, and 2 panelists were neutral on this statement. Most panelists acknowledged the risk of sensitive data leakage with GenAI, but some questioned whether this is a “data quality” issue per se or more of a security/governance issue. The consensus on the core risk was high but tying it to DQ frameworks drew mixed feelings.

Common Feedback Themes:

- Sensitive Data Leakage is Real: Almost everyone agreed that GenAI/LLM adoption

introduces new privacy and confidentiality risks. Examples were noted of employees unintentionally pasting proprietary data into ChatGPT and thereby exposing it. The panel widely recognized this challenge.

- Role of DQ Frameworks: Where opinions diverged was on whether examining/enhancing DQ frameworks is the appropriate solution. Those agreeing felt that traditional DQ frameworks might not sufficiently address data sensitivity and should be updated. One wrote “this is where frameworks really help” – presumably in establishing guidelines for what data can be used in AI and how to handle sensitive fields. Others highlighted focusing on data sensitivity as part of data quality governance.

- Security vs. Quality: Both dissenting panelists argued that preventing leakage is more of a security and policy matter than a data quality matter. One said, “information leakage would be best addressed by solid security,” not necessarily by DQ processes. Another expert similarly stated this falls under “security... not directly quality of data,” except insofar as some DQ frameworks include security as a dimension. Essentially, they don’t disagree that leakage is a challenge, but they question linking it to DQ framework enhancements. They likely interpreted “data quality frameworks” in a partial sense that does not cover data security controls.

Question 12 First Round Summary Analysis

Statement: *“Inconsistent accuracy based on training context is an emerging issue due to adoption of GenAI and LLMs, hence the need to examine data quality practices and frameworks.”*

Responses & Consensus: 13 panelists (86%) agreed that GenAI can exhibit inconsistent accuracy across different contexts/domains and that this issue merits reexamining DQ practices. None disagreed; however, 1 panelist gave a neutral response, and 1 panelist did not know how to answer. The high agreement suggests the phenomenon was understood by most, but the phrase “inconsistent accuracy” confused a couple of experts.

Common Feedback Themes:

- Understanding the Issue: Those who agreed interpreted “inconsistent accuracy based on training context” to mean that a model’s accuracy varies across different subject areas or scenarios depending on how well the training data covers those contexts. Panelists concurred this is indeed a problem: GenAI might be very accurate in domains it has lots of data for, and poor in others. “The accuracy of GenAI outputs varies if data quality and context shift,” one noted. In other words, a model can be inconsistently accurate – great on some inputs, terrible on others – due to uneven training data quality.

- Context & Coverage as DQ Concerns: Experts pointed out that this ties back to data quality practices like ensuring comprehensive and representative training data. If certain contexts are not covered (or the data in them is lower quality), accuracy in those areas will suffer. One panelist commented on “training context,” and suggested that they read it as domain coverage and agreed that it’s a DQ concern if coverage is patchy.

- Importance of Context Metadata: Some related this to the need for context and metadata management in DQ. If organizations do not manage what context their data covers, they will not know where the model might fail. “Context is always an essential part of understanding data,” one panelist remarked, supporting that DQ frameworks should explicitly consider context to ensure consistent accuracy.

Question 13 First Round Summary Analysis

Statement: *“With poor data quality management practices, industries will be facing the ‘Digital Inbreeding Risk’ – meaning that AI-trained data or AI-generated data of low quality can degrade over time, leading to challenges like bias, hallucination, and misinformation.”*

Responses & Consensus: This statement garnered near-unanimous agreement (14 of 15, 93.33%). Every panelist agreed except one who did not provide a clear agree/disagree but instead commented that the question “needs rephrasing or further explanation.” Thus, the concept of “digital inbreeding” resonated strongly, though the wording was evidently confusing to at least one person.

Common Feedback Themes:

- Strong Recognition of the Risk: Panelists broadly agreed that if AI systems start training on data that includes outputs from other AI (especially if those outputs are low-quality), there is a compounding degradation effect – essentially a feedback loop of poor quality. One described that “LLMs training on LLM-generated data tends to severely degrade the quality of the resulting model.” Another worried that “if the questionable output of GenAI becomes input to train another, the degradation of results can happen quite quickly.” This captures the crux of “digital inbreeding”: AI feeding on its own imperfect progeny and getting worse.

- Amplification of Errors: Panelists noted this leads to amplified biases and errors. Over generations, biases can compound, and misinformation can self-reinforce. One expert explicitly linked it to observed output concerns: “I have already been concerned that it (GenAI output) looks good, sounds good, but it is not right... when that output is used as input to another model, it quickly degrades results.” This scenario exemplifies why strong DQM is needed to prevent tainted outputs from contaminating the training pool.

- Agreement on Terminology (once explained): Many panelists essentially rephrased the statement in their own words, indicating they grasped it. For example, “the models increasingly learn from poor data which over time produce unreliable output” or “training a model on its own outputs leads to all kinds of risks.” This shows that after parsing; the panel overwhelmingly concurred with the idea.

Question 14 First Round Summary Analysis

Statement: *“Generative AI models (e.g., LLMs, GANs, Diffusion Models) rely on large, diverse datasets.”*

Responses & Consensus: 11 panelists (73.33%) agreed, 2 panelists disagreed, and 2 panelists did not answer. Essentially everyone accepted this statement as a factual premise, aside from one dissent emphasizing that not all models require such scale or diversity. The high agreement suggests it was seen as a mostly self-evident truth (with a caveat for smaller

models).

Common Feedback Themes:

- **General Agreement:** Most panelists simply agreed without additional comment, some noting “this is a fact.” It is broadly understood that large-scale models like GPT-4 are trained on enormous, diverse corpora. Panelists concurred that diversity and volume of data are part of what gives these models their power.
- **Nuance on Model Size:** One expert pointed out that while “large general purpose models...do require such large datasets,” there exist “smaller, more focused models” that may not need as diverse a dataset. They added that even for smaller models, having larger pools of high-quality data helps accuracy. This nuance is valid: not every ML model is an LLM; some narrow-domain models train on limited data. However, the statement was phrased generally about “Generative AI models (e.g., LLMs)” which by definition implies large scale, so the disagreement likely stems from the panelist considering exceptions (like fine-tuned small models).
- **Relevance to DQ:** Although not many wrote feedback, one panelist mentioned this reliance on diverse data “by definition...if they are to have general utility, the datasets must be diverse”. Some also brought up that there are indeed approaches like LoRA that allow working with smaller datasets for specific LLMs. In terms of DQ management, the implication is that ensuring the quality and representativeness of these large, diverse datasets is paramount.

Question 15 First Round Summary Analysis

Statement: *“There is a growing need for improved data quality frameworks in industry to effectively support data quality management amid the adoption of GenAI and LLMs.”*

Responses & Consensus: The panel showed virtually unanimous agreement (100%). All 15 panelists agreed (some with minor commentary). This strong consensus underscores that experts believe current industry DQ frameworks need upgrading to handle the challenges introduced by GenAI.

Common Feedback Themes:

- Resounding Yes: Panelists widely affirmed that industries need better DQ frameworks to cope with GenAI. The adoption of AI is multiplying opportunities for poor data to cause harm, thus creating urgency for stronger DQ structures. One noted, “AIs are providing many more opportunities for poor DQ to have negative impacts in multiple domains,” hence improved frameworks are required. Others simply agreed without comment, implying the statement was self-evident.

- Fit-for-Purpose Frameworks: Some emphasized that new or improved frameworks must be “fit for purpose” for GenAI, which differs significantly from traditional structured data environments. LLMs deal with unstructured data, so frameworks should incorporate semantic quality checks, contextual evaluations, etc., not just traditional dimensions.

- Volume, Speed, New Data Types: Panelists mentioned factors driving this need: the amount, speed, and types of data in GenAI are unprecedented. Traditional DQ frameworks were not built with continuously generated AI outputs or massive unstructured training corpora in mind, hence they require evolution. For example, frameworks may need to address how to validate model outputs (which is a new angle for DQ).

- Building on Existing Models: One expert advised that improving DQ frameworks for GenAI doesn’t mean reinventing the wheel from scratch: “Assume you will build on existing models, not throw them away”. Many/most traditional DQ practices still apply; they can be refined or expanded to include GenAI considerations. This aligns with earlier sentiments that we should extend rather than discard established DQM principles.

- Panelist Reservations: No one disagreed, and one expert commented that they do not know of any practitioner calling for them (improved frameworks).” This implies that while the need exists in theory, the demand has not been strongly voiced in industry yet. It could be an emerging realization - one this study aims to validate and bring to light. It also suggests an action item: raising awareness among practitioners about the importance of updating DQ frameworks for AI.

Question 16 First Round Summary Analysis

Statement: *“Data professionals should advocate for urgent review of existing data quality frameworks in organizations adopting GenAI and LLMs to identify gaps and implement necessary improvements.”*

Responses & Consensus: 12 panelists (80%) agreed with this prescriptive statement, 1 panelist disagreed, and 2 panelists gave neutral/uncertain comments. The majority supported the call to action that data professionals take the lead in reviewing and updating DQ frameworks for AI. The few non-agreeing responses provided thoughtful nuances about who would conduct such reviews and the realism of organizations doing it independently.

Common Feedback Themes:

- **Strong Advocacy Role:** Those who agreed championed the idea that data professionals (DQ managers, data stewards, etc.) must proactively push updating DQ frameworks when their orgs embrace GenAI. They saw it as part of professional responsibility to ensure AI projects do not outpace data quality. One said it’s not just about advocating – “it’s a necessity at this point”, indicating how critical this review is.

- **Gaps to Address:** Panelists acknowledged that a thorough review would likely reveal needed additions such as provenance/lineage tracking, bias detection measures, semantic consistency checks, etc. (some of which were mentioned in feedback to other questions). In essence, existing frameworks might lack these GenAI-specific considerations, hence professionals should identify and fill those gaps.

- **Leverage Existing Foundations:** A theme emerged that while review and improvement are urgent, it should build on what exists. For example, one expert noted if core traditional dimensions (like Wang and Strong’s top dimensions) are addressed, “almost all requirements for GenAI will be resolved, except provenance/lineage/traceability.” This implies current frameworks are a starting point and mostly adequate, with just a few critical enhancements needed for AI contexts (especially around traceability).

- Realism and Responsibility: The panelist who disagreed did so not because review isn't needed, but because they doubt organizations actually have DQ frameworks to review or will do it. "I don't think organizations adopting AI employ such frameworks," they said, suggesting many companies lack a formal DQ framework altogether. They did mention they personally called for an independent review of foundations (indicating support for the idea in principle). Another panelist agreed with the need but argued it is "unrealistic to think individual organizations will do this review" - instead, industry leaders (associations, consultants, researchers) should do it and publish guidance. These points reflect that many companies might not have the expertise or will to self-audit their DQ frameworks for AI, so data professionals may need external support or frameworks.

- Neutral View: One panelist did not explicitly agree or disagree but commented that reviewing existing frameworks is part of figuring out solutions, though it's uncertain if any current framework can serve as basis for a new one. Another expert provided a nuanced take without agreeing or disagreeing. These highlight considerations on how such reviews should be conducted (collaboratively, industry-wide perhaps).

Question 17 First Round Summary Analysis

Statement: *"To ensure trustworthy GenAI and LLM outcomes and maintain data quality integrity, industries must adopt enhanced data quality frameworks rather than relying solely on traditional ones."*

Responses & Consensus: A majority agreed (12 of 15, 80%) with this statement, 3 panelists disagreed. The split indicates that while most experts think enhanced frameworks are needed for trust in GenAI, a notable minority cautioned against the phrasing that might undervalue traditional frameworks. Indeed, their comments stressed a balanced approach.

Common Feedback Themes:

- General Agreement: Those in favor argued that modernized or enhanced DQ frameworks are essential to keep GenAI outputs trustworthy. Traditional frameworks alone, if used "as is," may

not cover aspects like real-time monitoring, semantic quality, or AI-specific risks. One panelist wrote, “Trust in GenAI outputs depends on modernized DQM frameworks integrating dynamic quality measures, metadata curation, ongoing audits,” etc. This exemplifies the enhancements envisioned.

- Areas of Enhancement: Panelists suggested enhanced frameworks should include things like semantic analysis for unstructured data, explicit linking of DQM to model lifecycle stages, and other practices beyond what traditional data quality programs do. The general vibe was that clinging strictly to old DQ methods without evolution would be inadequate for AI – some new components are “necessary” (though not sufficient).

- Not Either/Or: The dissenting voices pushed back on the phrasing “rather than relying solely on traditional ones.” They cautioned that traditional practices are still valuable and should not be thrown out. For instance, one expert said even enhanced frameworks will not ensure trustworthy outcomes by themselves; good data is necessary but “not sufficient.” One expert provided a comment suggesting to “be careful of assuming that none of the traditional practices apply... assume many do apply and can be refined/modified, with additional practices for GenAI.” Two panelists noted that if an organization is not adopting GenAI, traditional methods suffice, which also show a desire to balance new with old. In summary, even among those who agreed, there is an implicit understanding that hybrid approaches are best - use enhanced frameworks but also leverage proven traditional DQ practices.

- Need to Adopt Enhanced Frameworks: Despite the above nuance, no one disputed that some enhancements are needed. The debate is more about messaging. “Must adopt enhanced frameworks rather than solely traditional” was slightly polarizing; refining the wording to “augment traditional frameworks with enhancements” might bring full consensus. Essentially all panelists accept the need for improvement (as seen in Q15 and Q16), so the key is to not imply an abandonment of the past.

3.3.4 Prioritization of Semantic Data Quality Dimensions – (Questions 18–23)

In this section, the Delphi panel evaluated statements comparing semantic data quality dimensions (focused on meaning, context, etc.) versus syntactic dimensions (structure, format, etc.) in current practice. The goal was to validate observations that semantic DQ is often neglected and to gauge support for emphasizing semantic factors in the GenAI era. The panel strongly affirmed the importance of semantic DQ, though a few members provided caution that syntactic fundamentals remain important and that measurability of semantics is challenging.

Question 18 First Round Summary Analysis

Statement: *“Based on this research’s preliminary analysis and your expertise, do you agree that semantic data quality dimensions (meaning and context) are often overlooked and not adequately measured in current practice?”*

Responses & Consensus: An overwhelming 14 of 15 panelists (93.33%) agreed that semantic DQ dimensions are indeed often overlooked and under-measured in today’s data quality practice. This confirms our preliminary finding that meaning/context get short shrift in real-world DQM compared to more technical aspects.

Common Feedback Themes:

- Widespread Agreement with Premise: The majority of experts echoed that organizations focus on syntactic data quality (valid codes, formats, completeness) and pay far less attention to semantic aspects like coherence, context relevance, or interpretability. One stated plainly: “Most organizations focus on resolving syntactic issues... while essential aspects such as coherence, meaning, and context are much less prioritized.” Another noted their organization “doesn’t know the difference between semantic and syntactic... semantic isn’t part of our data quality vocabulary.” These comments exemplify the oversight of semantic DQ in practice.

- Challenges in Defining/Measuring: Panelists agreed it is happening, and some gave reasons why semantic dimensions are neglected. They are “more challenging to define and measure,” often requiring good metadata management (which many companies lack) and can

be expensive or hard to quantify. In contrast, syntactic checks can be automated (e.g., format checks), so organizations default to those. Essentially, semantics fall by the wayside because they're harder.

- Metadata & Foundations: It was pointed out that if basics (technical quality) aren't right, semantic quality is hard to achieve – “if you don't have the technical aspects right, it's really hard to have the semantic and contextual pieces right”, and those pieces depend on metadata which is “painfully lacking” in many orgs. This suggests a maturity model: some organizations have not even mastered syntactic DQ, so they never progress to semantic DQ. However, those that manage their metadata well tend to also do better on data meaning/context.

- Dissenting View: One expert seems to “disagree that semantic dimensions of DQ are overlooked,” noting that it is well-known that providing context in AI is essential. This disagreement suggests that not all corners of industry ignore semantics; some may be paying attention. But given the weight of other responses, those are likely exceptions.

- Panelist Insights: Others added that while overlooked in many places, the importance of semantic DQ is rising due to GenAI. Data curation for LLM success was cited. Also, one panelist cross-referenced our provided frequency counts table and said, “I have to agree” it shows semantic dims are not prioritized - “they are not ALWAYS overlooked,” they noted, but often. This nuance acknowledges there are cases or organizations where semantics are considered but generally supports the statement's truth.

Question 19 First Round Summary Analysis

Statement: *“Do you agree that syntactic data quality dimensions (structure and format) are prioritized over semantic dimensions in practice?”*

Responses & Consensus: 13 panelists (86.67%) agreed that in practice, organizations give more attention to syntactic aspects of data quality than to semantic ones and 2 panelists disagreed. The strong agreement further validates the pattern that format/accuracy checks take precedence, while meaning/context gets less attention.

Common Feedback Themes:

- Syntactic is Easier, So It's the Focus: Many pointed out that syntactic DQ is “easier to automate and measure, so it receives most attention in data quality programs”, whereas semantic issues (misclassification, ambiguity, context) are “harder to quantify and often left out of routine checks”. This summarizes why organizations prioritize syntactic: it is low-hanging fruit. Panelists concur that because you can readily run scripts for validity, completeness, consistency, etc., those get done, and semantic consistency or relevance checks (which might require human judgment or complex algorithms) often don't.

- Evidence in Practice: One panelist remarked on exactly this: “In practice, syntactic DQ dimensions are more often the focus because they are ‘easier’ to measure and manage”. Another said “this is the easier of the two (syntactic) but doesn't address correctness or meaningfulness” - implying organizations know they are focusing on the easy part and potentially ignoring the actual meaning. The consensus is that efficiency drives emphasis - companies tackle what they can with existing tools (formats, schema checks) and shy away from the messy semantic territory.

- Reinforcing Q18 Reasons: Panelists referenced earlier points: lack of metadata, difficulty, cost. Experts emphasized that semantic DQ dimensions are put on the back burner due to operational challenges and lack of know-how, leaving syntactic as the mainstay of DQ efforts.

- Disagreeing Opinions: The two who disagreed provided interesting perspectives. One said, “some dimensions fall under both semantic and syntactic sets,” hinting that the dichotomy is not always clear-cut – e.g., an issue like completeness has both semantic and syntactic facets. This could be interpreted as: in practice, professionals might be addressing some semantic elements without labeling them as such because of overlap. It is possible in these experts' experiences, certain organizations do prioritize semantics (for instance, where data meaning is critical, like medical data, some semantic controls exist). Still, 13 out of 15 agreeing is a decisive majority.

- Unified View: Combining Q18 and Q19 feedback, the panel clearly sees a priority gap

favoring syntactic DQ. This sets the stage for treating semantic dimensions as essential and promoting them.

Question 20 First Round Summary Analysis

Statement: *“Do you consider semantic data quality dimensions (DQ dimensions centered on meaning and context) as essential measurable factors for defining data quality requirements and evaluating industry data quality performance and risks in this age of GenAI and LLMs adoption?”*

Responses & Consensus: All panelists agreed (100%) that semantic DQ dimensions are essential factors for defining DQ requirements and assessing performance/risk in the GenAI era - with a slight caveat from one panelist about measurability. This unanimous endorsement underlines that the experts see semantic quality not as a luxury, but as a necessity now.

Common Feedback Themes:

- Semantic Dimensions = Vital in GenAI Context: The panel uniformly affirmed that meaning- and context-related dimensions (e.g., accuracy of meaning, relevance, completeness of context, etc.) are crucial metrics for data quality when dealing with AI. Comments included: “Semantic DQ dimensions are vital for defining requirements and evaluating data risks, especially with GenAI... models are highly sensitive to subtle shifts in meaning and context.” Also “Semantic is equally if not more important than syntactic... because it’s the business that drives the semantic aspects of data quality” - highlighting that ultimately it’s the business use (meaning and context) that matters most for data fitness.

- Acknowledging Measurability Challenge but Necessity: One panelist wrote they agree semantic dimensions are important, “Disagree that they are measurable” easily. They note “DQ measurement is always tough.” Data does not possess physical properties.” - implying that while essential, semantic quality often requires subjective or complex evaluation. Despite this, they still marked “Agree” to the overall question. This view suggests yes, we must include semantic dimensions as requirements, even if measuring them is challenging, because

excluding them would ignore key aspects of quality. Another panelist also said “Yes, these are measurable; however, measuring them takes a different kind of work than syntactic dimensions”, indicating it’s possible but perhaps through qualitative assessments or specialized techniques.

- Examples of Semantic DQ Dimensions Emphasized: Panelists enumerated specific semantic factors: factual accuracy, relevance, consistency of meaning, interpretability, etc., as things that need to be considered now. One looked forward to “more techniques and measures for semantic DQ dimensions” becoming available, showing that the field is evolving to quantify semantics (through NLP, perhaps human-in-loop evaluations). Another stressed that simply telling organizations to apply semantic measures is not enough - you have to guide them “how to do so and be more detailed.” This underscores that while everyone agrees on importance, there is a need for practical guidance on implementation.

- GenAI Makes Semantics Mandatory: The context of GenAI was repeatedly mentioned. Because LLMs work with natural language (which is loaded with context and nuance), ignoring semantic quality can directly lead to model failures (hallucinations, irrelevance, etc.). Panelists conveyed that industries must incorporate semantic quality checks now to ensure AI outputs are fit for use. The consensus here aligns with the research’s objective: bridging IQ theory and practice by injecting semantic IQ into everyday DQM.

Question 21 First Round Summary Analysis

Statement: *“Should data quality professionals promote prioritizing semantic data quality dimensions within DQM practices in this era of GenAI and LLMs adoption?”*

Responses & Consensus: 10 of 15 panelists (66.67%) agreed that DQ professionals should indeed push for greater priority on semantic DQ dimensions now, 3 disagreed, and 2 were neutral/unsure. The majority favors championing semantic DQ, but a sizable minority cautioned that priorities should be situation-dependent and not at the expense of other dimensions.

Common Feedback Themes:

- General Support for Emphasizing Semantics: Most panelists think DQ practitioners need to lead a shift in focus. “Yes, semantic data quality is especially important in LLMs,” one wrote plainly. Others elaborated that ensuring accurate meaning and context will “minimize the risk of harmful or misleading outcomes” and is advocated by emerging best practices (mentioning tools like data catalogs, knowledge graphs to support semantic understanding). This indicates many experts believe it is part of a DQ professional’s role today to elevate the discussion around semantic quality (which historically has been undervalued).

- Not Ignoring Other Dimensions: Four dissenting panelists each argued against prioritizing semantics to the exclusion or downgrading of other dimensions. One expert said DQ practitioners should pick dimensions based on context/purpose - “quality dimensions [should be] based on who will use the AI and for what purposes,” rather than a blanket prioritization of semantics. Another expert said, “all dimensions should be treated equally and as important.” Another panelist echoed that both syntactic and semantic are needed, “would not prioritize one over the other... semantic without syntactic doesn’t work.” These responses warn against swinging the pendulum too far - you cannot abandon basic data integrity (syntactic) in favor of semantics; they are complementary. Their disagreement seems rooted in the semantics of the word “prioritizing.” They likely agree semantics needs more attention than currently, but they resist any implication that semantic dimensions outrank or replace others in all cases.

- Conditional Agreement: Even among some who agreed, there was nuance. One expert provided a comment: “Not sure how to answer because it depends on the situation. Promoting semantic DQ generally is helpful, but not always at the expense of other DQ dimensions. Semantic dimensions should become part of the ‘menu’ of DQ dimensions available... organizations can determine which are most applicable to various situations.” This aligns with the dissenters - context matters, and DQ professionals should champion semantics by adding them to the toolkit and highlighting them but not necessarily making everything all about

semantics all the time. One expert (who agreed) also noted “prioritizing these dimensions is not an end in itself”, implying they must be part of a solid framework and solve real problems, not just a buzzword.

- Overall Insight: The panel clearly wants greater inclusion of semantic DQ in practice, but many prefer a balanced approach. The question’s phrasing “promote prioritizing” might have been interpreted as “promote above all else.” Wording like “promote increased focus on semantic DQ dimensions” might capture the intent without implying that syntactic or other aspects should be neglected. After all, the previous answers show semantics have been under-prioritized historically, so “prioritizing” could be read as “bringing them to equal footing” rather than “making them top dog.”

Question 22 First Round Summary Analysis

Statement: *“Do you agree that measuring accuracy of meaning, relevance, and context requires comparison with an Authoritative Data Source (ADS) or official System of Record (SOR)?”*

Responses & Consensus: 11 panelists (73.33%) agreed that to assess a data element’s meaning/relevance accuracy, you generally need an authoritative reference. 3 panelists disagreed and 1 panelist was neutral. The majority aligns with the idea that you need some ground truth (ADS/SOR) to judge semantic accuracy, but a notable minority pointed out other ways or complexities in evaluating meaning and relevance.

Common Feedback Themes:

- Agreement – ADS/SOR as Benchmark: Those in agreement argued that without a trusted reference point, subtle errors in meaning or context can go undetected. “Without a trusted reference, subtle errors... can persist undetected and undermine AI results,” one stated. Another said, “Semantic accuracy will be challenging even with an ADS but extremely hard without.” These comments support using authoritative data sources or knowledge bases as a yardstick for truth when evaluating if an AI output’s content is correct in context. In practice, this

might mean if an LLM says “X is Y”, you check against an official database or source of truth to verify that fact (meaning).

- Source of Truth & DQ: Panelists highlighted that a big difficulty is often identifying what the authoritative source is for a given piece of information. One noted their organization struggles to determine the ADS and responsible owner for data, implying that establishing authoritative sources is itself a data governance challenge. Nonetheless, having that ADS (and ensuring it's of high quality) is seen as key to measuring semantic correctness.

- Human-in-Loop Alternatives: One expert gave a perspective that meaning/relevance might be tied to the purposes and business processes, and if ADS/SOR are used, then those systems should include high-quality metadata to judge context reliability. They implied other methods like human judgment or business rules might also play a role, not strictly ADS comparison.

- Disagreements - Other Ways to Measure or Not Always Possible: The dissenting panelists provided two main counterpoints: (a) Subjectivity of meaning: One said “what if there is no standard? I do not think one is even possible when it comes to ‘meaning’ - things can mean different things to different people.” They argue meaning is sometimes inherently subjective or context-dependent, so an official source may not exist. (b) Alternative approaches: Another disagreed even though they acknowledge ADS/SOR can help - “I disagreed even though meaning, relevance, context could benefit from ADS comparison. However, they are often subjective, which means other measurements have to be employed. This suggests using methods like human-in-the-loop evaluations, user feedback, or AI-based heuristics to judge semantic quality. Similarly, another panelist disagreed and noted “other ways are also possible, e.g., human-in-the-loop or AI-in-the-loop approaches... ADS or SOR is possible but not necessarily required.” They envision leveraging expert review or even another AI to cross-check, rather than strictly needing an official dataset.

- Takeaway: Most believe having an authoritative reference is the safest and most objective way to measure semantic accuracy. The disagreement highlights fair points stating that the

ADS/SOR approach may not always be feasible (if no authority exists) and that we might need to develop other evaluative techniques.

Question 23 First Round Summary Analysis

Statement: *“Do you believe the practice of assuming validated data is inherently accurate without a proper comparison of data with an approved Authoritative Data Source should end, especially with the rise of GenAI and LLMs?”*

Responses & Consensus: 10 of 15 panelists (66.67%) agreed that blindly trusting “validated” data without cross-checking against an ADS should stop, particularly in the GenAI context. 1 panelist disagreed, but notably 4 panelists were neutral or unsure. This indicates that while a majority sees unverified trust in data as risky, a considerable number found the question unclear or needed more context to answer.

Common Feedback Themes:

- General Agreement – Don’t Assume, Always Verify: Those who agreed voiced that just because data passed some validation rules or came from a supposedly validated source, one shouldn’t assume it is 100% accurate. This is increasingly true with AI in the mix, where stakes and scrutiny are higher. One wrote “assuming ‘validated data’ is inherently accurate is dangerous!” Another noted it is “a slippery and dangerous slope” to believe validated = accurate without independent verification. The consensus here is that contextual and semantic validation (e.g., comparing to an ADS) is needed, not just internal validation checks. Especially as GenAI can amplify any errors, not double-checking against a source of truth can propagate inaccuracies widely.

- Complexity and Cost Considerations: A few agreeing panelists acknowledged that while ideal, requiring ADS comparison for every case might be impractical. One said, “while we should never assume validated data is accurate... without context it’s hard to determine since requiring an ADS in every case could make many tasks impossible due to cost.” So, they agree in principle but note real-world constraints – you have to pick battles (some data might be “good

enough” after standard validation if further checking is too costly). This suggests a nuanced approach: where possible, verify against ADS, but weigh the effort vs risk.

- Trusted ADS/SOR: Panelists commented that trust in data depends on validation criteria, and ADS comparison is one way, but even ADS must be trusted and rules appropriate. They conclude if people trust validations they do not understand, that’s bad – which aligns with the idea that blind trust is dangerous. Similarly, another panelist commented “... it is increasingly risky to assume ‘validated’ data is accurate without confirming against a formally approved ADS,” effectively agreeing but phrasing it as an observation. Another expert gave an example about definitions of “validated” and “accurate” - illustrating that something could be technically valid (e.g., code in a list) but still not accurate contextually. This insight supports the statement: just because data passes a validity check doesn’t ensure it’s accurate in the real-world sense.

- Disagreement: One panelist disagreed outright, and referenced back to Q16 nuance: essentially saying sometimes using validated data without checking an ADS is okay depending on context, and requiring an ADS always could be overkill. They advocate context-based decision - sometimes “validated” data is sufficient if it meets known criteria. This is a moderate dissent focusing on practicality rather than principle.

4.3.5 First Round Analysis Questions 24 - 30

Question 24 First Round Summary Analysis

Question: *What do you consider the most important semantic data quality dimensions for GenAI and LLMs?*

Key themes suggested by panelists:

- **Accuracy / factual correctness** – Nine panelists emphasized the need for data to truthfully represent the intended concept and avoid hallucinations.
- **Relevance / context / value add** – Nine panelists noted data must be pertinent to the specific use case and add value.
- **Completeness and consistency** – Three panelists mentioned ensuring all required concepts are present and semantically consistent across sources.

- **Believability / trustworthiness** – Four panelists noted that users must be able to trust the data's meaning and provenance.
- **Timeliness** – One panelist added that the semantic quality should be maintained over time.
- **Availability of ontologies/dictionaries** – One panelist emphasized the ability to resolve terms through shared dictionaries and glossaries.
- **Equal importance across dimensions** – Three panelists argued that no single dimension should dominate, underscoring the need to incentivize adoption of DQM practices.

Question 25 First Round Summary Analysis

Question: *What do you consider good measures for what you believe to be the top semantic Data Quality (DQ) dimensions?*

Key themes suggested by panelists:

- **Dictionary/ontology coverage and resolvability** – Two panelists suggested counting how many data elements are present in a publicly available dictionary and are resolvable to specific meanings.
- **Accuracy validation** – Five panelists recommended factual validation through triangulation, authoritative data sources, audit sampling, and lineage tracing.
- **Contextual consistency checks** – Two panelists proposed automated tools that compare meaning across sources and verify relevance to the use case.
- **Human and value-based measures** – Two panelists stated that business impact and human assessment (“are we accomplishing the mission?”) are necessary indicators.
- **Trustworthiness/fairness** – Two panelists noted trust metrics (e.g., percentage of data matched to an authoritative source) and fairness auditing as relevant but underdeveloped measures.
- **Lack of established metrics** – Three panelists remarked that there are currently few comprehensive metrics for semantic quality.

Question 26 First Round Summary Analysis

Question: *Any thoughts from the experts as to how to efficiently gather these measures?*

Key themes suggested by panelists:

- **Automated validation and auditing** – Seven panelists advocated automating the gathering of semantic metrics by embedding validation into pipelines, creating vector stores for retrieval-augmented generation, and running regular

completeness/timeliness audits.

- **Use of public dictionaries and ontologies** – Two panelists highlighted ensuring the presence of public dictionaries and resolvable elements as a first step.
- **Surveys and feedback from users** – Four panelists suggested using surveys, human feedback, and cross-functional discussions to collect qualitative measures where automated checks are insufficient.
- **Human-in-the-loop and organizational collaboration** – Three panelists stressed speaking with stakeholders across the organization and incorporating human reviews, despite the cost and time required.
- **Clear definition of measures and rules** – One panelist argued that efficient collection depends on well-defined, fine-grained rules and domain-specific metrics.

Question 27 First Round Summary Analysis

Question: *What do you consider the most challenging aspects of Data Quality (DQ)*

when it comes to GenAI and LLMs?

Key themes suggested by panelists:

- **Ensuring accuracy and context** – Six panelists pointed to inaccurate data, wrong context, and hallucinated outputs as core challenges.
- **Lack of ground truth and traceability** – Four panelists emphasized difficulties in validating outputs without a ground truth and tracing errors back to the source.
- **Bias and fairness** – Two panelists noted that detecting, mitigating, and auditing bias in semantic data is complex.
- **Unstructured and semi-structured data** – One panelist highlighted that GenAI models break unstructured data into pieces, making it hard to measure completeness and relevance; two panelists added that the pace of data creation outstrips current DQ processes.
- **Rapid technological change** – Two panelists observed that the speed of GenAI evolution makes traditional DQ practices lag behind.
- **Organizational skills and adoption** – Four panelists remarked on the unwillingness or inability of organizations to invest in deep DQ work and adopt necessary practices.

Question 28 First Round Summary Analysis

Question: *What components of Data Quality Management practice need the most enhancement to keep up with the rapid adoption of GenAI and LLMs?*

Key themes suggested by panelists:

- **Output validation and real-time monitoring** – Five panelists called for better methods to evaluate and monitor generated content in real time.
- **Automated tools for semantic checks and deduplication** – Five panelists suggested developing and deploying automated tools for semantic similarity, relevance scoring, deduplication, and metadata modeling.
- **Education, awareness, and organizational capability** – Four panelists emphasized the need to educate stakeholders and build organizational capacity to address semantic DQ.
- **Speed and agility of DQ processes** – Two panelists noted that DQM methods must operate at the speed of GenAI outputs.
- **Governance, metadata modeling, and stewardship** – Three panelists urged strengthening DQ governance structures, metadata modeling, and data stewardship.

Question 29 First Round Summary Analysis

Question: *What are your overall comments or suggestions for improving traditional data quality management frameworks and practices to ensure trustworthy and reliable outputs from GenAI and LLMs in current industry use?*

Key themes suggested by panelists:

- **Extend frameworks to include semantic and output monitoring** – Seven panelists argued that traditional frameworks must expand beyond input validation to include continuous output monitoring, semantic similarity checks and feedback loops.
- **Incorporate AI-powered tools and human-in-the-loop** – Three panelists recommended using AI to automate semantic checks, with human oversight and domain-specific adaptation.
- **Adapt frameworks rather than discard them** – Three panelists noted that frameworks should be treated as adaptable guides rather than rigid rules.
- **Broaden to model-ready dimensions** – Four panelists suggested integrating dimensions such as bias, representation, reproducibility, and fairness.
- **Define fine-grained, domain-specific rules** – Two panelists highlighted the need for granular semantic rules and ontologies.
- **Clarify semantics and educate stakeholders** – Two panelists emphasized education around semantic vs. syntactic dimensions.

Question 30 First Round Summary Analysis

Question: *What are your overall recommendations for prioritizing semantic data quality dimensions to help industries effectively define, assess, and monitor data quality requirements*

and performance in the era of GenAI and LLMs?

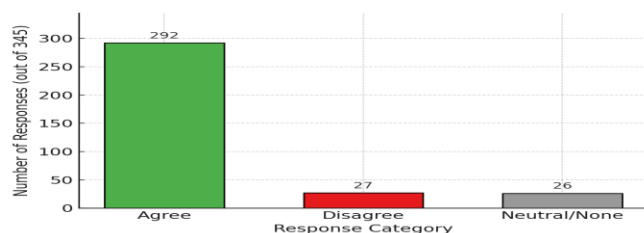
Key themes suggested by panelists:

- **Develop shared ontologies and dictionaries** – Three panelists stressed using publicly accessible dictionaries and ontologies to align meaning across systems.
- **Prioritize accuracy and factual correctness** – Six panelists highlighted verification against authoritative sources and maintaining accurate, complete data sets.
- **Combine automated profiling with human review** – Two panelists recommended combining automated semantic profiling (e.g., NLP entity recognition, summarization) with human review, crowdsourcing, and fairness audits.
- **Education and governance** – Four panelists called for investing in education, embedding semantic metrics into dashboards, and formalizing semantic DQ policies and standards.
- **All dimensions matter** – Two panelists suggested that no single dimension should dominate; incentives and research are needed to develop actionable measures.

4.3.6 Summary Analysis First Round

Figure 1

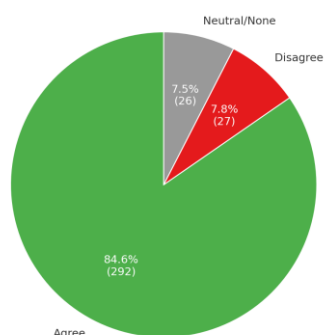
First Round Overall Agreement Distribution Questions 1-23



The bar chart summarizes the overall panelist agreement distribution across all 23 questions. Each bar represents the total number of responses (out of 345) that fell into Agree, Disagree, or Neutral/None categories. The Agree category dominates with 292 responses, indicating about 85% of all responses were agreements, whereas Disagree (27 responses) and Neutral/None (26 responses) are only around 7–8% each. This highlights a strong overall consensus among panelists, with relatively few disagreements or neutral responses.

Figure 2

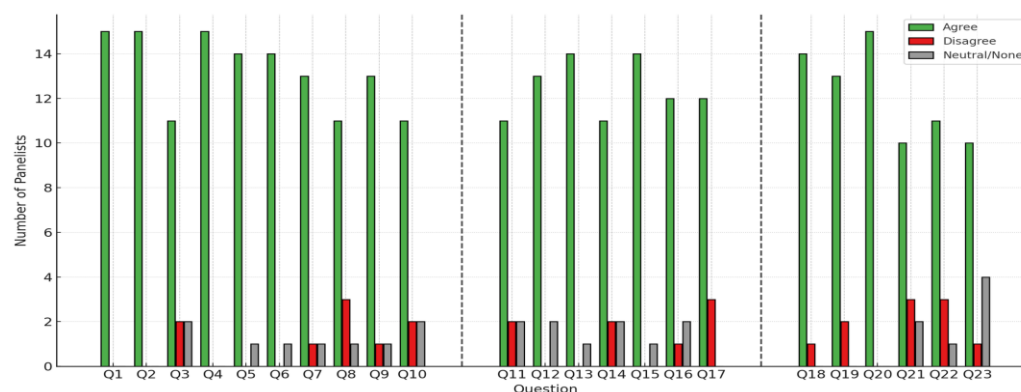
First Round Pie Chart Overall Agreement Distribution Questions 1-23



Pie chart illustrating the same overall agreement distribution across all 23 questions. The green slice (Agree) constitutes approximately 84.6% of the pie (292 out of 345 responses), reflecting the overwhelming majority. In contrast, the red slice (Disagree) and gray slice (Neutral/None) are very small (each about 7–8%), corresponding to the 27 disagree and 26 neutral/none responses, respectively. The pie visualization reinforces that nearly all panelists' answers were "Agree," with only a minor fraction of responses being dissenting or neutral, thus visually emphasizing the panel's broad consensus.

Figure 3

Per-Question Agreement Distribution First Round (Questions 1–23)

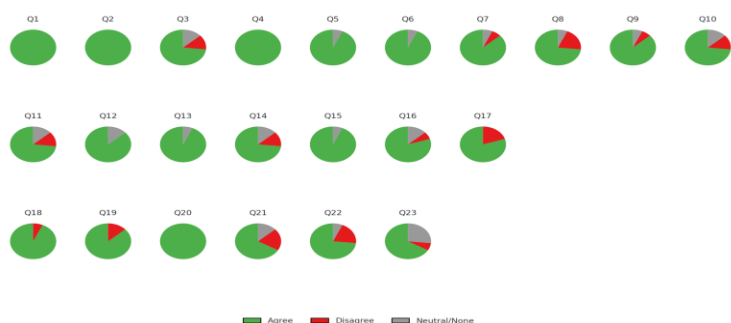


Consolidated bar chart showing the agreement distribution for each question (Q1–Q23). For each question, three adjacent bars represent the counts of Agree (green), Disagree (red),

and Neutral/None (gray) responses from the 15 panelists. Questions are arranged along the x-axis (Q1 through Q23), and we have inserted dashed vertical lines to separate the three question groupings (Q1–10, Q11–17, Q18–23) for clarity. This chart reveals that most questions had a predominant majority of “Agree” responses (green bars reaching the maximum height of 15), with very few red or gray bars visible indicating a few disagreements or neutral opinions from the panelists.

Figure 4:

Panel of Pie Charts Agreement Distribution Per Question Round 1(Q1–Q23)



Panel of pie charts showing the agreement distribution per question (Q1–Q23). Each small pie chart corresponds to one question (labeled Q1, Q2, ..., Q23) and is divided into green, red, and gray segments representing the proportion of Agree, Disagree, and Neutral/None responses for that question. The pies are arranged in three rows: the top row contains Q1–Q10, the middle row Q11–Q17, and the bottom row Q18–Q23. This panel provides a quick visual scan of consensus on each item. Green segments dominate most pies, especially in Q1–Q10 (top row) where many pies are nearly full green circles (indicating unanimous or near-unanimous agreement). The middle-row pies (Q11–Q17) also show large green portions with only small slivers of red or gray, meaning these statements also enjoyed strong agreement with minimal dissent. In the bottom row (Q18–Q23), some pies have more noticeable red and gray slices — for example, Q21, Q22, and Q23 pies include visible red (disagreement) segments and gray segments, signifying that these later questions saw a few panelists disagreeing or remaining

neutral. Overall, this pie chart panel visually reinforces that agreement was high for every question, with only a handful of questions (mostly in the last group) eliciting any substantial disagreement or neutrality from the panelists.

Figure 5:

First Round Agreement Distribution Questions 1-10

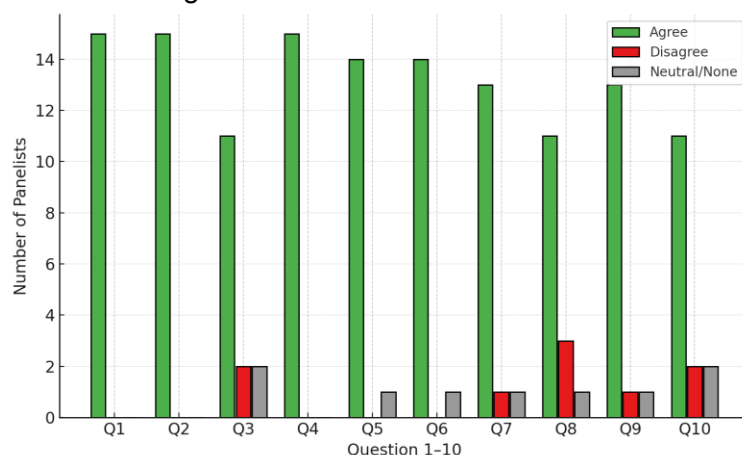


Figure: Bar chart of agreement distribution for Questions 1–10. Each question (Q1 through Q10 on the x-axis) has up to three bars showing counts of Agree (green), Disagree (red), and Neutral/None (gray) responses. We see that for the majority of these foundational questions, all 15 panelists chose “Agree”, the green bars for Q1, Q2, Q4 reach the full height (15), with no red or gray bars at all. For a number of questions most panelists still agreed (green bar), but a couple of panelists disagreed or were neutral (small red and gray bars are visible). Overall, the questions Q1–Q10 exhibits overwhelming consensus, as evidenced by the dominance of green bars, indicating that nearly everyone agreed on these initial statements.

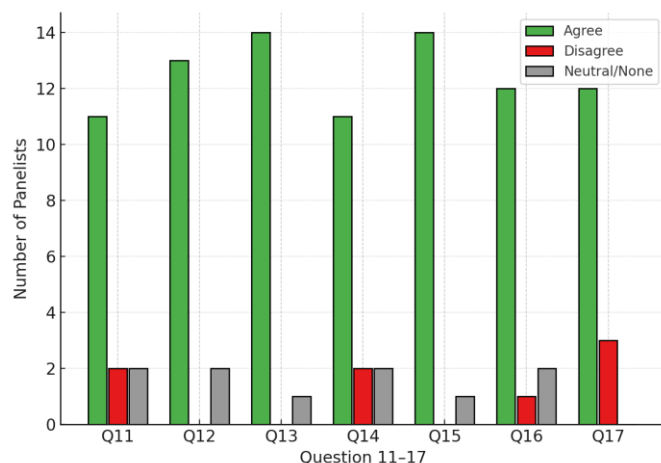
Figure 6:**Agreement Distribution Questions 11-17 Round 1**

Figure: Bar chart of agreement distribution for Questions 11–17. These questions continue to show strong agreement trends. Green bars (Agree counts) are very high for each question Q11 through Q17, often reaching 13–15 panelists in agreement. A few questions show minor dissent: for instance, Q11 and Q14 each have a small red bar (indicating a couple of panelists Disagree), and some couple questions have a small gray bar for Neutral/None (e.g., one panelist chose neither agree nor disagree on Q12 and Q15). Nonetheless, agree responses still overwhelmingly predominate in this set. The pattern suggests that even for these questions, the expert panel was largely in consensus, with only one or two panelists occasionally voicing uncertainty or disagreement on certain points.

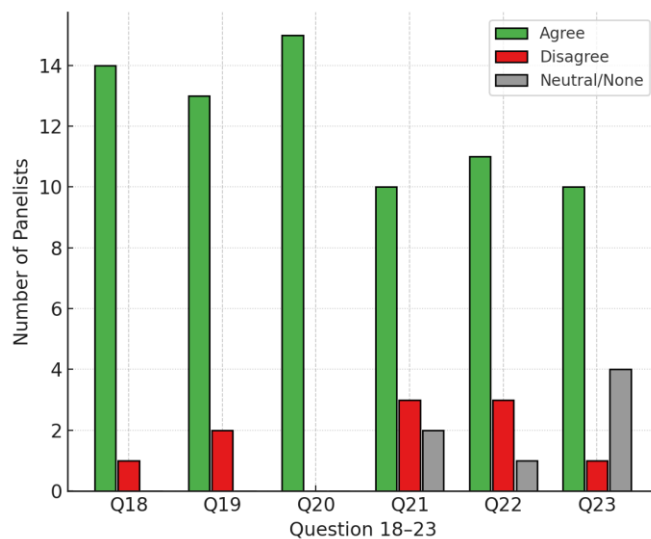
Figure 7:**Agreement Distribution Questions 18-23 Round 1**

Figure: Bar chart of agreement distribution for Questions 18–23. For these questions as well “Agree” (green) remains the majority for each. We observe that questions like Q18, Q19, and Q20 have nearly unanimous agreement (tall green bars). However, in Q21, Q22, and Q23 the green bars, while still dominant, are a bit shorter compared to earlier questions (indicating not all 15 panelists agreed). In particular, Q21 (“promote prioritizing semantic DQ dimensions”) and Q22 (“measuring accuracy of meaning requires ADS/SOR comparison”) each have a few red bars (disagreements) and gray bars (neutral responses), reflecting a couple of panelists who did not agree with those statements. Q23 similarly shows a notable gray segment (a couple of panelists indicated uncertainty). This chart highlights that although consensus remained high even in Q18–23, these questions sparked the most divergence of opinion: they are more contentious or nuanced, leading to a small but visible minority of the panel expressing disagreement or no opinion.

4.3.7 Summary Feedback Questions 24-30

The first-round responses reveal several recurring themes in the open-ended questions

24-30:

1. Accuracy and relevance emerge as the dominant semantic DQ dimensions, with completeness, consistency and trustworthiness also viewed as critical. Panelists repeatedly stressed that data must be both factually correct and contextually appropriate for GenAI/LLM use cases.

2. There is a lack of widely accepted measures for semantic quality. Participants called for metrics based on ontologies, authoritative sources, contextual consistency, and trustworthiness, yet noted that current frameworks provide few concrete measures.

3. Automation is seen as necessary but insufficient. Many panelists advocated automating validation and auditing processes, while also acknowledging the need for human review, surveys, and cross-functional collaboration to interpret meaning and business value.

4. Verifying outputs without ground truth and detecting bias are major challenges. The absence of definitive “correct” answers makes it difficult to evaluate GenAI outputs, and biases embedded in training data are hard to detect and mitigate.

5. Traditional DQM frameworks need to evolve, not be replaced. The consensus is to extend existing frameworks to include semantic and output-focused processes, broaden dimensions to model-ready factors (bias, reproducibility), and adopt fine-grained, domain-specific rules.

6. Education and organizational capability are key. Many responses highlighted the need to educate stakeholders, build data stewardship skills, and incentivize adoption of semantic DQ practices.

4.4 Second Round Delphi Study Data Collection and Analysis

4.4.1 Second Round Delphi Study Instructions

- This Second Round of the Delphi study focuses primarily on ranking the semantic data quality dimensions for the open-ended questions (Questions 24–30), because there was a broad consensus on questions 1-23 (see summarized analysis for each question).
- For this Second Round, I am requesting that you rank the responses for questions 24–30 from Round One using the ranking scale and tables for each question below. The area where I am requesting your input is highlighted in **Red Font**.
- OPTIONAL - If you want to provide additional comments for Questions 1–23, please enter your responses in the area highlighted in **Green Font**.
- Note: In the First Round we had a total of fifteen anonymous experts who participated in the study.

4.4.2 Second Round Questions 24 - 30

Second Round Question 24

“Please rate in the table below the importance of the following semantic data quality dimensions for GenAI and LLM applications.” On a 5-point scale (1 = Not important, 5 = Extremely important).

Semantic DQ Dimensions	Importance Rate (1-5)
Accuracy	
Relevance	
Trustworthiness/Believability	
Completeness	
Consistency	
Timeliness	

4.4.3 Semantic Data Quality Dimensions Analysis

Table 11

Semantic Data Quality Dimensions Ratings

R2 Panelists	Accuracy	Relevance	Trustworthiness/Believability	Completeness	Consistency	Timeliness
#1	5	3	1	5	3	1
#2	5	1	4	4	5	3
#3	5	5	4	3	3	2
#4	4	5	5	4	3	3
#5	5	4	5	3	4	3
#6	5	5	5	5	4	4
#7	5	5	4	4	4	3
#8	5	5	4	4	5	5
#9	5	5	5	4	4	3
#10	5	5	5	5	5	4
#11	5	5	4	4	4	3
#12	5	5	5	3	3	3
#13	4	4	4	2	2	3
#14	5	5	5	5	5	5

This table shows second round experts' ratings for question 24

Background and Methodology

A second-round Delphi survey was conducted with 14 domain experts to assess the importance of six semantic data quality (DQ) dimensions for generative AI (GenAI) and large language model (LLM) applications. Participants rated each dimension on a 5-point Likert scale (1 = Not Important, 5 = Extremely Important). Measures of central tendency (mean, median, mode) and dispersion (standard deviation, inter-quartile range, min–max) were computed. For ordinal Likert data, medians and IQRs provide a more robust measure of consensus than means.

Descriptive statistics are summarized in the table below:

Table 12

Descriptive Statistics Semantic DQ Dimensions

Dimension	Mean	Median	Mode	Std Dev	IQR	Min–Max
Accuracy	4.86	5	5	0.36	0	4–5
Relevance	4.43	5	5	1.16	0.75	1–5
Trustworthiness/Believability	4.29	4.5	5	1.07	1	1–5
Completeness	3.93	4	4	0.92	1.50	2–5
Consistency	3.86	4	4	0.95	1.75	2–5
Timeliness	3.21	3	3	1.05	0.75	1–5

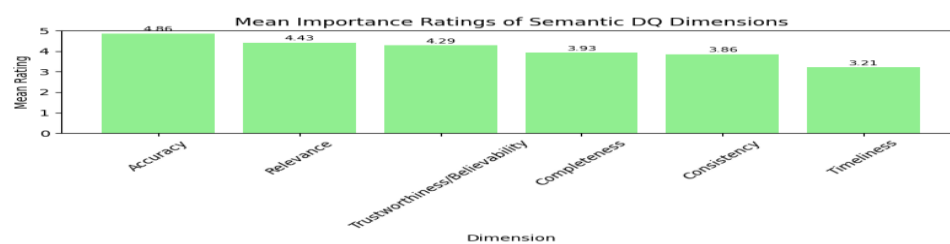
Key Insights

- Accuracy received the highest importance rating with near-perfect consensus (IQR = 0).
- Relevance and Trustworthiness were rated highly but showed greater variability.
- Completeness and Consistency were of moderate importance and exhibited wider IQRs.
- Timeliness was the lowest rated dimension and displayed the greatest variability.

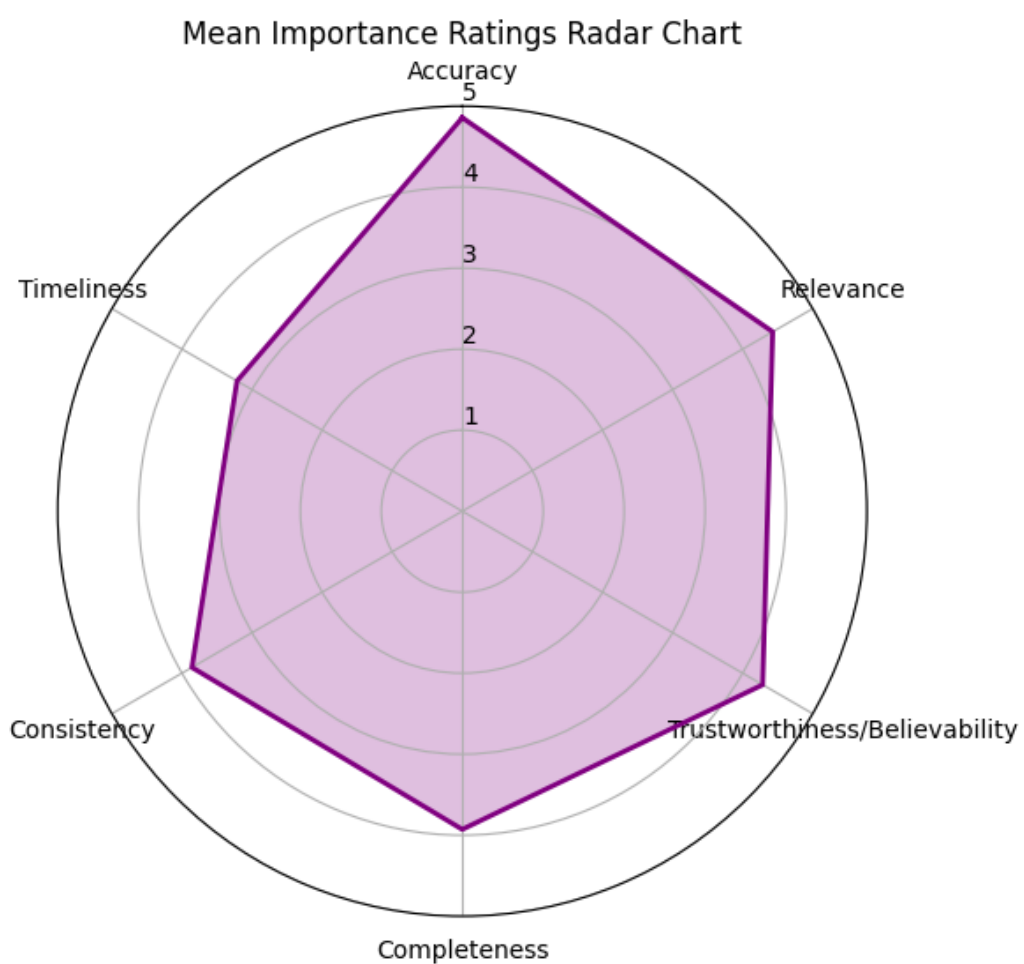
Figure summarizes the mean importance scores, while Figure 2 visualizes the relative importance across dimensions.

Figure 8

Mean Importance Ratings of Semantic DQ Dimensions

**Figure 9**

Mean Importance Ratings Radar Chart



Discussion and Implications

Accuracy as the bedrock: Experts overwhelmingly agree that ensuring accurate data and verifying sources is foundational for GenAI/LLM systems.

Relevance & Trustworthiness: High ratings indicate that context and credibility are vital. Governance mechanisms such as semantic layers help enforce consistent definitions across dashboards and AI agents.

Completeness & Consistency: Moderate importance suggests that missing or inconsistent metadata can hamper model performance, but improvements in processes and tooling are achievable.

Timeliness: The lowest rating reflects challenges in keeping training data up-to-date; real-time checks and continuous monitoring can mitigate this.

Human & Culture: Training programmes and human-in-the-loop approaches are crucial for cultivating a data quality culture.

Conclusions & Recommendations

- **Prioritise data accuracy:** Implement rigorous validation against authoritative sources and enforce mandatory fields to minimise errors.
- **Strengthen relevance & trustworthiness:** Adopt semantic layers and ontology-driven models to standardise definitions and maintain credible, context-rich data.
- **Enhance completeness & consistency:** Automate quality checks to detect missing or inconsistent values and engage domain experts to resolve issues.
- **Improve timeliness:** Introduce real-time monitoring, regular refresh cycles and feedback loops to keep training data current.
- **Build a data quality culture:** Invest in training, education and human-in-the-loop processes to foster accountability and continuous improvement.

Second Round - Question 25:

“Considering the importance of semantic quality, please rate the effectiveness of the following measures for assessing accuracy and relevance.” Use a 5-point effectiveness scale, 1 = Less effective, 5 = Very effective.

Measure for Assessing Accuracy and relevance	Effectiveness Rate (1-5)
(a) coverage and resolvability via shared dictionaries/ontologies	

(b) cross-source validation and triangulation	
(c) automated contextual consistency checks	
(d) human evaluation of value add	
(e) fairness and trust audits	

4.4.4 Assessing Measures for Accuracy & Relevance

Table 13

Measure for Assessing Accuracy and relevance Ratings

R2 Panelists	(a) coverage	(b) cross-source	(c) automated	(d) human	(e) fairness
#1	5	3	3	3	1
#2	4	5	4	3	5
#3	4	5	4	5	3
#4	5	4	3	4	4
#5	2	5	3	4	5
#6	5	2	2	2	2
#7	3	4	3	4	3
#8	4	5	5	4	5
#9	5	4	5	2	3
#10	4	5	5	5	5
#11	5	5	4	5	5
#12	1	3	2	4	1
#13	5	3	5	4	5
#14	2	1	2	3	4

This table shows second round experts' ratings for question 25

Introduction

In the second round of our Delphi study, 14 domain experts assessed the effectiveness of five measures for evaluating accuracy and relevance in generative AI and large language model (LLM) applications. The measures assessed were: coverage and resolvability via shared

dictionaries/ontologies, cross-source validation and triangulation, automated contextual consistency checks, human evaluation of value add, and fairness and trust audits.

Methodology

Experts rated each measure on a 1–5 Likert scale. Following recommended Delphi methodology, we computed descriptive statistics—mean, median, mode—and measures of dispersion such as standard deviation and interquartile range to summarise collective judgements. Central tendency alone provides an incomplete picture; variability must be reported alongside to understand consensus and divergence among experts. Table below summarizes these statistics.

Table 14

Descriptive Statistics Assessing Measures for Accuracy & Relevance

Measure	Mean	Median	Mode	Std Dev	IQR
Coverage & resolvability	3.86	4	5	1.35	1.75
Cross-source validation & triangulation	3.86	4	5	1.29	2.00
Automated contextual consistency checks	3.57	3.5	3 & 5	1.16	1.75
Human evaluation of value add	3.71	4	4	0.99	1.00
Fairness & trust audits	3.64	4	5	1.50	2.00

Findings

Coverage & resolvability and cross-source validation both received the highest mean rating

(3.86), indicating strong consensus. Human evaluation of value add scored 3.71 with the lowest standard deviation (0.99), reflecting consistent support. Fairness & trust audits (mean = 3.64) exhibited greater variability (IQR = 2.00), suggesting divergent perspectives. Automated contextual checks recorded the lowest mean (3.57) and showed mixed responses.

Figure 10

Average Effectiveness Rating by Accuracy/Relevance Measure

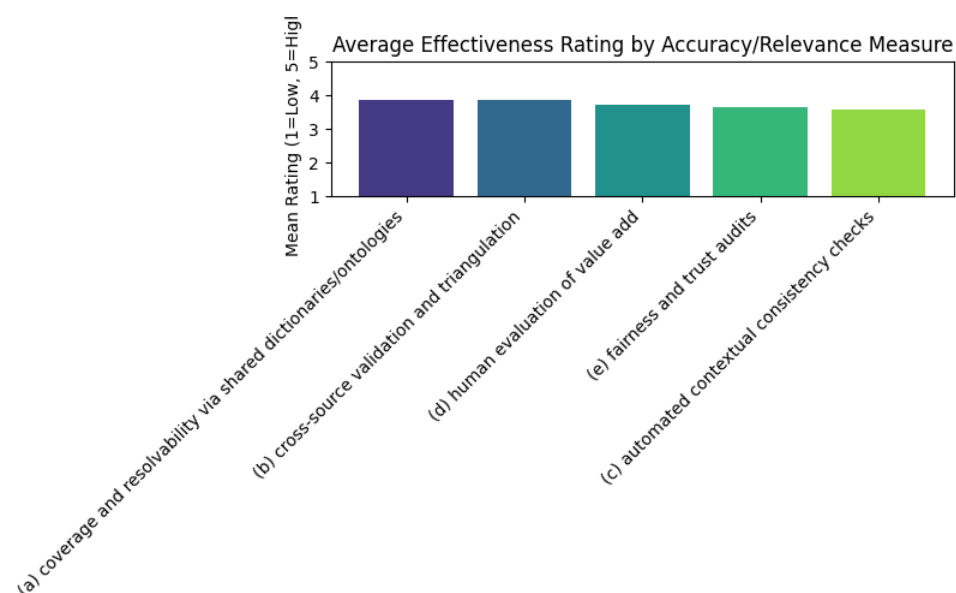
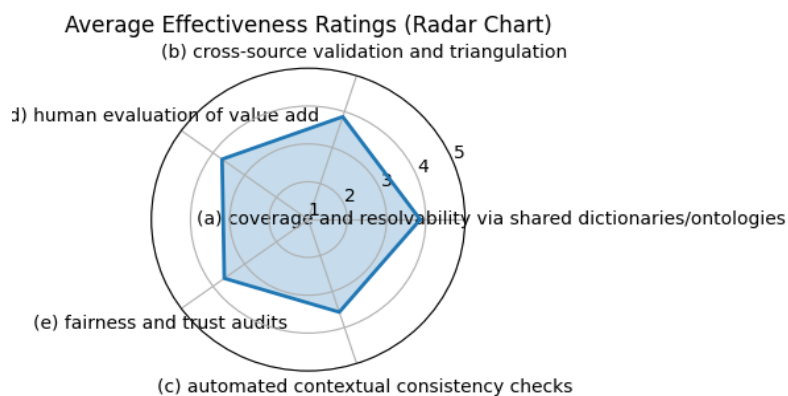


Figure 11

Average Effectiveness Ratings (Radar Chart)



Radar chart of mean ratings highlighting overall importance ranking.

Discussion & Recommendations

The results highlight strong agreement on the importance of coverage & resolvability and cross-source validation for evaluating accuracy and relevance. Human evaluation remains a critical complement to automated methods. Fairness and trust audits were rated lower and showed greater variability, suggesting the need for clearer communication about their value and integration earlier in evaluation processes. We recommend combining human and automated checks, prioritising high-consensus measures in quality frameworks, and consistently reporting measures of dispersion in future Delphi rounds.

Second Round - Question 26:

“Which approaches are most efficient for gathering semantic DQ measures?” Please indicate your efficiency rate (1–5) in the below table, (1 = Not Efficient, 5 = Most Efficient).

Approaches for Gathering Semantic DQ Measures	Efficiency Rate (1-5)
(a) automated validation using authoritative sources and ontologies	
(b) automated audits and dashboards	
(c) surveys and qualitative feedback	
(d) human review and cross-functional collaboration	
(e) hybrid AI/human approaches	

4.4.5 Efficient Approaches for Gathering Semantic Data Quality Measures

Table 15

Approaches for Gathering Semantic DQ Measures Ratings

R2 Panelists	(a) automated validation	(b) automated audits	(c) surveys	(d) human review	(e) hybrid AI/human
#1	5	1	1	3	3
#2	5	4	1	3	5
#3	5	4	2	3	5
#4	4	3	4	5	5
#5	5	4	3	5	2
#6	5	2	3	3	4
#7	5	4	3	3	4
#8	5	5	4	4	4
#9	5	5	3	4	5
#10	5	5	4	4	4
#11	5	5	2	1	1
#12	5	4	2	3	5
#13	2	2	5	4	5

This table shows second round experts' ratings for question 26

This section presents the results of the second round of the Delphi study focused on identifying the most effective approaches for gathering semantic data quality (DQ) measures. Thirteen experts rated each approach using a 1–5 Likert scale. The goal of this analysis is to summarise their ratings using measures of central tendency and dispersion, illustrate the distribution of responses, and extract key insights for inclusion in the dissertation.

Methodology

Participants: Fourteen experts participated in the second round of the Delphi study, however, for this question only thirteen experts provided their ratings as one participant declared to not be qualified enough to answer this question. Each participant rated five predefined approaches for gathering semantic DQ measures on a scale from 1 (least efficient) to 5 (most efficient).

Analysis: For each approach, we computed the mean, median and mode as measures of central tendency, and the standard deviation and inter-quartile range (IQR) to describe

dispersion. Reporting both central tendency and variability follows the Delphi literature's recommendation to show consensus and spread among experts. Measures of central tendency alone are insufficient; variability indicates how far ratings deviate from the center.

Table 16

Descriptive Statistics Approaches for Gathering Semantic Data Quality Measures

Approach	Mean	Median	Mode	Std. Dev.	IQR
Automated validation (a)	4.69	5.00	5.00	0.85	0.00
Automated audits/dash boards (b)	3.69	4.00	4.00	1.32	2.00
Surveys & feedback (c)	2.85	3.00	3.00	1.21	2.00
Human review & collaboration (d)	3.46	3.00	3.00	1.05	1.00
Hybrid AI/human (e)	4.00	4.00	5.00	1.29	1.00

The highest mean rating (4.69) and lowest variability (IQR = 0) were observed for automated validation using authoritative sources and ontologies, indicating strong consensus on its effectiveness. Hybrid AI/human approaches also scored highly (mean = 4.00) but showed more variability, reflecting different opinions on the optimal balance between automation and human insight. Automated audits and dashboards (mean = 3.69) and human review & cross-functional collaboration (mean = 3.46) received moderate ratings with wider dispersion, suggesting mixed views about their efficiency and consistency. Surveys and qualitative feedback received the lowest mean (2.85) and broadest spread, implying skepticism about the

effectiveness of purely qualitative methods.

Distribution of Ratings

Bar and pie charts illustrate how experts distributed their ratings for each approach. Automated validation and hybrid AI/human methods exhibit concentrated distributions at the higher end of the scale, whereas surveys and qualitative feedback show a more even spread across ratings. Human review & cross-functional collaboration and automated audits/dashboards occupy middle positions, with ratings spanning the full range.

Key Findings

Automated validation (authoritative sources and ontologies) emerged as the most efficient approach, with experts nearly unanimously rating it 5. This aligns with research showing that automated numeric checks and alerts improve data validity and reliability. Hybrid AI/human approaches received strong ratings, underscoring the value of combining machine efficiency with human judgement. Human–AI collaboration, or hybrid intelligence, leverages complementary strengths: AI excels at data analysis while humans excel at contextual reasoning.

Cross-functional collaboration (human review) was valued but rated lower than automated methods, highlighting the need to integrate diverse stakeholders for effective data governance. Surveys and qualitative feedback were least favored, indicating that experts see them as supplementary rather than primary measures.

Automated audits and dashboards yielded mixed opinions, suggesting that metrics and transparency should be carefully designed to build trust.

Recommendations

1. Prioritise automated validation using trusted ontologies and authoritative sources to ensure high semantic accuracy.
2. Combine human expertise with AI systems to harness complementary strengths and reduce automation bias.

3. Facilitate cross-functional collaboration among legal, IT and governance teams to develop coherent data policies and retention strategies.
4. Use automated audits and dashboards with clear, interpretable metrics to monitor data quality and foster trust.
5. Employ surveys and qualitative feedback as supplementary tools to capture user sentiment and contextual nuances.

Second Round - Question 27:

“For each of the following challenges in the table below, please rate its significance for ensuring data quality in GenAI/LLMs.” Use a 5-point scale rate, (1 = Not significant, 5 = Extremely significant).

Challenges for ensuring DQ in GenAI/LLMs	Challenge Significance Rate (1-5)
(a) verifying accuracy and relevance of training data and outputs	
(b) absence of ground truth and traceability	
(c) detecting and mitigating bias and fairness issues	
(d) managing unstructured and semi-structured data	
(e) rapid pace of technology change	
(f) organizational capacity and skills	

4.4.6 Challenges for Ensuring Data Quality in GenAI/LLMs

Table 17

Challenges for Ensuring Data Quality in GenAI/LLMs Ratings

R2 Panelists	(a) verifying accuracy	(b) absence of ground truth	(c) detecting and mitigating	(d) managing unstructured	(e) rapid pace of change	(f) organizational
#1	5	5	5	3	3	2
#2	5	4	5	2	4	5
#3	5	5	5	4	4	4
#4	5	4	4	3	3	5
#5	4	5	3	1	3	4
#6	5	5	4	5	5	5
#7	2	3	2	4	3	2
#8	5	4	4	4	5	5
#9	5	5	4	5	3	3
#10	5	5	5	5	5	5
#11	5	5	5	4	4	4
#12	5	4	4	5	2	5
#13	5	4	5	2	5	5
#14	4	3	4	4	5	5

This table shows second round experts' ratings for question 27

Introduction

A panel of fourteen experts participated in the second round of our Delphi study. They rated six key challenges related to ensuring data quality (DQ) in generative AI (GenAI) and large language model (LLM) applications on a 1–5 significance scale. The purpose of this analysis is to summarise the distribution of expert judgments, highlight areas of consensus and divergence, and provide recommendations for practice.

Methodology

The dataset contained complete responses for all experts and challenges; no imputation was required. Descriptive statistics were calculated for each challenge: mean, median, mode (to summarise central tendency), standard deviation, inter-quartile range (IQR), minimum and maximum (to summarise variability). Following Delphi guidelines, both measures of central tendency and dispersion were reported because central tendency alone provides an incomplete picture.

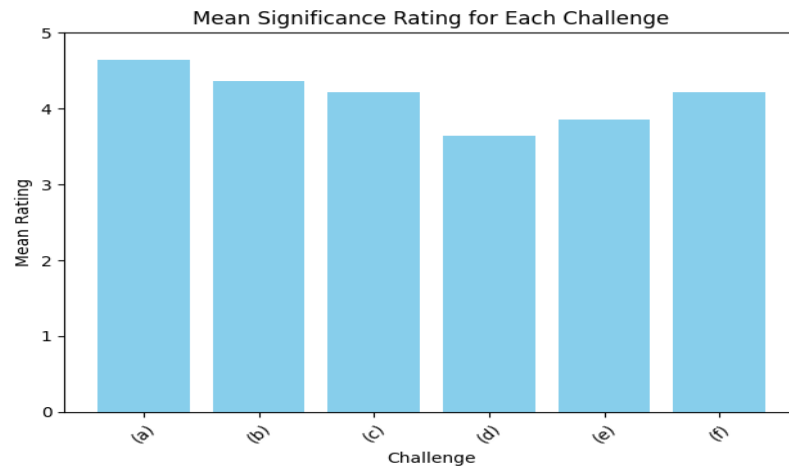
Table 18

Descriptive Statistics Challenges for Ensuring Data Quality in GenAI/LLMs

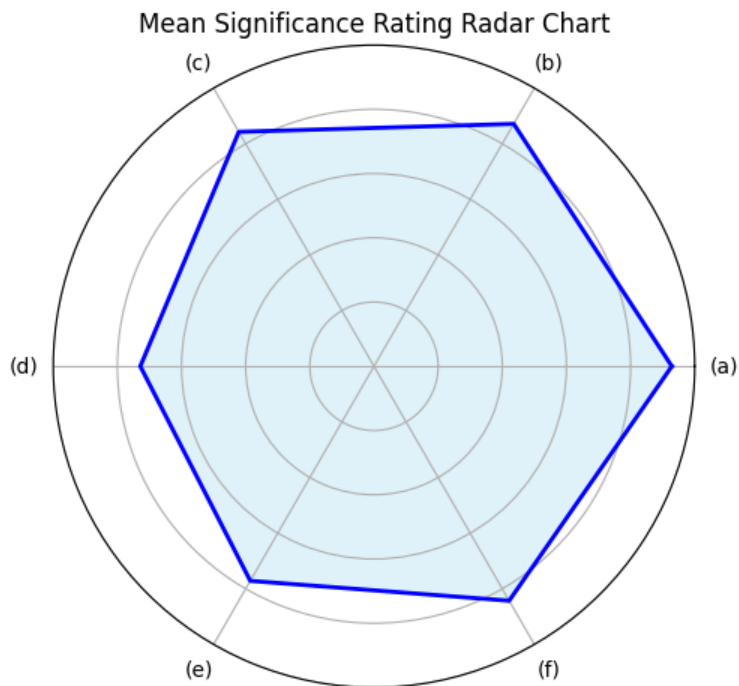
Challenge	Mean	Median	Mode	Std Dev	IQR	Min	Max
(a) verifying accuracy and relevance of training data and outputs	4.64	5.0	5.0	0.81	0.0	2.0	5.0
(b) absence of ground truth and traceability	4.36	4.5	5.0	0.72	1.0	3.0	5.0
(c) detecting and mitigating bias and fairness issues	4.21	4.0	4.0	0.86	1.0	2.0	5.0
(d) managing unstructured and semi-structured data	3.64	4.0	4.0	1.23	1.75	1.0	5.0
(e) rapid pace of technology change	3.86	4.0	3.0	0.99	2.0	2.0	5.0
(f) organizational capacity and skills	4.21	5.0	5.0	1.08	1.0	2.0	5.0

Figure 12

Mean significance ratings across challenges

**Figure 13**

Radar chart of mean significance ratings



Discussion

Highest mean ratings were recorded for verifying accuracy & relevance and organisational

capacity, indicating these challenges are considered most critical. The absence of ground truth and bias & fairness issues also ranked highly. Lower mean ratings and greater variability for managing unstructured data and rapid technology change suggest more divergent views among experts. Several studies show that poor training data quality can significantly reduce model precision, and noise or inconsistencies introduced during fine-tuning degrade performance. Challenges related to data provenance and traceability reflect the high cost and difficulty of creating ground-truth datasets; however, such datasets build confidence in model performance. Bias and fairness concerns echo the widespread recognition that AI systems can inherit systemic biases; selection bias and automation bias arise when training data are unrepresentative, highlighting the need for high-quality, diverse sources. Managing unstructured and semi-structured data poses unique data quality challenges. Unstructured data collected for non-research purposes often lack systematic collection and contain missing or inconsistent values, making them difficult to use without extensive curation. Organisations frequently face infrastructure limitations and skills shortages when linking these data sources, and scaling storage and processing can raise ethical and privacy concerns. The rapid pace of technological change and streaming data surges create opportunities for mistakes and inaccuracies. Exponentially improving AI capabilities demand agile governance frameworks and continuous policy updates. Finally, organisational capacity and skills limitations remain a barrier to AI adoption, with many businesses citing lack of skills as a key obstacle.

Conclusions & Recommendations

Experts prioritised rigorous data curation and cross-source validation to ensure model accuracy. Institutions should invest in ground-truth frameworks and consider leveraging expert review and LLM-as-judge methods when labels are unavailable. Diversifying training data and conducting fairness audits help mitigate bias. To handle unstructured data, organisations should establish metadata standards, invest in data cleaning and enrichment tools, and build scalable infrastructures. Agile data governance frameworks are needed to keep pace with technological change, and programmes to enhance organisational skills and cross-functional collaboration are essential for responsible AI initiatives.

Second Round - Question 28:

“Please rate the priority for enhancing the following DQM components to support GenAI and LLM adoption.” Use a 5-point priority scale in the table below, (1 = Less Priority, 5 = High Priority).

DQM Components Needing Enhancement to support GenAI/LLMs adoption	Enhancement Priority Rate (1-5)
(a) output validation and real-time monitoring	
(b) automated tools for semantic similarity and deduplication	

(c) education and organizational capability building	
(d) speed and agility of DQM processes	
(e) governance, metadata modeling and stewardship	

4.4.7 Enhancement Priorities for DQM to Support GenAI/LLM Adoption

Table 19

DQM Components Needing Enhancement to support GenAI/LLMs adoption Ratings

R2 Panelists	(a) output validation	(b) automated tools	(c) education	(d) speed	(e) governance,
#1	5	5	4	4	5
#2	2	4	2	4	5
#3	5	4	4	4	5
#4	5	3	5	4	4
#5	5	3	4	5	5
#6	5	3	4	5	5
#7	3	4	2	4	2
#8	5	4	4	5	5
#9	5	4	3	1	2
#10	5	5	5	5	5
#11	5	4	4	5	5
#12	4	2	5	3	1
#13	5	5	5	5	5
#14	1	2	4	3	5

This table shows second round experts' ratings for question 28

Overview

This document summarises findings from the second round of a Delphi study where fourteen experts rated the priority of enhancements needed for various Data Quality Management (DQM) components to support adoption of Generative AI and Large Language Models (GenAI/LLMs). The components assessed include output validation and real-time monitoring, automated tools for semantic similarity and deduplication, education and organisational capability building, speed

and agility of DQM processes, and governance, metadata modelling and stewardship. The goal is to identify which enhancements experts view as most critical for safe and effective GenAI/LLM adoption.

Methodology

The Delphi method uses structured rounds of surveys to elicit expert judgement and build consensus. In this second round, fourteen experts rated each DQM component on a 5-point scale (1 = low priority, 5 = high priority). Basic statistics were computed for each component, including mean, median, mode, standard deviation (variability), inter-quartile range (IQR), and minimum/maximum values. Bar and pie charts were generated to visualise the distribution of ratings, and summary charts (mean bar and radar) were used to compare priorities across components. Reporting both measures of central tendency and dispersion provides a complete picture of expert agreement and variability.

Table 20

Descriptive Statistics Enhancement Priorities for DQM to Support GenAI/LLM Adoption

Comp onent	Count	Mean	Media n	Mode	Std De v	IQR	Min	Max
(a) output validat ion and real-ti me monito ring	14	4.29	5.00	5.00	1.33	0.75	1	5
(b) autom ated tools for seman tic similar ity and dedupl ication	14	3.71	4.00	4.00	0.99	1.00	2	5

(c) education and organizational capability building	14	3.93	4.00	4.00	1.00	0.75	2	5
(d) speed and agility of DQM processes	14	4.07	4.00	5.00	1.14	1.00	1	5
(e) governance, metadata modeling and stewardship	14	4.21	5.00	5.00	1.42	0.75	1	5

Summary Charts

The bar and radar charts below summarise the mean priority ratings across components, highlighting their relative importance.

Figure 14

Mean Significance Ratings across DQM Components

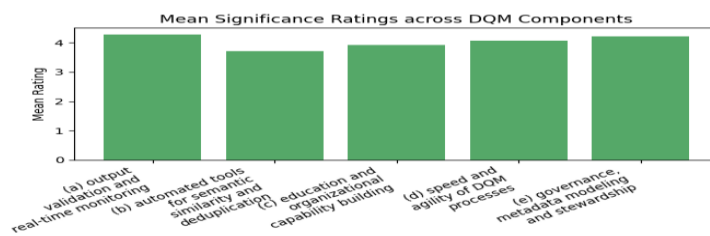
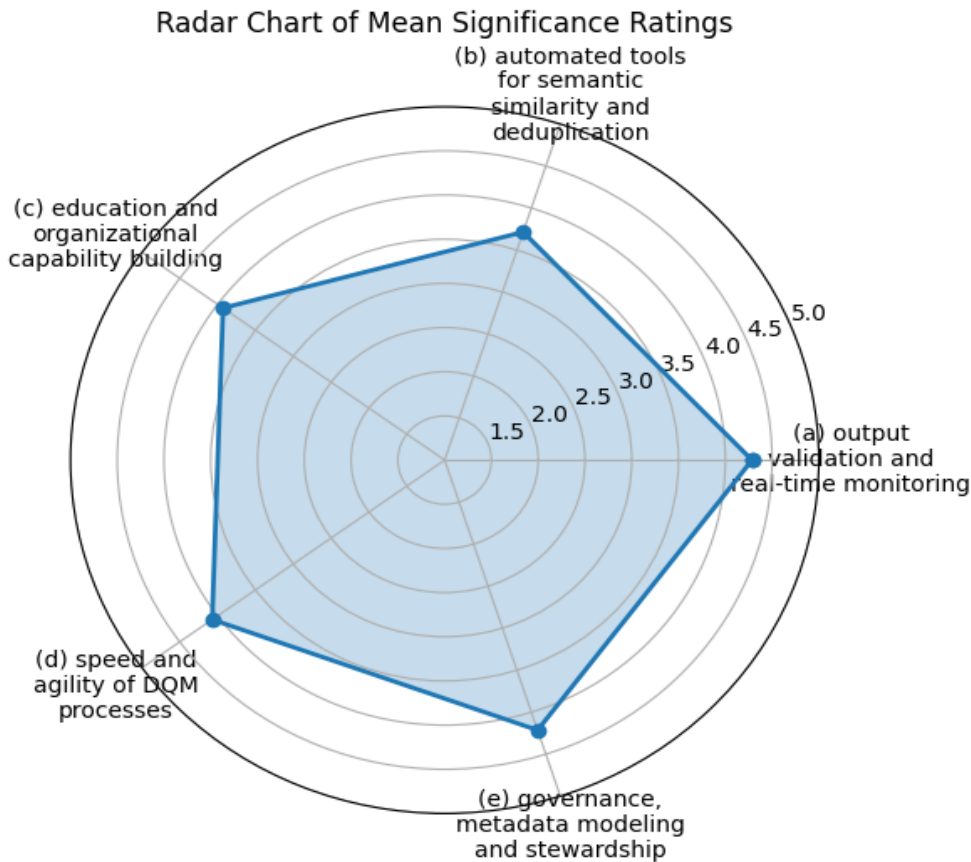


Figure 15

Radar Chart of Mean Significance Ratings



Key Findings and Discussion

- **Output validation & real-time monitoring** received the highest mean rating (4.29) and a low IQR, indicating strong consensus that continuous validation is essential. Experts emphasised the need to standardise data, embed contextual metadata, maintain semantic consistency and validate inputs in real time.

- **Automated semantic deduplication tools** had the lowest mean (3.71), but experts still recognised the value of automating the removal of duplicate or similar records. Manual coding (SQL or Python) is time-consuming and cannot handle deduplication at scale, whereas no-code tools perform deduplication in seconds with ten-times higher accuracy and improve

operational efficiency by about 60%.

- **Education and organisational capability building** earned a mean of 3.93. Experts noted that training employees on data governance best practices and providing ongoing education about new technologies are vital for sustaining data quality and reliability.

- **Speed and agility of DQM processes** scored a mean of 4.07. Flexible and agile DQM practices require assessing existing data, processes, and tools against accuracy, completeness, consistency and timeliness measures. Maintaining agility in data quality frameworks enables organisations to respond to volatile markets and ensures their digital businesses remain viable.

- **Governance, metadata modelling and stewardship** achieved a mean of 4.21, reflecting the high priority experts place on establishing robust metadata management and stewardship practices. Metadata provides context and meaning to data, clarifies definitions, documents lineage, and captures consent, forming the foundation for effective AI governance and enabling transparency and compliance.

- **Skills gaps and organisational capacity** were noted as underlying factors influencing the adoption of data quality practices. Over half of companies report inadequate AI skills and many struggle to find skilled employees, underscoring the need for upskilling and training programmes.

Recommendations

- Implement continuous monitoring and real-time validation processes to detect data drift, anomalies and context mismatches before they compromise GenAI outputs.
- Adopt automated, no-code deduplication tools to efficiently resolve semantic duplicates and free technical staff to focus on higher-value tasks.
- Invest in data quality training programmes and build organisational capabilities to ensure staff understand governance practices and evolving technologies.
- Streamline data pipelines and promote agile DQM practices to accelerate response times and improve accuracy, completeness, consistency and timeliness.
- Establish robust metadata management and stewardship frameworks to provide

context, traceability and ownership of data throughout its lifecycle.

Conclusion

The second round of the Delphi study suggests that enhancing output validation, governance and real-time monitoring should be prioritised to ensure the reliability of GenAI/LLM systems. Automating deduplication, investing in education, and maintaining agility across DQM processes will further strengthen organisational readiness for AI adoption. Robust metadata management and stewardship are essential to provide context, transparency and compliance in AI-driven environments.

Second Round - Question 29:

“To improve traditional DQM frameworks for GenAI and LLMs, please indicate your agreement with these proposals in the below table.” Use a 5-point agreement scale to indicate your level of agreement (1 = Disagree, 5 = Strongly agree).

Proposals to improve traditional DQM frameworks for GenAI/LLMs	Agreement Level (1-5)
(a) extend frameworks to include continuous output monitoring and semantic checks	
(b) integrate AI-powered tools and human-in-the-loop validation	
(c) adapt frameworks flexibly to the context and mission rather than treat them as rigid	
(d) broaden DQ dimensions to include bias, representation and reproducibility	
(e) develop fine-grained, domain-specific rules and ontologies	

4.4.8 Analysis of Proposals to Improve DQM Frameworks for GenAI/LLMs

Table 21

Proposals to improve traditional DQM frameworks for GenAI/LLMs Ratings

R2 Panelists	(a) extend frameworks	(b) integrate AI-powered tools	(c) adapt frameworks	(d) broaden DQ dimensions	(e) develop fine-grained,
#1	5	4	3	2	2
#2	3	5	4	5	2
#3	5	5	5	5	5
#4	3	5	4	4	3
#5	3	4	4	3	3
#6	5	4	5	5	3
#7	3	4	3	3	3
#8	5	5	4	4	5
#9	5	5	4	5	5
#10	5	5	5	5	5
#11	5	5	5	5	5
#12	5	3	4	4	2
#13	5	5	3	5	5
#14	2	4	4	5	5

Table shows second round experts' ratings for question 29

Introduction

In the second round of a Delphi study, 14 experts evaluated a set of proposals designed to enhance traditional data quality management (DQM) frameworks to better support generative artificial intelligence (GenAI) and large language models (LLMs). Each expert rated the importance of five improvement proposals using a 5-point Likert scale. This report summarises the descriptive statistics, visualisations and key findings from these expert ratings.

Methodology

The dataset comprises 14 responses for each of five proposals: (a) continuous output monitoring and semantic checks, (b) integration of AI-powered tools with human-in-the-loop validation, (c) flexible and context-specific frameworks, (d) expansion of data quality dimensions to encompass bias, representation and reproducibility, and (e) development of fine-grained, domain-specific rules and ontologies. Descriptive statistics were computed for each proposal, including the mean, median, mode, standard deviation and inter-quartile range (IQR). These measures summarise central tendency and variability, which are essential for understanding

consensus and dispersion in Delphi studies.

Table 22

Descriptive Statistics Proposals to Improve DQM Frameworks for GenAI/LLMs

Proposal	N	Mean	Median	Mode	Std Dev	IQR	Min	Max
(a) extend frameworks to include continuous output monitoring and semantic checks	14	4.21	5.00	5.0	1.12	2.00	2.0	5.0
(b) integrate AI-powered tools and human-in-the-loop validation	14	4.50	5.00	5.0	0.65	1.00	3.0	5.0
(c) adapt frameworks flexibly to the context and mission rather than	14	4.07	4.00	4.0	0.73	0.75	3.0	5.0

treat them as rigid								
(d) broaden DQ dimensions to include bias, representation and reproducibility	14	4.29	5.00	5.0	0.99	1.00	2.0	5.0
(e) develop fine-grained, domain-specific rules and ontologies	14	3.79	4.00	5.0	1.31	2.00	2.0	5.0

Summary

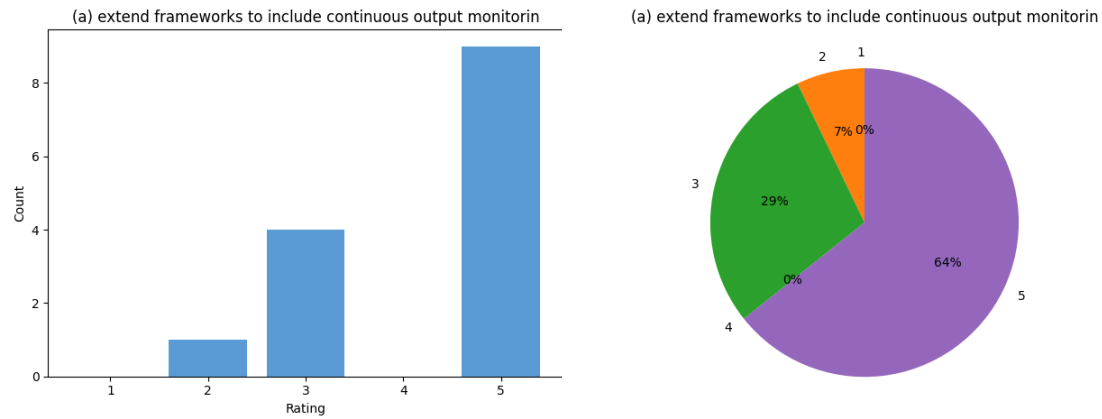
Proposal (b) – integration of AI-powered tools and human-in-the-loop validation – achieved the highest mean (4.50) and the lowest standard deviation (0.65), indicating strong consensus.

Proposal (e) concerning domain-specific rules exhibited the lowest mean (3.79) and the largest variability (IQR = 2), suggesting mixed views. Continuous monitoring (a) and expanded DQ dimensions (d) also scored highly, whereas flexible frameworks (c) received moderate support.

Distribution of Ratings

Figure 16

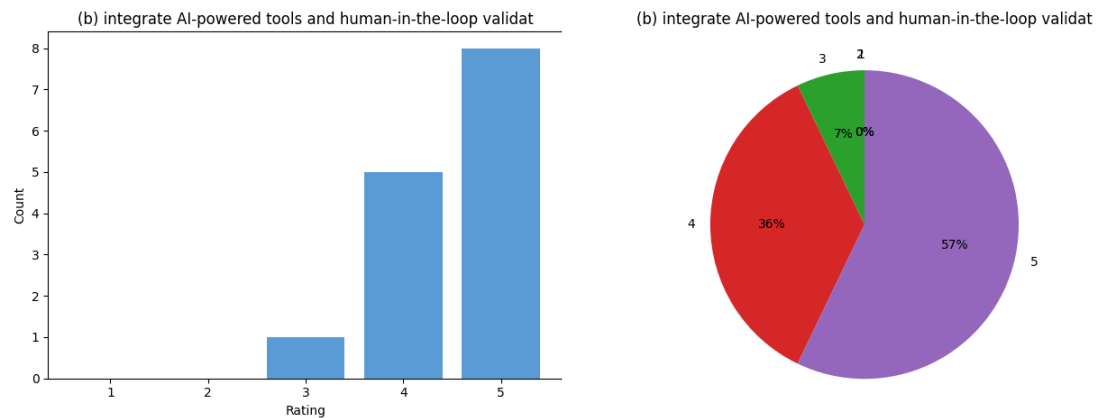
Proposal A: Continuous monitoring & semantic checks



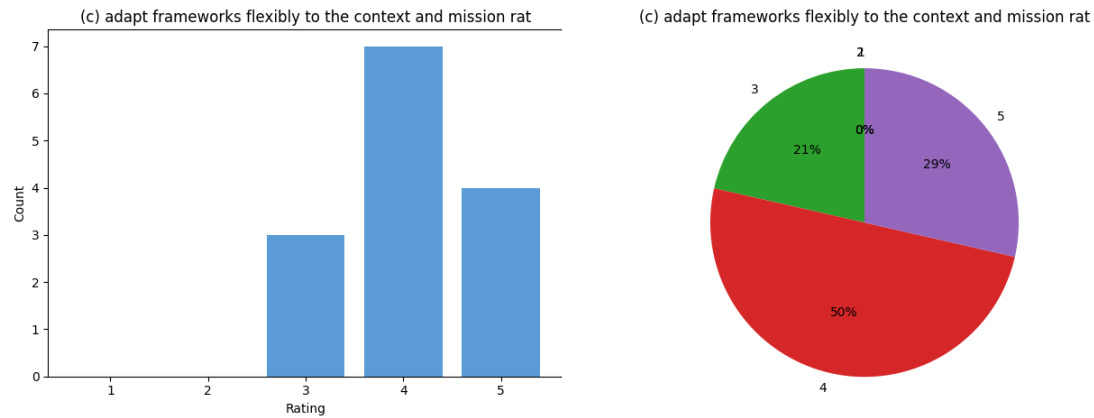
Most experts rated the continuous monitoring proposal highly, with a majority awarding scores of 4 or 5. There is still notable variability (IQR = 2).

Figure 17

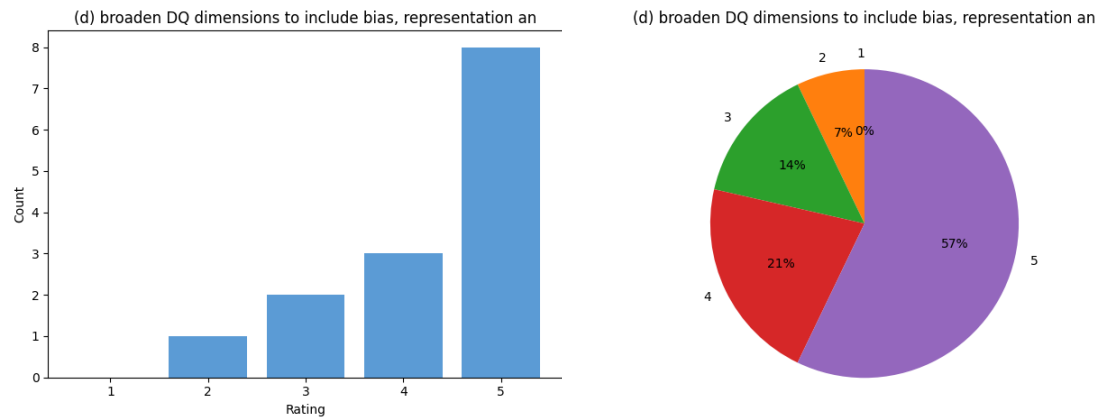
Proposal B: AI-powered tools & human-in-the-loop



This proposal received the highest consensus; over 70 % of ratings were 5, reflecting strong support for hybrid validation.

Figure 18**Proposal C: Flexible, context-specific frameworks**

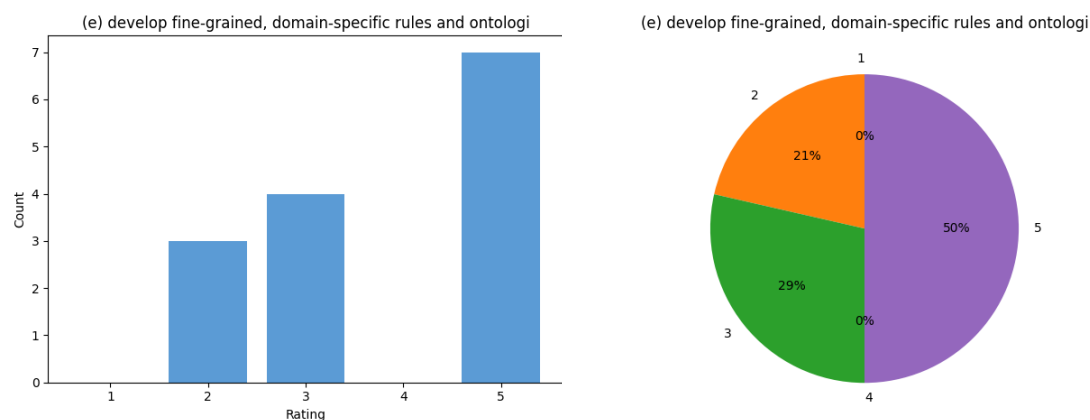
Ratings for flexible frameworks were more evenly distributed around the mean, indicating moderate consensus.

Figure 19**Proposal D: Broaden DQ dimensions**

Expanding DQ dimensions obtained high scores, but some lower ratings show varied perceptions of necessity.

Figure 20

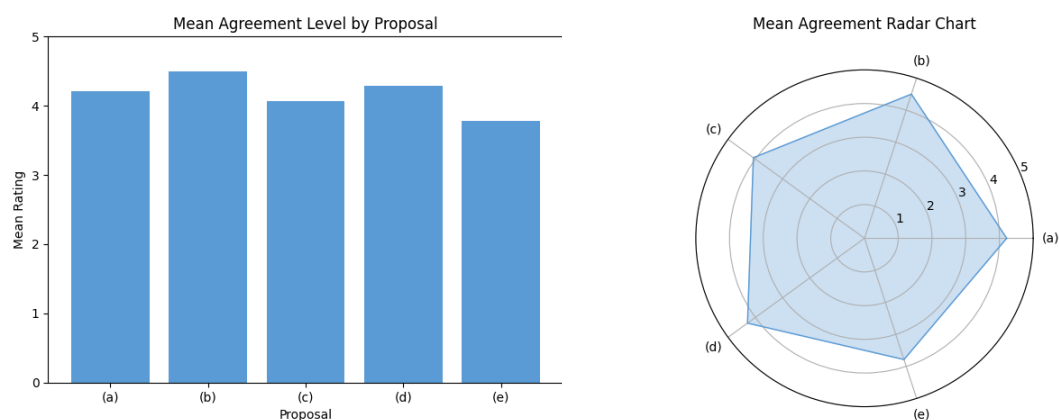
Proposal E: Domain-specific rules & ontologies



Developing domain-specific rules had the widest distribution of ratings, suggesting hesitancy due to complexity and resources.

Figure 21

Summary Rankings Mean Agreement Level and Radar Chart by Proposal



The summary mean bar chart and radar plot highlight that Proposal (b) has the highest average rating, followed by Proposals (d) and (a). Proposals (c) and (e) rank lower; Proposal (e) shows greater dispersion.

Discussion

- **Continuous monitoring & semantic checks:** Continuous output monitoring, contextual metadata integration and real-time validation are necessary to prevent AI from generating misleading or harmful outputs.
- **AI-powered tools & human-in-the-loop:** Hybrid approaches combining machine intelligence and human judgement yield high-quality outcomes and reduce cost and

time; fully automated methods alone are insufficient.

- **Flexible, context-specific frameworks:** Data quality frameworks need to be modular and flexible to accommodate diverse data sources and evolving requirements; rigid frameworks impede adoption and adaptability.
- **Broaden DQ dimensions:** Expanding DQ dimensions to include fairness, representation and reproducibility helps prevent biased outcomes and ensures AI systems respect fundamental rights.
- **Domain-specific rules & ontologies:** Ontologies formalise domain knowledge and provide contextual guardrails for agents, guiding decisions aligned with enterprise goals; however, building ontologies requires significant expertise and resources.

Conclusion & Recommendations

This Delphi round reveals strong consensus on integrating AI-powered tools with human oversight, highlighting the importance of human judgement in ensuring data quality. Continuous monitoring and semantic checks, along with broadened data quality dimensions, are widely viewed as essential enhancements to existing frameworks. Flexible architectures and domain-specific ontologies are recognised as valuable but secondary priorities due to implementation complexity.

Recommendations: Institutions should invest in real-time monitoring and automated validation tools, adopt hybrid AI-human evaluation mechanisms, design modular data quality frameworks adaptable to changing contexts, incorporate fairness and reproducibility as core dimensions, and develop domain ontologies to guide AI decision-making. Further research should assess the cost-benefit trade-offs of these enhancements and explore best practices for constructing domain-specific ontologies.

Second Round - Question 30

“Please rate the importance of the following recommended actions for prioritizing semantic data quality.” Use a 5-point importance scale rate in the table below to show your agreement with these recommendations provided from the First Round, (1 = Not Important, 5 = Very Important).

Recommended Actions (from Round 1) for prioritizing semantic DQ	Importance Rate (1-5)
(a) develop and adopt shared ontologies and publicly accessible dictionaries	
(b) verify accuracy against authoritative sources and maintain complete data sets	
(c) embed semantic metrics into dashboards and governance processes	
(d) combine automated semantic profiling with human oversight and feedback	
(e) invest in education and research on semantic DQ	

4.4.9 Analysis of Recommended Actions to Prioritize Semantic Data Quality

Table 23

Recommended Actions (from Round 1) for prioritizing semantic DQ Ratings

R2 Panelists	(a) develop and adopt	(b) verify accuracy	(c) embed semantic	(d) combine automated	(e) invest in education
#1	5	5	2	2	3
#2	3	4	4	5	3
#3	5	5	5	5	4
#4	5	4	3	4	5
#5	3	5	3	3	4
#6	5	5	5	5	5
#7	4	3	3	4	2
#8	5	4	4	5	5
#9	5	5	5	5	4
#10	5	5	5	5	5
#11	5	5	5	5	4
#12	2	5	1	3	5
#13	5	5	5	5	5
#14	5	5	5	4	5

Table below show second round experts' ratings for question 30

Introduction & Methodology

This report summarises the second round of a Delphi study in which fourteen experts

rated the importance of five recommended actions for prioritising semantic data quality in GenAI/LLM applications. Participants rated each action on a five-point Likert scale (1 = unimportant, 5 = very important). To analyse the collective judgements, descriptive statistics were calculated for each action, including the median, mode, mean, standard deviation, inter-quartile range (IQR), minimum and maximum values. The median and IQR are recommended measures for assessing consensus in Delphi studies because they are less influenced by outliers and better suited to ordinal data. Bar and pie charts were generated to visualise the distribution of ratings for each action, and summary mean, and radar charts compare the overall ranking of actions.

Table 24

Descriptive Statistics Recommended Actions to Prioritize Semantic Data Quality

Action	Mean	Median	Mode	Std Dev	IQR	Min–Max
Shared ontologies & dictionaries (a)	4.43	5.00	5.00	1.02	0.75	2–5
Accuracy & completeness (b)	4.64	5.00	5.00	0.63	0.75	3–5
Semantic metrics & governance (c)	3.93	4.50	5.00	1.33	2.00	1–5
Automated profiling + HITL (d)	4.29	5.00	5.00	0.99	1.00	2–5
Education & research (e)	4.21	4.50	5.00	0.97	1.00	2–5

The statistics show that validating accuracy and maintaining complete data sets (action b) received the highest mean rating (4.64) and a narrow IQR of 0.75, indicating strong consensus. Developing shared ontologies (a) and combining automated profiling with human oversight (d)

follow closely with means around 4.4. Embedding semantic metrics into dashboards and governance processes (c) has the lowest mean (3.93) and the largest variability (IQR = 2.00), suggesting divergent views on its urgency. Investing in education and research on semantic data quality (e) enjoys broad support with a mean of 4.21 and moderate dispersion.

Figure 22

Mean Importance Ratings Recommended Actions for prioritizing semantic DQ

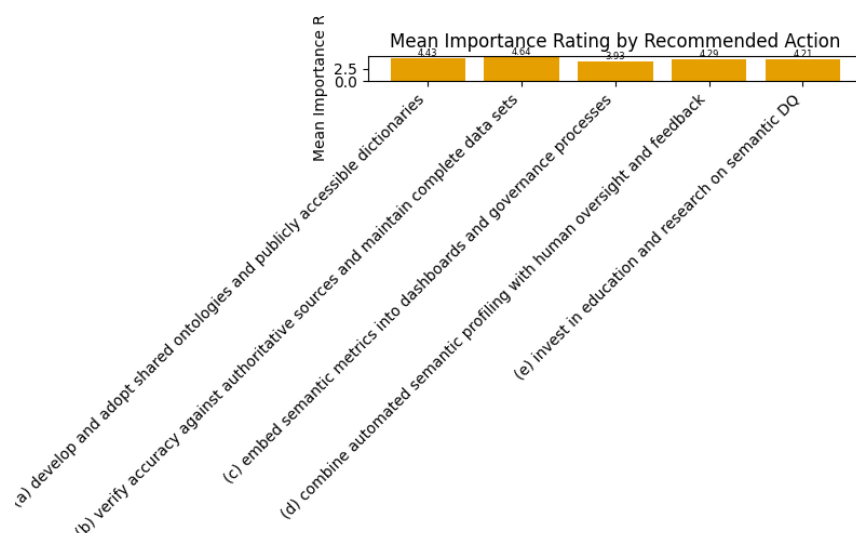
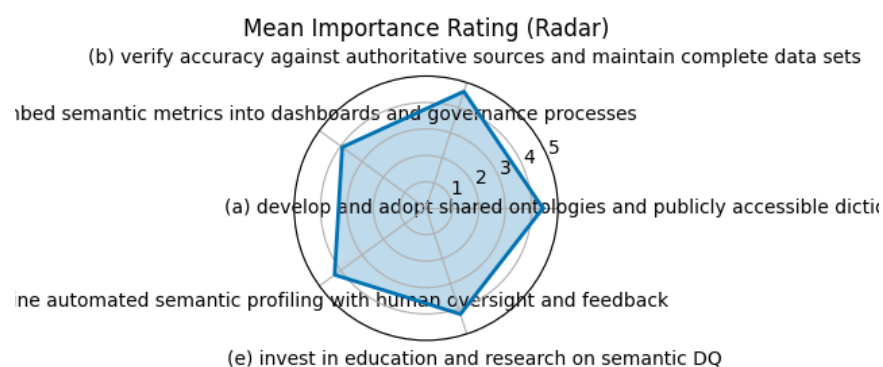


Figure 23

Recommended Actions for prioritizing semantic DQ (Radar Chart)



Discussion of Recommended Actions

Shared ontologies & public dictionaries (a): Building an ontology-based single source of truth defines business concepts and relationships, standardises processes and provides

consistent field names for AI agents.

Accuracy & completeness (b): Data must be validated against trusted sources and cross-referenced to ensure accuracy. Missing or incomplete data hinders analysis and bias results, so mandatory fields and systematic checks are essential.

Semantic metrics & governance (c): A semantic layer embeds governance policy directly into the query path, ensuring that all dashboards, notebooks and AI agents use consistent definitions and automatically inherit updates when definitions change.

Automated profiling + human oversight (d): Human-in-the-loop approaches combine automated systems with targeted human oversight to catch semantic errors, resolve ambiguity and adapt metadata as definitions and use cases evolve.

Education & research (e): Training is crucial for fostering a data-quality culture. Employees should understand their roles in creating and maintaining accurate data, and training programmes should cover basic data entry practices through advanced data governance principles.

Conclusions & Recommendations

The Delphi panel agreed most strongly on verifying accuracy and maintaining complete data sets, highlighting the importance of validating data against authoritative sources and ensuring no critical fields are missing. Developing shared ontologies and publicly accessible dictionaries is nearly as important, providing the foundational guardrails for agents and enabling interoperability across systems. Embedding semantic metrics into dashboards is viewed as less critical at present, possibly because organisations vary in their readiness to adopt semantic governance technologies. Combining automated semantic profiling with human oversight and investing in education and research were both rated highly, emphasising the need for hybrid approaches and capacity building. We recommend prioritising actions (b) and (a) in the near term; concurrently building capacity for actions (d) and (e); and piloting semantic metrics integration (c) in contexts where governance maturity is sufficient.

4.4.10 Second Round Questions 1 - 23

For questions 1 - 23 in the second round, responses from experts were optional because there was a broad consensus in the first round and resulting in no significant comments on these questions from the experts. They all reached an overwhelming consensus in the first round, and during the second round they only focused on rating questions 24-30. See first round analysis and result summary for the expert's consensus for each question 1-23.

4.4.11 Chapter Summary

The first round of the Delphi study assessed expert agreement on twenty-three statements about data quality management practices and gathered qualitative feedback on seven open-ended questions. The quantitative results revealed overwhelming consensus on the need to revisit DQM practices in light of GenAI and LLM adoption. Panelists unanimously agreed that the rise of AI amplifies existing data quality issues, with 100 % agreement on statements underscoring the urgent need for updated frameworks. They also concurred that GenAI/LLMs introduce novel challenges—including hallucinations, prompt sensitivity, context-window limitations, bias amplification and privacy risks—beyond the scope of traditional DQ methods.

The panelists highlighted several reasons why traditional DQM may fall short without enhancements. Experts noted that LLMs rely on unstructured, probabilistic inputs rather than structured databases; they stressed that garbage-in–garbage-out dynamics are magnified at scale, and that governance and oversight for AI training data are underdeveloped. Many organizations lack awareness of basic data quality concepts, with unclear ownership of DQ requirements and processes. A few participants suggested that GenAI itself could support data quality tasks, for example by automating data cleaning or monitoring.

Qualitative feedback on open-ended questions identified accuracy/factual correctness and relevance/value add as the most important semantic data quality dimensions.

Completeness and consistency were seen as critical by some, while believability/trustworthiness and timeliness were also highlighted. Panelists emphasized the need for common dictionaries and ontologies to resolve terms and ensure contextual understanding. Measures for semantic quality should include ontology coverage, authoritative source validation, contextual consistency checks, business impact assessments and trust metrics. Automated validation pipelines, vector stores for retrieval-augmented generation, and regular completeness and timeliness audits were recommended, alongside human-in-the-loop reviews and cross-functional collaboration.

Regarding challenges, experts pointed to the difficulty of ensuring accuracy and context in AI outputs, the lack of ground truth and traceability, the complexity of mitigating bias and ensuring fairness, the proliferation of unstructured data, the rapid evolution of AI technologies, and the scarcity of organizational skills and adoption. To enhance DQM practices, panelists prioritized output validation and real-time monitoring, development of automated semantic checks and deduplication tools, education and stakeholder awareness, agility in DQM processes, and strengthened governance, metadata modeling, and stewardship. They advocated extending existing frameworks rather than discarding them, broadening dimensions to include bias and reproducibility, and defining fine-grained, domain-specific rules. Consensus was strong on the need to prioritize semantic dimensions and develop actionable measures, though opinions differed on the best mechanisms for requiring comparison to authoritative data sources (ADS) for accuracy verification.

Overall, the first round established a baseline of agreement: GenAI and LLMs necessitate revised data quality practices, with semantics at the forefront. Experts agreed that semantic dimensions—accuracy, relevance, completeness, believability, timeliness and ontological clarity—must be prioritized. They also recognized that automation must be coupled with human judgement, and that organizations need education and governance structures to sustain improved practices.

The second round of the Delphi study refined consensus on key semantic data quality

dimensions and evaluated proposals for enhancing DQM frameworks. Fourteen panelists ranked six semantic dimensions on a 5-point Likert scale. Accuracy and factual correctness received the highest mean rating (indicating strongest agreement), followed closely by relevance/value add and completeness/consistency. Believability/trustworthiness also scored high, whereas timeliness and ontology/dictionary availability were rated somewhat lower but still deemed important. The distribution of ratings showed moderate variability, reflecting differing views on the relative importance of timeliness and ontologies. Nonetheless, the panel coalesced around accuracy and relevance as the top semantic dimensions for GenAI/LLM applications.

Participants were then asked to rank five proposals for improving traditional DQM frameworks. Proposal (b) – integrating AI-powered tools with human-in-the-loop validation – achieved the highest mean score (4.50) and the lowest standard deviation, demonstrating strong consensus that hybrid validation mechanisms are essential. Proposals (d) – broadening data quality dimensions to include bias, representation and reproducibility – and (a) – continuous output monitoring and semantic checks – also received high ratings. Proposal (e) – developing domain-specific rules and ontologies – showed the lowest mean rating and the greatest dispersion, indicating mixed views on the feasibility and priority of ontology development. Proposal (c) – flexible, context-specific frameworks – received moderate support.

In the discussion of proposals, experts underscored the importance of continuous monitoring and semantic checks to prevent the generation of misleading or harmful outputs. They advocated hybrid approaches combining machine intelligence with human judgement, noting that fully automated systems alone are insufficient to ensure quality. Flexible frameworks were seen as necessary to accommodate diverse data sources and evolving requirements, but their adoption depends on organizational readiness and resource availability. Expanding data quality dimensions to include fairness, representation and reproducibility was considered crucial for preventing biased outcomes. Domain ontologies were recognised as valuable for formalising domain knowledge and providing contextual guardrails, though panelists acknowledged the

significant expertise and effort required to build them.

When asked to rate five recommended actions for prioritising semantic DQ dimensions, participants again prioritised accuracy and completeness (action b), followed closely by building shared ontologies and public dictionaries (action a) and combining automated profiling with human oversight (action d). Embedding semantic metrics and governance processes (action c) received lower ratings, suggesting that organizations vary in readiness to integrate semantic governance into dashboards and pipelines. Education and research on semantic DQ (action e) garnered broad support, reflecting recognition of the need to cultivate data-quality culture.

The second round conclusions emphasise that the highest priority actions for organizations are: (1) validating data against authoritative sources and ensuring completeness; (2) developing shared ontologies to align meaning and promote interoperability; (3) adopting hybrid validation mechanisms that integrate AI-powered tools with human oversight; and (4) expanding DQ dimensions to encompass fairness, representation and reproducibility. Continuous monitoring and semantic checks were regarded as essential for safeguarding AI outputs. Flexible frameworks and domain-specific ontologies, while important, were considered secondary priorities due to complexity and resource demands.

Together, both rounds of the Delphi study reveal a coherent narrative: current DQM practices must evolve to include rigorous semantic quality measures, and expert consensus supports a balanced combination of automated tools and human judgement, enriched by ontological structure and broader ethical dimensions. Addressing these needs will enhance trustworthiness and reliability of GenAI and LLM outputs and bridge the gap between information quality theory and practical application.

Chapter 5: Discussion, Conclusion, and Recommendations

5.1 Introduction

This chapter interprets the study findings in relation to the dissertation problem, the research questions, and the existing literature on Information Quality and Data Quality Management. By synthesizing the documentary framework analysis and the Delphi results, the chapter presents the study's conclusions, practical implications for DQM in the context of GenAI and LLM adoption, and directions for future research.

This final section of this dissertation interprets the findings of the two-round Delphi study in relation to the central problem addressed in this dissertation: the persistent gap between Information Quality theory and Data Quality Management practice in the era of Generative Artificial Intelligence and Large Language Models. Chapter 4 established that expert panelists largely agreed that existing DQM practices remain more mature in their treatment of syntactic dimensions than semantic dimensions, even though GenAI and LLM applications depend heavily on meaning, relevance, contextual accuracy, trustworthiness, and authoritative correctness. The present chapter moves beyond description and considers what those findings mean for theory, practice, organizational decision making, and future scholarship.

The discussion in this chapter is intentionally anchored in the Delphi evidence rather than in general impressions. For that reason, each major interpretation is tied back to specific parts of the Delphi process. Round 1, Questions 1-23, provided the study's foundational consensus statements regarding the adequacy of current DQM practices, the risks introduced by GenAI and LLM adoption, the importance of revisiting current frameworks, and the need to distinguish validation from true accuracy. Within that first round, Questions 18-23 were particularly important because they focused directly on whether semantic data quality dimensions are overlooked, whether syntactic dimensions are prioritized in practice, whether semantic dimensions are now essential, and whether authoritative data sources are necessary to assess meaning and

contextual accuracy. Round 1, Questions 24-30, then supplied qualitative insight on the specific dimensions, measures, challenges, framework enhancements, and recommended actions that experts regarded as most important.

Round 2 transformed those qualitative themes into a more refined prioritization exercise. Question 24 asked experts to rate the importance of six semantic data quality dimensions, thereby giving this study a ranked answer to the issue raised in Research Question 2. Question 25 rated possible measures for assessing semantic quality. Question 26 evaluated the efficiency of approaches for gathering semantic DQ measures. Question 27 addressed the significance of challenges facing organizations implementing GenAI and LLMs. Question 28 focused on which components of DQM practice need the most enhancement. Question 29 rated proposals for improving traditional DQM frameworks. Finally, Question 30 asked experts to rate recommended actions for prioritizing semantic data quality. Because the second round followed an already strong first-round consensus, the findings offer both validation and prioritization, which gives the concluding chapter a stronger foundation than a single-round expert exercise would have provided.

This chapter therefore has four broad purposes. First, it synthesizes the whole study and clarifies how the documentary framework analysis and Delphi results fit together. Second, it answers the three research questions specified for the dissertation's final framing: the extent to which current DQM practices incorporate semantic data quality dimensions for GenAI and LLM applications perceived as most critical; and the actions organizations should take to prioritize semantic DQ. Third, it translates the Delphi results into theoretical and practical implications. Fourth, it offers recommendations, limitations, and future research directions grounded in the actual evidence provided by the expert panel rather than generic prescriptions. In this sense, Chapter 5 serves as the interpretive bridge between the detailed evidence presented in Chapter 4 and the final scholarly and practical contributions of the dissertation.

5.2 Summary of the Study

This dissertation examined whether widely adopted Data Quality Management frameworks adequately addressed the semantic data quality dimensions required for trustworthy Generative AI and Large Language Model applications. Through a comparative analysis of prominent DQM frameworks and a two-round Delphi study, the research demonstrated that prevailing practices emphasized syntactic dimensions while underprioritizing semantic attributes related to meaning, relevance, context, and trust, thereby validating the existence and significance of this gap for contemporary data governance practice.

This dissertation examined whether contemporary Data Quality Management practices sufficiently address the semantic data quality dimensions required to support trustworthy Generative AI and Large Language Model applications. The study was motivated by the recognition that current DQM frameworks are strong in areas such as validity, completeness, consistency, timeliness, and structural conformity, yet may underemphasize the dimensions that matter most when systems must preserve meaning, context, and trust. Guided by Information Quality theory, particularly the multidimensional view of data quality associated with Wang and Strong, the dissertation compared the coverage of data quality dimensions in widely used practice-oriented DQM frameworks with the richer theoretical treatment of semantic quality found in IQ literature.

The study used a qualitative, multi-phase design. In the first phase, a comparative framework analysis reviewed prominent DQM frameworks and standards to determine how extensively they address the full set of data quality dimensions advanced in IQ theory. This analysis indicated that practice frameworks generally give stronger and more explicit attention to syntactic dimensions than to semantic dimensions such as accuracy of meaning, value added, relevance, objectivity, believability, reputation, security, and appropriate amount of data. The framework analysis also highlighted a related conceptual problem: many organizations continue to equate validation with accuracy, even though validation only shows conformance to predefined

rules and does not establish whether data or AI outputs are semantically correct when compared with an authoritative source.

The second phase used a two-round Delphi study to validate and refine those documentary findings. In Round 1, fifteen subject-matter experts responded to twenty-three structured statements and seven open-ended questions. The structured statements tested whether experts agreed that existing DQM practices should be reexamined in light of GenAI and LLM adoption, whether current practices are overly focused on syntactic quality, whether semantic dimensions are essential in the current environment, and whether established practices are sufficient without enhancement. The open-ended questions then asked the panelists to identify the most important semantic dimensions, suggest practical measures, identify efficient ways to gather those measures, describe the most important challenges, identify which elements of DQM need the greatest enhancement, propose ways to improve traditional frameworks, and recommend concrete actions for prioritizing semantic data quality.

Round 2 narrowed the focus to prioritization. Fourteen experts participated in the second round, where they rated semantic DQ dimensions, measures, efficient approaches, major challenges, DQM enhancement priorities, framework-improvement proposals, and recommended actions derived from Round 1. The quantitative results provided central tendency and dispersion statistics that allowed the study to identify where expert opinion was strongest and where more variability remained. For example, Round 2 Question 24 showed that accuracy received the highest mean importance score of 4.86, followed by relevance at 4.43 and trustworthiness/believability at 4.29, while timeliness ranked lowest at 3.21. Round 2 Question 30 similarly showed that the strongest recommended action was verifying accuracy against authoritative sources while maintaining complete data sets, with a mean of 4.64, followed by shared ontologies and dictionaries at 4.43 and automated profiling combined with human oversight at 4.29.

Taken together, the study documented a coherent story. The comparative framework analysis suggested that DQM practice underrepresents semantic dimensions. Round 1 then validated that this underrepresentation is visible to experienced practitioners, especially in Questions 18-23. Round 1 Questions 24-30 further showed that experts consistently pointed to accuracy, relevance, trustworthiness, context, authoritative validation, human judgment, and governance capability as central issues. Round 2 then demonstrated that these themes were not only conceptually persuasive but also rank-ordered in a way that can support practice recommendations. The study therefore contributes not by proposing an entirely new framework, but by clarifying what existing DQM practice must elevate, operationalize, and govern if organizations are to use GenAI and LLMs in a trustworthy manner.

Table 25

Alignment of Research Questions with Delphi Evidence

Research Question	Primary Delphi Evidence	Key Result	Interpretive Meaning
RQ1. To what extent do current DQM practices incorporate semantic data quality dimensions for GenAI and LLM applications?	Round 1 Q18-Q20; Round 1 Q15-Q17; Round 1 Summary of Q1-Q23	93.33% agreed semantic dimensions are often overlooked; 86.67% agreed syntactic dimensions are prioritized; 100% agreed semantic dimensions are essential.	Current DQM practices incorporate semantic DQ only partially and inconsistently; the expert panel sees a clear underprioritization gap.
RQ2. What semantic DQ dimensions are perceived as most critical for GenAI/LLMs?	Round 1 Q24 themes; Round 2 Q24 / Table 10	Accuracy (4.86), Relevance (4.43), and Trustworthiness/Believability (4.29) received the strongest support.	Experts view semantic correctness, contextual fit, and credibility as the central quality requirements for trustworthy GenAI/LLM outputs.
RQ3. What actions should organizations take to prioritize semantic DQ?	Round 1 Q25-Q30 themes; Round 2 Q25-Q30 / Tables 12, 14, 16, 18, 20, and 22	Top actions included authoritative validation and complete data sets (4.64), shared ontologies and dictionaries (4.43), and automated profiling combined with human oversight (4.29).	Organizations should operationalize semantic DQ through authoritative references, semantic assets, hybrid review, stronger governance, output monitoring, and capability building.

5.3 Discussion of Major Findings

The findings indicate that current Data Quality Management practices are substantially stronger in enforcing syntactic conformance than in ensuring semantic adequacy, a gap that is increasingly problematic with the adoption of GenAI and Large Language Models. The results show that trustworthy AI outcomes require greater prioritization of semantic dimensions—such as meaning, context, accuracy of interpretation, and trust—supported by a combination of automated controls, human judgment, and cross-functional governance.

The first major finding of this study is that the semantic gap identified in the documentary review is not merely theoretical; it is recognized by practitioners and experts who work with data quality and governance in real organizational settings. Round 1 provides especially strong evidence for this interpretation. Question 18 found that 93.33 percent of the panel agreed that semantic data quality dimensions centered on meaning and context are often overlooked and not adequately measured in practice. Question 19 found that 86.67 percent agreed that syntactic dimensions are prioritized over semantic dimensions. Question 20 then moved beyond diagnosis and showed full consensus that semantic dimensions are essential measurable factors for defining data quality requirements and evaluating data quality performance and risk in the age of GenAI and LLM adoption. These responses are important because they confirm that the theoretical gap identified in the framework analysis is not an abstract scholarly concern; it is a practical deficiency recognized by experts who understand current organizational realities.

The second major finding is that accuracy occupies a foundational position in semantic data quality, but the expert panel used the concept of accuracy in a richer and more demanding sense than ordinary validation practice. Round 1 Questions 22 and 23 are particularly significant here. In Question 22, 73.33 percent of panelists agreed that measuring meaning, relevance, and context often requires comparison with an Authoritative Data Source or System of Record. In Question 23, 66.67 percent agreed that the practice of assuming validated data is inherently accurate without comparison to an approved authoritative source should end, especially in the

context of GenAI and LLM adoption. These first-round insights were reinforced by Round 2. Question 24 rated accuracy as the highest-priority semantic dimension with a mean of 4.86 and an interquartile range of 0, indicating exceptionally strong consensus. Question 25 also showed that coverage and resolvability as well as cross-source validation and triangulation both received the highest mean effectiveness ratings of 3.86 among candidate measures for assessing accuracy and relevance. Together, these findings show that the study's emphasis on the distinction between validation and true accuracy is not a marginal argument; it is central to how experts believe semantic DQ should be understood in AI-enabled environments.

A third major finding is that organizations cannot rely on automation alone to manage semantic quality. The Delphi evidence consistently points toward hybrid evaluation and governance approaches. Round 1 Question 26 revealed that experts favor automated validation and auditing, but they also emphasized surveys, feedback, cross-functional collaboration, and human-in-the-loop review where meaning and business value cannot be resolved algorithmically. Round 2 gives this conclusion further precision. In Question 26, automated validation using authoritative sources and ontologies achieved the highest mean efficiency score of 4.69, but hybrid AI/human approaches also received a strong mean of 4.00, while purely human review and collaboration remained valued at 3.46. Round 2 Question 25 similarly showed that human evaluation of value add received a mean of 3.71 and the lowest standard deviation among the measures considered, which indicates relatively stable support across experts. The implication is not that automation is unimportant; rather, the panel sees automation as necessary for scale but insufficient for semantic adequacy, especially where ambiguity, use context, fairness, and interpretive meaning are involved.

A fourth major finding is that the most serious GenAI and LLM risks identified by experts are tightly connected to semantic quality problems rather than to purely structural data defects. Round 1 Questions 7-13 showed high agreement that bias, hallucination, misinformation, lack of transparency, inadvertent leakage of sensitive information, inconsistent accuracy based on

training context, and digital inbreeding are all pressing concerns in AI-enabled environments. The strongest support appeared in Question 13, where 93.33 percent agreed that poor DQM practices expose organizations to digital inbreeding risk, and in Questions 7, 9, and 12, where 86 percent or more agreed that poor DQM makes bias mitigation, misinformation control, and consistent accuracy harder to achieve. Round 2 Question 27 provided a ranked version of those same risks: verifying the accuracy and relevance of training data and outputs received the highest challenge score at 4.64, absence of ground truth and traceability followed at 4.36, and detecting and mitigating bias and fairness issues was rated at 4.21. These patterns show that the study's concern with semantic DQ is justified because the core failure modes of GenAI and LLMs are strongly tied to meaning, truth, context, provenance, and interpretive reliability.

A fifth major finding is that experts do not advocate abandoning traditional DQM; instead, they recommend extending and augmenting it. Round 1 Question 17 is especially informative because 80 percent agreed that industries must adopt enhanced DQ frameworks rather than relying solely on traditional ones, yet the qualitative feedback showed repeated caution against framing the matter as an either/or choice. Round 1 Question 29 reinforced this interpretation when experts recommended extending frameworks to include semantic and output monitoring, using AI-powered tools with human oversight, adapting frameworks flexibly rather than treating them as rigid, broadening DQ dimensions to include model-ready factors such as bias and reproducibility, and educating stakeholders on semantics. Round 2 Question 29 then prioritized these proposals, with integrating AI-powered tools and human-in-the-loop validation receiving the highest mean of 4.50, broadening DQ dimensions to include bias, representation, and reproducibility receiving 4.29, and extending frameworks to include continuous output monitoring and semantic checks receiving 4.21. This is one of the most important conclusions in the chapter: the study does not support replacing DQM foundations, but it clearly supports expanding them so that semantic DQ, output quality, and AI-specific governance risks become explicit responsibilities within the DQM program.

5.4 Discussion by Research Question

Research Question 1 asked: To what extent do current DQM practices incorporate semantic data quality dimensions for GenAI and LLM applications? The combined evidence of the documentary framework review and the Delphi study indicates that current DQM practices incorporate semantic data quality dimensions only partially and inconsistently. The documentary review established the initial pattern by showing that the selected practice frameworks more explicitly emphasized validity, completeness, consistency, timeliness, and other syntactic or operational dimensions than they did dimensions related to meaning, context, believability, or relevance. That conclusion was then validated by the Delphi panel. In Round 1, Question 18 found that 93.33 percent of panelists agreed that semantic dimensions are often overlooked and not adequately measured in current practice, while Question 19 found that 86.67 percent agreed that syntactic dimensions are prioritized over semantic ones. Question 15 also showed unanimous agreement that there is a growing need for improved DQ frameworks to support GenAI and LLM adoption. These results indicate that current DQM practice includes some semantic awareness, but not at a level that experts consider adequate for AI-enabled environments.

The Round 1 results also help clarify why this limited incorporation persists. The panel repeatedly noted that semantic dimensions are more difficult to define, quantify, standardize, and automate than syntactic dimensions. Several comments in Questions 18 and 19 emphasized that semantic quality depends on metadata maturity, context, shared vocabulary, and interpretive understanding, all of which are less developed in many organizations than routine structural checks. That insight is important for interpreting RQ1 because it suggests the gap is not simply a matter of negligence; it is also a matter of measurement difficulty, inconsistent terminology, and low organizational maturity. Even so, Round 1 Question 20 established unanimous agreement that semantic dimensions are now essential measurable factors for DQ requirements and performance evaluation in the GenAI and LLM era. Therefore, while current practice incorporates

semantic DQ to some degree, the extent of that incorporation remains insufficient when judged against expert expectations and contemporary AI risk conditions.

Research Question 2 asked: What semantic DQ dimensions are perceived as most critical for GenAI/LLMs? The answer emerges from both the qualitative themes in Round 1 Question 24 and the formal ratings in Round 2 Question 24. In Round 1, the strongest recurring themes were accuracy or factual correctness, relevance or contextual value, trustworthiness or believability, completeness, and consistency. Round 2 confirmed and ranked these dimensions. Accuracy received the highest mean importance score at 4.86, with an interquartile range of 0, indicating near-perfect consensus. Relevance ranked second at 4.43, and trustworthiness or believability ranked third at 4.29. Completeness and consistency followed with means of 3.93 and 3.86 respectively, while timeliness ranked lowest at 3.21. These findings provide a direct answer to RQ2: experts perceive semantic accuracy, contextual relevance, and trustworthiness as the most critical semantic dimensions for GenAI and LLM applications, with completeness and consistency still important but somewhat less central, and timeliness more dependent on context and use case.

The deeper significance of this answer lies in the way the panel conceptualized these dimensions. Accuracy was not treated as mere rule conformance but as factual and semantic correctness relative to authoritative references. Relevance was understood as fitness for the specific task, domain, or decision context. Trustworthiness or believability referred to whether the data and the resulting outputs can be regarded as credible and dependable by users. Completeness and consistency were important because missing or inconsistent data can distort interpretation and degrade downstream model performance. Timeliness was rated lower not because it is irrelevant, but because the urgency of recency varies across use cases. Thus, RQ2 is answered not only by a ranked list, but also by a clearer understanding that the most critical semantic dimensions are those most directly tied to truth, contextual fit, and user confidence in AI outputs.

Research Question 3 asked: What actions should organizations take to prioritize semantic DQ? The strongest answers come from Round 1 Question 30 and Round 2 Question 30, but they are also supported by Questions 25 through 29. In Round 1, panelists recommended developing shared ontologies and dictionaries, prioritizing accuracy and factual correctness through authoritative verification, combining automated profiling with human review, and investing in education and governance. Round 2 translated these ideas into prioritized actions. Verifying accuracy against authoritative sources while maintaining complete data sets received the highest mean of 4.64. Developing shared ontologies and publicly accessible dictionaries followed at 4.43. Combining automated semantic profiling with human oversight and feedback received 4.29, and investing in education and research on semantic data quality received 4.21. Embedding semantic metrics into dashboards and governance processes received 3.93, still positive but with greater variability across experts.

These recommended actions are further reinforced by adjacent Round 2 items. Question 25 showed strong support for coverage and resolvability as well as cross-source validation and triangulation as measures for semantic quality. Question 26 showed that automated validation using authoritative sources and ontologies is the most efficient gathering approach, while hybrid AI/human approaches also received strong support. Question 27 showed that the most serious challenge is verifying the accuracy and relevance of training data and outputs, which makes the top-ranked action in Question 30 especially logical. Question 28 showed that output validation and real-time monitoring, governance and stewardship, and agility of DQM processes need substantial enhancement. Question 29 showed that improving frameworks should involve AI-powered tools with human-in-the-loop validation, broadening DQ dimensions, and extending frameworks to include output monitoring and semantic checks. Accordingly, the answer to RQ3 is that organizations should not pursue a single isolated intervention. Rather, they should pursue a coordinated package of actions built around authoritative verification, semantic standardization,

hybrid evaluation, governance integration, output monitoring, and organizational capability building.

5.5 Implications for Theory

This study contributes to Information Quality theory by reaffirming the enduring relevance of Wang and Strong's multidimensional framework in the context of Generative AI and Large Language Models. The findings demonstrate that rather than diminishing the value of IQ theory, the adoption of GenAI and LLMs heightens the importance of distinguishing between syntactic and semantic data quality and underscores the need to reconnect theory with contemporary practice.

The findings of this study have several implications for Information Quality theory and for the continuing relationship between IQ theory and DQM practice. First, the results strengthen the argument that classic IQ frameworks remain conceptually relevant in contemporary AI settings. The dimensions emphasized by Wang and Strong and related IQ scholarship were not invalidated by GenAI and LLM adoption; if anything, the Delphi panel's responses suggest that these dimensions have become even more important. Round 1 Questions 18-20 clearly showed that experts view semantic dimensions as overlooked in practice yet essential in AI-enabled contexts. Round 2 Question 24 then elevated accuracy, relevance, and trustworthiness to the top of the semantic hierarchy. This means that the semantic richness of IQ theory has not been superseded by modern AI; rather, the arrival of GenAI and LLMs has made the older theoretical insight about multidimensional fitness for use newly urgent.

Second, the study supports a more explicit theoretical distinction between syntactic quality and semantic quality. Traditional DQM practice often treats quality as if technical conformance and semantic adequacy lie on the same continuum, but the Delphi evidence suggests these are related yet analytically distinct. Round 1 Questions 18 and 19 showed that experts perceive a structural imbalance in which syntactic dimensions are easier to measure and therefore more

likely to be prioritized. Round 1 Question 20, however, demonstrated unanimous support for semantic dimensions as essential evaluative factors. The implication for theory is that models of data quality in AI-enabled environments should more clearly separate what can be established through structural conformance from what must be judged through meaning, relevance, context, and authoritative correctness. This distinction does not weaken existing theory; it sharpens it by clarifying that different classes of data quality require different kinds of evidence and governance.

Third, the study contributes to theory by extending the interpretation of accuracy. In many operational settings, accuracy is treated interchangeably with validity, but the Delphi results repeatedly challenged that simplification. Round 1 Questions 22 and 23, along with Round 2 Questions 24 and 25, show that experts understand accuracy as something more demanding than internal rule compliance. It requires comparison with trusted references, cross-source triangulation, and, in many contexts, human interpretation. This is theoretically significant because it restores the semantic and epistemic meaning of accuracy. For AI-enabled systems, where fluent outputs can be factually incorrect, the theory of data quality must recognize that semantic truth and formal validation are not substitutes. The dissertation therefore contributes to theory by making the validation-versus-accuracy distinction more explicit and by grounding it in expert consensus rather than in conceptual argument alone.

Fourth, the study suggests that IQ theory should be more intentionally connected to theories of governance, risk, and sociotechnical evaluation. The Delphi panel repeatedly linked semantic DQ to downstream risks such as hallucination, misinformation, bias, opacity, privacy leakage, and loss of trust. Round 1 Questions 7-13 and Round 2 Question 27 show that semantic quality cannot be understood only as a technical attribute of data; it is also a condition that influences ethical, legal, operational, and reputational outcomes. This broadens the theoretical significance of semantic DQ. It is not merely part of a consumer-oriented definition of quality; it is also a determinant of AI trustworthiness and organizational risk exposure. Consequently, future

theoretical work should situate semantic DQ not only within IQ taxonomies but also within larger governance and model-risk frameworks.

Finally, the study has theoretical implications for how consensus-based research can refine emerging data quality constructs. The Delphi design allowed the study to move from conceptual diagnosis to prioritized judgment. By connecting Round 1 exploratory themes to Round 2 rating and ranking results, the dissertation demonstrates that theory building around semantic DQ can benefit from iterative expert validation, especially in domains where measurement remains unsettled. The implication is that IQ theory can remain stable in its conceptual foundation while still being refined, translated, and operationalized through Delphi-based expert interpretation in emerging technological environments.

5.6 Implications for Practice

From a practical perspective, the findings suggest that organizations should extend rather than replace existing Data Quality Management frameworks, retaining traditional syntactic controls while explicitly incorporating semantic quality expectations. In AI-intensive environments, fitness for purpose must be broadened to encompass contextual appropriateness, factual grounding, source credibility, and the potential risks associated with misleading or biased outputs.

The practice implications of this study are substantial because the Delphi results translate abstract concerns about semantic data quality into concrete organizational priorities. The most direct implication is that organizations adopting GenAI and LLMs should no longer assume that traditional DQM routines are sufficient if those routines focus mostly on structural correctness, completeness of fields, and business-rule conformance. Round 1 Questions 15, 16, and 17 showed strong support for reviewing and improving existing frameworks, and Round 2 Question 29 identified the most useful forms of that improvement. In practical terms, this means organizations need to extend DQM beyond input-level control and treat output quality, semantic

consistency, contextual fit, and trustworthiness as legitimate objects of governance and monitoring.

A second implication is that authoritative verification must become a more visible part of practice wherever semantic accuracy matters. The panel repeatedly returned to this point. In Round 1 Questions 22 and 23, experts argued that validation without authoritative comparison can create a false sense of quality. Round 2 then reinforced the point by assigning the highest action score to verifying accuracy against authoritative sources while maintaining complete data sets. For practice, this means DQ programs should strengthen reference data management, systems of record governance, lineage discipline, and policies for when AI outputs must be checked against trusted sources. In regulated, customer-facing, safety-critical, or high-impact contexts, this recommendation is especially important because plausible but incorrect outputs can cause real harm.

A third practical implication is that semantic standardization assets should be treated as core infrastructure rather than optional documentation. Round 1 Questions 24, 25, and 30 repeatedly highlighted dictionaries, ontologies, and shared vocabularies, and Round 2 Question 30 ranked shared ontologies and dictionaries as the second most important action. This suggests that organizations need to invest more deliberately in glossaries, business terms, semantic models, metadata stewardship, and ontology-like structures that allow both humans and AI systems to interpret data consistently. Without these assets, semantic ambiguity persists across reports, dashboards, models, agents, and operational workflows. With them, organizations improve interoperability, reduce interpretive conflict, and create more stable conditions for trustworthy AI use.

A fourth implication is that hybrid quality evaluation should become standard practice for GenAI and LLM use cases. The Delphi panel did not reject automation; in fact, Round 2 Question 26 rated automated validation using authoritative sources and ontologies as the most efficient measure-gathering approach. However, the panel also rated hybrid AI/human approaches highly

and consistently reinforced the role of human judgment in Round 1 Questions 25, 26, and 29. For practice, this means organizations should design DQM processes in which automated checks handle scale, speed, repeatability, and known rules, while human reviewers address ambiguity, contextual fitness, business value, fairness concerns, and exceptional cases. This is particularly important when outputs influence decisions, communications, or public-facing interactions where semantic nuance matters.

A fifth implication concerns organizational capability and governance design. Round 1 Question 28 emphasized the need for education, awareness, stewardship, and stronger governance structures, and Round 2 Question 28 showed that output validation and real-time monitoring, governance and metadata stewardship, and agility of DQM processes are among the top enhancement priorities. For organizations, this means the DQM function cannot remain isolated from AI governance, data science, security, privacy, risk, and business teams. Semantic DQ is inherently cross-functional because meaning, context, and trust are distributed across technical systems and organizational roles. Therefore, practice must evolve toward shared accountability, stronger stewardship, targeted training, and operating models where DQM and AI governance are mutually reinforcing rather than separate control silos.

5.7 Overall Conclusions

The central conclusion of this dissertation is that the adoption of Generative AI and Large Language Models has revealed a significant and practice-relevant gap between Information Quality theory and prevailing Data Quality Management practices. While widely adopted DQM frameworks remain valuable, expert consensus confirmed that trustworthy AI outcomes require elevating the priority of semantic data quality dimensions and aligning governance, measurement, and accountability mechanisms accordingly.

The overall conclusion of this dissertation is that current Data Quality Management practice does not sufficiently incorporate semantic data quality dimensions to meet the needs of

GenAI and LLM applications, even though those dimensions are now central to trustworthy AI performance. The framework analysis established the initial diagnosis, and the Delphi study provided robust expert validation. Round 1 Questions 18 and 19 confirmed that semantic dimensions are often overlooked and that syntactic dimensions remain dominant in practice. Round 1 Question 20 then demonstrated unanimous agreement that semantic dimensions are essential in the current environment. These findings, combined with the strong Round 1 support for revisiting DQM in response to GenAI and LLM adoption, justify the study's core conclusion that DQM practice must evolve if it is to remain relevant and effective in AI-enabled environments.

A second conclusion is that the highest-priority semantic dimensions for GenAI and LLM applications are accuracy, relevance, and trustworthiness. Round 1 Question 24 surfaced these dimensions qualitatively, and Round 2 Question 24 confirmed their rank order quantitatively, with mean scores of 4.86, 4.43, and 4.29, respectively. This is a meaningful conclusion because it identifies where organizations should begin when resources, governance maturity, and measurement capacity are limited. Rather than attempting to operationalize every possible quality dimension at once, organizations can focus first on whether data and AI outputs are accurate in meaning, relevant to the task and context, and trustworthy from the user's perspective.

A third conclusion is that the semantics of accuracy require authoritative comparison, cross-source verification, and contextual judgment rather than mere validation. This conclusion arises from the strong Round 1 responses to Questions 22 and 23, and from the high Round 2 ratings for accuracy-related measures and actions. The dissertation therefore concludes that traditional validation should be regarded as necessary but insufficient wherever GenAI and LLM applications are expected to produce reliable outcomes. This conclusion directly supports the dissertation's critique of the widespread assumption that validated data is inherently accurate. In semantic contexts, especially those involving generated language or synthesized responses, that assumption is not defensible.

A fourth conclusion is that organizations should not replace traditional DQM frameworks but should augment them. Round 1 Question 17 and Round 1 Question 29 made this clear conceptually, while Round 2 Questions 28, 29, and 30 showed which enhancements experts consider most important. The strongest practical priorities include authoritative validation, shared ontologies and dictionaries, automated semantic profiling combined with human oversight, output monitoring, governance integration, and education. Accordingly, the dissertation concludes that the solution to the semantic gap is not a wholesale rejection of existing DQM foundations, but a structured expansion of those foundations to include semantic quality and AI-specific control needs.

The final overall conclusion is that semantic data quality is now inseparable from AI governance, risk management, and organizational trust. The Delphi panel linked weak semantic DQ to hallucination, misinformation, bias, opacity, inconsistent accuracy, and digital inbreeding. This means semantic DQ can no longer be treated as a specialized or secondary concern. In the era of GenAI and LLM adoption, it is part of the core control environment that determines whether organizations can rely on AI outputs, explain them, govern them, and justify them. That conclusion is perhaps the most important contribution of the dissertation because it ties semantic data quality directly to the broader institutional question of trustworthy AI.

5.8 Overall Recommendations

Based on the findings of this study, the following recommendations outline how organizations can enhance existing Data Quality Management practices to better support trustworthy and reliable GenAI and Large Language Model deployments. These recommendations emphasize elevating semantic data quality, strengthening governance integration, and aligning technical controls with organizational capability and risk management needs in AI-intensive environments.

Based on the Delphi evidence, the first overall recommendation is that organizations should establish authoritative-reference-based validation routines for high-impact AI use cases. This recommendation is grounded in Round 1 Questions 22 and 23 and is strongly reinforced by Round 2 Question 30, where verifying accuracy against authoritative sources and maintaining complete data sets received the highest rating. In operational terms, this recommendation means organizations should define which systems of record, approved reference datasets, curated knowledge stores, or trusted external sources are appropriate for specific domains and then ensure that validation workflows use those sources consistently. This recommendation is especially important for customer communications, regulatory reporting, financial decisions, healthcare, and any other use cases where plausible but incorrect outputs could create significant harm.

The second recommendation is that organizations should build and govern shared semantic assets, including ontologies, glossaries, dictionaries, metadata standards, and controlled vocabularies. This recommendation follows directly from Round 1 Questions 24, 25, and 30 and from Round 2 Question 30, where shared ontologies and dictionaries received a mean score of 4.43. It is also indirectly supported by Round 2 Question 26, where automated validation using authoritative sources and ontologies was rated the most efficient measure-gathering approach. The practical value of this recommendation is that shared semantic assets reduce ambiguity across systems, make cross-functional interpretation more consistent, and provide guardrails for both human analysts and AI agents.

The third recommendation is that DQM programs should implement hybrid quality evaluation processes that combine automation with human review. This recommendation arises from Round 1 Questions 25, 26, and 29 and from Round 2 Questions 25 and 26. Organizations should automate what is repeatable, high-volume, and rules-driven, such as consistency checks, coverage assessments, and cross-source comparisons. At the same time, they should preserve deliberate human review for ambiguous, high-risk, or high-context cases where meaning,

relevance, fairness, and value cannot be determined mechanically. This recommendation does not imply inefficiency; rather, it suggests designing targeted human intervention points so that scarce expert attention is used where it adds the most value.

The fourth recommendation is that organizations should redesign DQM governance to include output validation, real-time monitoring, stewardship, and AI-aligned escalation paths. This recommendation is grounded in Round 1 Question 28 and supported by Round 2 Question 28, where output validation and real-time monitoring, governance and stewardship, and speed and agility of DQM processes all received high enhancement-priority scores. Practically, this means the DQM function should help monitor generated outputs, define escalation criteria for semantic failures, maintain metadata and stewardship responsibilities, and ensure that recurring semantic issues are analyzed and resolved rather than treated as isolated model incidents. This recommendation also suggests a tighter operating relationship between DQM, model-risk teams, data governance councils, and business owners.

The fifth recommendation is that organizations should invest in semantic DQ education, stewardship capability, and cross-functional literacy. Round 1 Questions 28, 29, and 30 emphasized education and capability building, and Round 2 Questions 28 and 30 confirmed that these remain meaningful priorities. Training should not be limited to technical teams. Business users, data stewards, AI product owners, governance personnel, and decision makers should understand why semantic DQ matters, how it differs from structural validation, how authoritative data supports trust, and when human judgment is required. Over time, this recommendation should help build a culture in which semantic quality is not treated as an abstract theory topic but as a practical requirement for trustworthy digital operations.

Table 26
Priority Action Roadmap for Strengthening Semantic Data Quality

Priority Area	Delphi Source	Evidence of Priority	Near-Term Organizational Response
Authoritative validation and completeness	Round 1 Q22-Q23; Round 2 Q24, Q25, Q30	Accuracy ranked first (4.86); cross-source validation and triangulation rated strongly; 'Accuracy & completeness' received the highest action mean (4.64).	Identify authoritative sources and systems of record, define verification rules, and require reference-based checks for high-impact AI use cases.
Shared ontologies, dictionaries, and metadata assets	Round 1 Q24-Q26 and Q30; Round 2 Q26 and Q30	Shared ontologies and dictionaries ranked second among actions (4.43); automated validation using authoritative sources and ontologies rated most efficient (4.69).	Expand glossary, ontology, and metadata stewardship practices so humans and AI systems use consistent concepts, terms, and relationships.
Hybrid automation plus human oversight	Round 1 Q25-Q26 and Q29-Q30; Round 2 Q25-Q26 and Q29-Q30	Human evaluation of value add remained important (3.71); hybrid AI/human approaches scored 4.00; automated profiling with human oversight scored 4.29.	Automate scalable checks but design targeted human review points for ambiguity, fairness, context, and mission-critical interpretation.
Output monitoring, governance, and agile DQM operations	Round 1 Q28-Q29; Round 2 Q27-Q29	Top challenges included verifying output accuracy (4.64) and lack of traceability (4.36); output validation and real-time monitoring scored 4.29; governance/stewardship scored 4.21.	Treat generated outputs as governed quality objects; implement monitoring, escalation paths, stewardship ownership, and faster issue-response cycles.
Education, capability, and culture	Round 1 Q28-Q30; Round 2 Q28 and Q30	Education/capability building remained a meaningful enhancement priority (3.93) and education/research scored 4.21 among recommended actions.	Train stewards, analysts, product owners, and business users on semantic DQ, authoritative validation, and the governance implications of GenAI/LLM adoption.

5.9 Overall Discussions

The overall discussion emerging from this study is that the adoption of GenAI and LLMs has shifted the center of gravity in data quality discourse. For many years, organizations could often operate effectively with DQM programs that were heavily oriented toward structured data,

transactional correctness, and rule conformance. Those activities remain necessary, but the Delphi evidence shows they are no longer enough when systems generate language, synthesize information, and interact with users in ways that depend on meaning, context, and perceived credibility. What Chapter 4 revealed, and what this chapter now interprets, is that the semantic demands of AI expose the limits of a DQM approach that remains too narrowly tied to structural correctness alone.

This shift has both conceptual and operational consequences. Conceptually, it changes how quality itself must be understood. The panel repeatedly indicated that high-quality data for AI is not simply data that is valid, complete, and well-formed. It is data that supports semantically trustworthy outcomes. That means the meaning of quality becomes more relational and contextual. Data must be correct relative to authoritative truth, relevant to the purpose, interpretable within a domain, and credible to the users who depend on the resulting outputs. Operationally, this means DQM programs must pay greater attention to metadata, stewardship, semantic assets, output monitoring, and review processes that are sensitive to context rather than limited to format checks.

The study also reveals an important tension that should shape ongoing discussion. Experts clearly support prioritizing semantic dimensions, but some also warn against treating semantic DQ as though it can replace traditional DQ fundamentals. This tension was visible in Round 1 Questions 17 and 21, where several respondents explicitly argued that semantic and syntactic quality are complementary. This is an important caution. The dissertation does not argue that organizations should stop caring about validity, completeness, consistency, or timeliness. Rather, it argues that these dimensions should no longer be mistaken for a complete definition of quality in AI-enabled environments. The proper interpretation is additive and integrative: traditional DQ foundations remain necessary, but they must be supplemented with stronger semantic controls, evaluative routines, and governance mechanisms.

A further discussion point concerns the organizational maturity required to act on these findings. Some Delphi responses suggest that many organizations do not yet have sufficiently mature DQ frameworks, stewardship structures, or metadata practices even for traditional data environments. If that is true, then the challenge of prioritizing semantic DQ is not simply one of adding a few new metrics; it is also a matter of maturing the broader DQM operating model. This insight matters because it explains why some Round 2 items, such as embedding semantic metrics into dashboards and governance processes, received lower means and higher variability than other actions. Experts may not be rejecting those ideas; rather, they may be recognizing that organizations differ significantly in readiness. This interpretation encourages a phased approach to implementation rather than an assumption that every organization can move immediately to the most advanced semantic-governance practices.

Finally, the overall discussion reinforces the dissertation's central practical message: semantic data quality is no longer optional in environments shaped by GenAI and LLM adoption. Once organizations begin to rely on AI-generated or AI-mediated outputs, weaknesses in meaning, truth, context, and interpretability become visible quickly and can damage trust just as quickly. The study's discussion therefore returns to the same basic insight from a new angle: the gap between IQ theory and IQ practice is not merely an academic inconsistency. It is a practical vulnerability that becomes more consequential as AI adoption expands.

5.10 Limitations

This study is subject to limitations related to the scope of frameworks analyzed, the reliance on expert consensus rather than statistical generalization, and its focus on identifying semantic data quality gaps rather than testing or implementing new Data Quality Management frameworks in real-world organizational settings.

This study should be interpreted in light of several limitations. The first limitation concerns the scope of the framework review that informed the Delphi process. Although the documentary analysis included widely referenced DQM frameworks and standards, it did not include every framework, every industry-specific methodology, or every way in which organizations adapt formal frameworks in practice. As a result, the identified gap between semantic and syntactic emphasis reflects the selected frameworks and the explicit content they present. It is possible that some organizations operationalize semantic controls in ways that are not fully visible in published framework documents. This limitation does not negate the study's findings, but it does suggest caution in treating the framework analysis as a census of all DQM practice.

A second limitation concerns the nature of Delphi methodology itself. The Delphi method is designed to obtain informed expert judgment rather than statistically generalizable findings. Accordingly, the study's conclusions reflect consensus among a purposive panel of experts rather than representative inference about all data professionals, all organizations, or all industries. The strength of Delphi lies in the quality of expert insight, the anonymity of response, and the iterative refinement of judgment, not in broad population sampling. Therefore, the findings should be interpreted as carefully reasoned expert consensus that is highly relevant to practice, but not as a claim that all possible stakeholders would rank the same dimensions and actions in the same order.

A third limitation is that the study focused on identifying, validating, and prioritizing semantic DQ concerns rather than on building and testing a new framework. This was a deliberate design choice consistent with the purpose of the dissertation, but it means the study stops short of implementation testing. The dissertation does not provide a new semantic DQ maturity model, a fully operationalized measurement architecture, or a controlled evaluation of how the recommended actions change GenAI performance over time. Therefore, although the study offers strong evidence about what should be prioritized, it does not by itself prove the downstream effect size of each recommendation in live organizational environments.

A fourth limitation involves measurement maturity for semantic DQ itself. One of the recurring themes in Round 1 and Round 2 was that semantic dimensions are difficult to define and measure consistently. Some panelists explicitly noted that meaning, relevance, and context are not always reducible to the same type of metric used for structural data checks. This means that even though the study identifies priority dimensions and promising measures, the field still lacks the degree of measurement standardization that exists for many syntactic dimensions. Consequently, the dissertation contributes conceptual and priority-setting clarity, but it does not resolve all methodological challenges surrounding semantic DQ measurement.

A fifth limitation is contextual variability across domains and use cases. The importance of certain dimensions, measures, and actions may vary depending on whether GenAI and LLMs are used in regulated industries, consumer-facing applications, internal productivity settings, knowledge retrieval, or domain-specific advisory tasks. For example, timeliness ranked relatively lower overall in Round 2, but its importance may be much higher in settings where stale information creates immediate operational or compliance risks. Similarly, some organizations may require stronger fairness auditing or stricter transparency mechanisms than others depending on their exposure. This means the study's priorities should be applied thoughtfully and adapted to the context rather than treated as a rigid one-size-fits-all prescription.

5.11 Recommendations for Future Research

Future research could build on this study by operationalizing semantic data quality dimensions, evaluating their impact within organizational DQM programs, and examining how enhanced semantic controls influence GenAI outcomes, governance maturity, and decision quality over time and across different industries.

Future research should build on this dissertation by developing more explicit operational measures for the semantic dimensions identified as most important. The Delphi findings clearly support the priority of accuracy, relevance, and trustworthiness, but the field still needs stronger,

more standardized ways to measure these constructs in practice. Subsequent research could develop and test semantic DQ indicators based on authoritative reference matching, ontology coverage, contextual fit scoring, value-add assessment, and provenance quality. Such work would extend the current study by moving from priority identification to measurement design and validation.

A second important direction is implementation-based research. The present dissertation identifies what experts believe organizations should prioritize, but future work should test these recommendations in live DQM and AI governance environments. For example, researchers could examine whether integrating authoritative-source validation reduces hallucination rates, whether shared ontologies improve consistency across agents and dashboards, or whether hybrid review workflows improve decision quality in high-stakes GenAI use cases. Case studies, field experiments, or action research designs would be especially useful because they could show how semantic DQ controls operate under real organizational constraints.

A third direction is the development of maturity models for semantic DQ in AI-enabled organizations. One of the recurring themes in the Delphi results is that organizations differ in readiness. Some are still building foundational metadata and stewardship capabilities, while others may be prepared to embed semantic metrics into governance dashboards and automated AI control layers. Future research could therefore define maturity stages for semantic DQ, identify the capabilities associated with each stage, and propose phased implementation roadmaps. Such work would help practitioners translate the study's recommendations into more realistic transformation programs.

A fourth direction concerns domain-specific research. The semantic risks and priorities identified in this dissertation are broadly relevant, but their relative importance may differ across industries. Regulated environments such as banking, healthcare, insurance, defense, and public administration may require different semantic controls than marketing, retail, or internal knowledge-management applications. Sector-specific Delphi studies or comparative case studies

could examine whether the ranking of semantic dimensions changes by domain and whether different industries require different combinations of authoritative validation, ontology design, fairness auditing, transparency mechanisms, or human oversight.

A fifth direction is longitudinal research on institutionalization. The current study shows what experts think organizations should do, but less is known about how these practices become embedded over time. Future longitudinal studies could investigate how organizations sustain semantic DQ initiatives, how stewardship and governance roles evolve, how AI-related incidents reshape DQM priorities, and how semantic controls interact with organizational culture. This research would be valuable because it would move the discussion from initial adoption to long-term governance resilience and help explain why some organizations successfully integrate semantic DQ into their operations while others do not.

5.12 Overall Chapter Summary

This chapter demonstrated that Data Quality Management practice must evolve to address semantic data quality more explicitly in the era of GenAI and Large Language Models, documenting and validating a persistent gap between Information Quality theory and practice while offering actionable recommendations to enhance trust, transparency, and data integrity in AI-enabled environments.

This chapter interpreted the findings of the dissertation by bringing together the documentary framework analysis and the two rounds of Delphi evidence. It showed that the gap between Information Quality theory and DQM practice is not simply a conceptual observation from the literature review, but a gap strongly recognized by expert practitioners. Round 1 demonstrated broad agreement that current practice overemphasizes syntactic dimensions and underemphasizes semantic dimensions, while Round 2 established that accuracy, relevance, and trustworthiness are the most critical semantic dimensions for GenAI and LLM contexts. The chapter also showed that the panel's recommendations converge around authoritative validation,

shared ontologies and dictionaries, hybrid human/automation approaches, stronger governance, output monitoring, and organizational capability building.

The chapter further demonstrated a strong linkage between the dissertation's three research questions and the Delphi findings. RQ1 was answered through the convergence of the documentary review and Round 1 questions showing limited and inconsistent incorporation of semantic DQ in current practice. RQ2 was answered through the Round 1 qualitative themes and the Round 2 ranking of semantic dimensions, which placed accuracy, relevance, and trustworthiness at the top. RQ3 was answered through the strong alignment between Round 1 open-ended recommendations and Round 2 rated actions, measures, approaches, challenges, and enhancement priorities. In that sense, the chapter did not merely summarize findings; it showed how the Delphi study provides a traceable evidentiary path from expert opinion to dissertation conclusions and recommendations.

The broad conclusion of Chapter 5 is that DQM practice must evolve if organizations are to deploy GenAI and LLMs in a trustworthy, transparent, and risk-aware manner. Traditional DQ foundations remain valuable, but they are insufficient when treated as the whole of quality. Semantic data quality must become more visible, more governable, and more actionable within DQM programs. By documenting the existing gap, validating it through expert consensus, and identifying practical priorities for action, this dissertation contributes both to scholarship and to professional practice. The chapter therefore closes the dissertation by reaffirming that improving trust in AI-enabled systems requires not less attention to data quality, but more sophisticated attention to the semantic dimensions that determine whether data and AI outputs are actually meaningful, reliable, and fit for use.

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Appendix

Appendix A - First Round Questionnaire

Delphi Study

Examining Data Quality Management Practices in Light of the Adoption of LLMs and GenAI: Closing the Gap between IQ Theory and IQ Practice.

Introduction

This Delphi Study is designed to gather expert insight, validation, and consensus on the evolving practices of Data Quality Management (DQM) in the context of widespread adoption of Generative AI (GenAI) and Large Language Models (LLMs). The study focuses on addressing the gap between Information Quality (IQ) theory and its practical application, with particular attention to the often-overlooked semantic dimensions of data quality. To provide context and frame the research problem and objective to participants, a summary of preliminary findings related to these neglected Data Quality (DQ) dimensions in widely used DQM frameworks is presented below. Based on this summary, participants' expertise, knowledge and experience in these fields (data domain and/or adopted technologies), are invited to evaluate the below statements, providing their consensus, and feedback. Participants' responses will offer their level of agreement to support the development of expert-driven conclusions for the neglected data quality dimension critical in this emerging era of GenAI and LLMs adoption.

Preliminary Findings Summary

This section summarizes preliminary findings and highlights gaps in DQM frameworks' emphasis on semantic data quality, essential for GenAI and LLM adoption.

Data Quality Dimensions - Data Quality dimensions are measurable characteristics of data, providing a framework for defining Data Quality requirements and assessing Data Quality over time. Semantic data quality dimensions focus on meaning, relevance, and context. Syntactic data quality dimensions focus on structure, format, and adherence to rules.

Research Problem -Current Data Quality Management practices overlook Semantic Data Quality Dimensions, that are critical for GenAI and LLMs performance, exposing industries for increased risks such as:

- Regulatory fines, financial loss
- Loss of trust from members and customers

Research Objectives - This research aims to:

- Examine widely used Data Quality Management Frameworks.

- Review which semantic data quality dimensions each framework focuses on and identify gaps.
- Share discovered gaps in a Delphi panel to seek consensus and validation of research observations.
- Recommend prioritizing semantic data quality dimensions to boost the trust and reliability of GenAI and LLMs outputs.
- Contribute to bridging the gap between information quality (IQ) theory and information quality (IQ) practice.

Conceptual Data Quality Frameworks - Basis for Information Quality (IQ) Theory

Theoretical conceptual frameworks forming the basis of IQ theory outline both semantic and syntactic data quality dimensions, grouped into four categories: intrinsic, contextual, representational, and accessibility - that professionals can use to define, measure, and monitor data quality over time. Below are the theoretical frameworks:

- Beyond Accuracy Framework: What Data Quality Means to Data Consumers by Richard Y. Wang and Diane M. Strong
- A Product Perspective on Total Data Quality Management (TDQM) Framework by Richard Y. Wang

Several widely used practical data quality frameworks were reviewed to assess their emphasis on semantic dimensions compared to those in the conceptual frameworks above. Below is the list of commonly used frameworks that are used in this study:

- Data Management Body of Knowledge (DAMA-DMBOK)
- Conformed Dimensions of Data Quality (CDDQ)
- Data Management Capability Assessment Model (DCAM)
- Six Sigma Data Quality (SSDQ) Framework
- Data Quality Assessment Framework (DQAF)
- Three-Dimensional Analysis
- ISO 8000 Part 61 Standard
- Ten Steps to Quality Data and Trusted Information
- BCBS 239 (Basel Committee on Banking Supervision)

Overlooked Semantic Data Quality Dimensions in Practical Frameworks

A review of commonly used practical frameworks compared to theoretical ones reveals a clear gap: semantic data quality dimensions - those that assess meaning and context - are largely overlooked. Most practical frameworks focus primarily on syntactic or representational aspects of data. This research's preliminary findings underscore that gap. As organizations increasingly adopt GenAI and LLMs, it becomes essential to shift focus toward evaluating the meaning,

relevance, and context of data, not just its structure and format. Frequency Comparison of Semantic and Syntactic Data Quality Dimensions in Nine Evaluated Frameworks is shown in the below table.

Table 27

Frequency Comparison of Semantic versus Syntactic Data Quality Dimensions Preliminary Analysis

Semantic	Frequency Count	Frequency %		Frequency %	Frequency count	Syntactic
Believability	0	0%		77.8%	7	Validity
Objectivity	1	11.1%		66.7%	6	Completeness
Reputation	0	0%		77.8%	7	Consistent Representation (Standardization)
Value-added	0	0%		77.8%	7	Timeliness
Relevancy	2	22.2%		77.8%	7	Duplication (Uniqueness)
Appropriate amount of data	0	0%				
Access security	1	11.1%				
Accuracy	7	77.8%				

Preliminary analysis shows a greater emphasis on measuring syntactic over semantic data quality dimensions in practice.

Delphi Study Statements & Participants Agreement and Feedback

Instructions

- Read each statement and respond with **Agree** or **Disagree** based on your experience, expertise and the above research summary finding.
- Provide any additional comments, feedback or any other level of consensus to the **statement** in the "Additional Feedback" section beneath each **statement**.
- There are a total of 30 statements/questions in this Delphi study.
- **Your insights are highly valued.**

Section 1: GenAI, LLMs and Data Quality Management

These questions explore the impact of GenAI and LLMs on data quality management practices.

1. Data Quality Management professionals in both academia and enterprise need to examine current Data Quality Management practices given the rise of Generative AI (GenAI) and Large Language Models (LLMs).
 - o Response: **Agree / Disagree**
 - o **Additional Feedback:**

2. GenAI and LLMs introduce complex data challenges in practice.
 - o Response: **Agree / Disagree**
 - o **Additional Feedback:**

3. Traditional Data Quality Management practices may fall short due to the adoption of GenAI and LLMs for companies who are not enhancing their Data Quality management practices.
 - o Response: **Agree / Disagree**
 - o **Additional Feedback:**

4. Robust Data Quality Management frameworks are critical for Artificial Intelligence performance.
 - o Response: **Agree / Disagree**
 - o **Additional Feedback:**

5. Effective Data Quality Management practice would tremendously help industries minimize risk and maximize value through strong and robust data governance and data quality management programs in light of GenAI and LLMs adoption.
 - o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
6. Data Quality (Good or Poor) directly impacts GenAI and LLMs outputs such as “Accuracy, Reliability and Ethical integrity”.
 - o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
7. If Data quality management frameworks and practices are not examined and enhanced, mitigation of Biases in the GenAI and LLMs output is among the challenges industries will be facing.
 - o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
8. If Data quality management frameworks and practices are not examined and enhanced, mitigation of hallucination in the GenAI and LLMs output is among the challenges industries will be facing.
 - o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
9. If Data quality management frameworks and practices are not examined and enhanced, mitigation of misinformation in the GenAI and LLMs output is among the challenges industries will be facing.
 - o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
10. If Data quality management frameworks and practices are not examined and enhanced, there will be a lack of transparency of the GenAI and LLMs output.
 - o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)

11. Inadvertent leakage of sensitive information is an emerging challenge industries will be facing due to the adoption of GenAI and LLMs, hence the need to examine traditional data quality frameworks and practices.
- o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
12. Inconsistent accuracy based on training context is an emerging issue due adoption of GenAI and LLMs, hence the need to examine data quality practices and frameworks.
- o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
13. With poor data quality management practices, industries will be facing the “Digital Inbreeding Risk” Meaning that artificial intelligence trained data, or artificial intelligence generated data on low-quality can degrade overtime, leading companies or practices to face challenges of mitigation of bias, hallucination and misinformation.
- o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
14. Generative AI models (e.g., LLMs, GANs, Diffusion Models) rely on large, diverse datasets.
- o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
15. There is a growing need for improved data quality frameworks in industry to effectively support data quality management amid the adoption of GenAI and LLMs.
- o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
16. Data professionals should advocate for urgent review of existing data quality frameworks in organizations adopting GenAI and LLMs to identify gaps and implement necessary improvements.
- o Response: [Agree / Disagree](#)
 - o [Additional Feedback:](#)
17. To ensure trustworthy GenAI and LLM outcomes and maintain data quality integrity, industries must adopt enhanced data quality frameworks rather than relying solely on traditional ones.

- o Response: [Agree / Disagree](#)
- o [Additional Feedback:](#)

Section 2: Prioritization of Semantic Data Quality Dimensions

18. Based on this research's preliminary analysis and your expertise, do you agree that semantic data quality dimensions (meaning and context) are often overlooked and not adequately measured in current practice?

- o Response: [Agree / Disagree](#)
- o [Additional Feedback:](#)

19. Based on this research's preliminary analysis and your expertise, do you agree that syntactic data quality dimensions (structure and format) are prioritized over semantic dimensions in practice?

- o Response: [Agree / Disagree](#)
- o [Additional Feedback:](#)

20. Based on your expertise, do you consider semantic data quality dimensions (DQ dimensions centered on meaning and context) as essential measurable factors for defining data quality requirements and evaluating industry data quality performance and risks in this age of GenAI and LLMs adoption?

- o Response: [Agree / Disagree](#)
- o [Additional Feedback:](#)

21. Should data quality professionals promote prioritizing semantic data quality dimensions within data quality management practices in this era of GenAI and LLMs adoption?

- o Response: [Agree / Disagree](#)
- o [Additional Feedback:](#)

22. Do you agree that measuring accuracy of meaning, relevance, and context requires comparison with an Authoritative Data Source (ADS) or official System of Records (SOR)?

- o Response: [Agree / Disagree](#)
- o [Additional Feedback:](#)

23. Do you believe the practice of assuming validated data is inherently accurate without a proper comparison of data with an approved Authoritative Data Source should end, especially with the rise of GenAI and LLMs?

- o Response: [Agree / Disagree](#)

- o [Additional Feedback:](#)

Section 3: Overall Feedback

These questions provide an opportunity to share overall thoughts and recommendations.

24. What do you consider the most important semantic data quality dimensions for GenAI and LLMs?

25. What do you consider good measures for what you believe to be the top semantic Data Quality (DQ) dimensions?

26. Any thoughts from the experts as to how to efficiently gather these measures?

27. What do you consider the most challenging aspects of Data Quality (DQ) when it comes to GenAI and LLMs?

28. What components of Data Quality Management(DQM) practice that need the most enhancement to keep up with the rapid adoption of GenAI and LLMs?

29. What are your overall comments or suggestions for improving traditional data quality management frameworks and practices to ensure trustworthy and reliable outputs from GenAI and LLMs in current industry use?

- o [Overall Feedback:](#)

30. What are your overall recommendations for prioritizing semantic data quality dimensions - those focused on meaning, relevance, and context - to help industries effectively define, assess, and monitor data quality requirements and performance in the era of GenAI and LLMs?

- o [Overall Feedback:](#)

Thank you for your participation!

Appendix B - Second Round Questionnaires

DELPHI STUDY - SECOND ROUND

Examining Data Quality Management Practices in Light of the Adoption of LLMs and GenAI:
Closing the Gap between IQ Theory and IQ Practice.

Second Round Delphi Study Instructions

- This Second Round of the Delphi study focuses primarily on ranking the semantic data quality dimensions for the open-ended questions (Questions 24–30), because there was a broad consensus on questions 1-23 (see summarized analysis for each question).
- For this Second Round, I am requesting that you rank the responses for questions 24–30 from Round One using the ranking scale and tables for each question below. The area where I am requesting your input is highlighted in **Red Font**.
- OPTIONAL - If you want to provide additional comments for Questions 1–23, please enter your responses in the area highlighted in **Green Font**.
- Note: In the First Round we had a total of fifteen anonymous experts who participated in the study.

SECOND ROUND: QUESTIONS 24 - 30

Question 24 - First Round Summary:

Original question: What do you consider the most important semantic data quality dimensions for GenAI and LLMs?

Key themes suggested by panelists:

- **Accuracy / factual correctness** – Nine panelists emphasized the need for data to truthfully represent the intended concept and avoid hallucinations.
- **Relevance / context / value add** – Nine panelists noted data must be pertinent to the specific use case and add value.
- **Completeness and consistency** – Three panelists mentioned ensuring all required concepts are present and semantically consistent across sources.
- **Believability / trustworthiness** – Four panelists noted that users must be able to trust the data's meaning and provenance.
- **Timeliness** – One panelist added that the semantic quality should be maintained over time.
- **Availability of ontologies/dictionaries** – One panelist emphasized the ability to resolve terms through shared dictionaries and glossaries.
- **Equal importance across dimensions** – Three panelists argued that no single dimension should dominate, underscoring the need to incentivize adoption of DQM practices.

Second Round - Question 24:

“Please rate in the table below the importance of the following semantic data quality dimensions for GenAI and LLM applications.” On a 5-point scale (1 = Not important, 5 = Extremely important).

Semantic DQ Dimensions	Importance Rate (1-5)
Accuracy	
Relevance	
Trustworthiness/Believability	
Completeness	
Consistency	
Timeliness	

Question 25 – First Round Summary:

Original question: What do you consider good measures for what you believe to be the top semantic Data Quality (DQ) dimensions?

Key themes suggested by panelists:

- **Dictionary/ontology coverage and resolvability** – Two panelists suggested counting how many data elements are present in a publicly available dictionary and are resolvable to specific meanings.
- **Accuracy validation** – Five panelists recommended factual validation through triangulation, authoritative data sources, audit sampling, and lineage tracing.
- **Contextual consistency checks** – Two panelists proposed automated tools that compare meaning across sources and verify relevance to the use case.
- **Human and value-based measures** – Two panelists stated that business impact and human assessment (“are we accomplishing the mission?”) are necessary indicators.
- **Trustworthiness/fairness** – Two panelists noted trust metrics (e.g., percentage of data matched to an authoritative source) and fairness auditing as relevant but underdeveloped measures.
- **Lack of established metrics** – Three panelists remarked that there are currently few comprehensive metrics for semantic quality.

Second Round - Question 25:

“Considering the importance of semantic quality, please rate the effectiveness of the following measures for assessing *accuracy* and *relevance*.” Use a 5-point effectiveness scale, 1 = Less effective, 5 = Very effective.

Measure for Assessing Accuracy and relevance	Effectiveness Rate (1-5)

(a) coverage and resolvability via shared dictionaries/ontologies	
(b) cross-source validation and triangulation	
(c) automated contextual consistency checks	
(d) human evaluation of value add	
(e) fairness and trust audits	

Question 26 – First Round Summary:

Original question: Any thoughts from the experts as to how to efficiently gather these measures?

Key themes suggested by panelists:

- **Automated validation and auditing** – Seven panelists advocated automating the gathering of semantic metrics by embedding validation into pipelines, creating vector stores for retrieval-augmented generation, and running regular completeness/timeliness audits.
- **Use of public dictionaries and ontologies** – Two panelists highlighted ensuring the presence of public dictionaries and resolvable elements as a first step.
- **Surveys and feedback from users** – Four panelists suggested using surveys, human feedback, and cross-functional discussions to collect qualitative measures where automated checks are insufficient.
- **Human-in-the-loop and organizational collaboration** – Three panelists stressed speaking with stakeholders across the organization and incorporating human reviews, despite the cost and time required.
- **Clear definition of measures and rules** – One panelist argued that efficient collection depends on well-defined, fine-grained rules and domain-specific metrics.

Second Round - Question 26:

“Which approaches are most efficient for gathering semantic DQ measures?” Please indicate your efficiency rate (1–5) in the below table, (1 = Not Efficient, 5 = Most Efficient).

Approaches for Gathering Semantic DQ Measures	Efficiency Rate (1-5)
(a) automated validation using authoritative sources and ontologies	

(b) automated audits and dashboards	
(c) surveys and qualitative feedback	
(d) human review and cross-functional collaboration	
(e) hybrid AI/human approaches	

Question 27 – First Round Summary:

Original question: What do you consider the most challenging aspects of Data Quality (DQ) when it comes to GenAI and LLMs?

Key themes suggested by panelists:

- **Ensuring accuracy and context** – Six panelists pointed to inaccurate data, wrong context, and hallucinated outputs as core challenges.
- **Lack of ground truth and traceability** – Four panelists emphasized difficulties in validating outputs without a ground truth and tracing errors back to the source.
- **Bias and fairness** – Two panelists noted that detecting, mitigating, and auditing bias in semantic data is complex.
- **Unstructured and semi-structured data** – One panelist highlighted that GenAI models break unstructured data into pieces, making it hard to measure completeness and relevance; two panelists added that the pace of data creation outstrips current DQ processes.
- **Rapid technological change** – Two panelists observed that the speed of GenAI evolution makes traditional DQ practices lag behind.
- **Organizational skills and adoption** – Four panelists remarked on the unwillingness or inability of organizations to invest in deep DQ work and adopt necessary practices.

Second Round - Question 27:

“For each of the following challenges in the table below, please rate its significance for ensuring data quality in GenAI/LLMs.” Use a 5-point scale rate, (1 = Not significant, 5 = Extremely significant).

Challenges for ensuring DQ in GenAI/LLMs	Challenge Significance Rate (1-5)
(a) verifying accuracy and relevance of training data and outputs	
(b) absence of ground truth and traceability	
(c) detecting and mitigating bias and fairness issues	
(d) managing unstructured and semi-structured data	

(e) rapid pace of technology change	
(f) organizational capacity and skills	

Question 28 – First Round Summary:

Original question: What components of Data Quality Management practice need the most enhancement to keep up with the rapid adoption of GenAI and LLMs?

Key themes suggested by panelists:

- **Output validation and real-time monitoring** – Five panelists called for better methods to evaluate and monitor generated content in real time.
- **Automated tools for semantic checks and deduplication** – Five panelists suggested developing and deploying automated tools for semantic similarity, relevance scoring, deduplication, and metadata modeling.
- **Education, awareness, and organizational capability** – Four panelists emphasized the need to educate stakeholders and build organizational capacity to address semantic DQ.
- **Speed and agility of DQ processes** – Two panelists noted that DQM methods must operate at the speed of GenAI outputs.
- **Governance, metadata modeling, and stewardship** – Three panelists urged strengthening DQ governance structures, metadata modeling, and data stewardship.

Second Round - Question 28:

“Please rate the priority for enhancing the following DQM components to support GenAI and LLM adoption.” Use a 5-point priority scale in the table below, (1 = Less Priority, 5 = High Priority).

DQM Components Needing Enhancement to support GenAI/LLMs adoption	Enhancement Priority Rate (1-5)
(a) output validation and real-time monitoring	
(b) automated tools for semantic similarity and deduplication	
(c) education and organizational capability building	
(d) speed and agility of DQM processes	
(e) governance, metadata modeling and stewardship	

Question 29 – First Round Summary:

Original question: What are your overall comments or suggestions for improving traditional data quality management frameworks and practices to ensure trustworthy and reliable outputs from GenAI and LLMs in current industry use?

Key themes suggested by panelists:

- **Extend frameworks to include semantic and output monitoring** – Seven panelists argued that traditional frameworks must expand beyond input validation to include continuous output monitoring, semantic similarity checks and feedback loops.
- **Incorporate AI-powered tools and human-in-the-loop** – Three panelists recommended using AI to automate semantic checks, with human oversight and domain-specific adaptation.
- **Adapt frameworks rather than discard them** – Three panelists noted that frameworks should be treated as adaptable guides rather than rigid rules.
- **Broaden to model-ready dimensions** – Four panelists suggested integrating dimensions such as bias, representation, reproducibility, and fairness.
- **Define fine-grained, domain-specific rules** – Two panelists highlighted the need for granular semantic rules and ontologies.
- **Clarify semantics and educate stakeholders** – Two panelists emphasized education around semantic vs. syntactic dimensions.

Second Round - Question 29:

“To improve traditional DQM frameworks for GenAI and LLMs, please indicate your agreement with these proposals in the below table.” Use a 5-point agreement scale to indicate your level of agreement (1 = Disagree, 5 = Strongly agree).

Proposals to improve traditional DQM frameworks for GenAI/LLMs	Agreement Level (1-5)
(a) extend frameworks to include continuous output monitoring and semantic checks	
(b) integrate AI-powered tools and human-in-the-loop validation	
(c) adapt frameworks flexibly to the context and mission rather than treat them as rigid	
(d) broaden DQ dimensions to include bias, representation and reproducibility	
(e) develop fine-grained, domain-specific rules and ontologies	

Question 30 – First Round Summary:

Original question: What are your overall recommendations for prioritizing semantic data quality dimensions to help industries effectively define, assess, and monitor data quality requirements and performance in the era of GenAI and LLMs?

Key themes suggested by panelists:

- **Develop shared ontologies and dictionaries** – Three panelists stressed using publicly accessible dictionaries and ontologies to align meaning across systems.
- **Prioritize accuracy and factual correctness** – Six panelists highlighted verification against authoritative sources and maintaining accurate, complete data sets.
- **Combine automated profiling with human review** – Two panelists recommended combining automated semantic profiling (e.g., NLP entity recognition, summarization) with human review, crowdsourcing, and fairness audits.
- **Education and governance** – Four panelists called for investing in education, embedding semantic metrics into dashboards, and formalizing semantic DQ policies and standards.
- **All dimensions matter** – Two panelists suggested that no single dimension should dominate; incentives and research are needed to develop actionable measures.

Second Round - Question 30

“Please rate the importance of the following recommended actions for prioritizing semantic data quality.” Use a 5-point importance scale rate in the table below to show your agreement with these recommendations provided from the First Round, (1 = Not Important, 5 = Very Important).

Recommended Actions (from Round 1) for prioritizing semantic DQ	Importance Rate (1-5)
(a) develop and adopt shared ontologies and publicly accessible dictionaries	
(b) verify accuracy against authoritative sources and maintain complete data sets	
(c) embed semantic metrics into dashboards and governance processes	
(d) combine automated semantic profiling with human oversight and feedback	
(e) invest in education and research on semantic DQ	

SECOND ROUND: QUESTIONS 1 - 23

OPTIONAL - Providing additional comments for Questions 1–23 is optional as a strong overall consensus was achieved in the First Round. Please review the summarized feedback from panelists in the first round for each question below, and feel free to provide any additional comments you may have if any.

Question 1 – First Round Summary:

Original Statement: “Data Quality Management professionals in both academia and enterprise need to examine current Data Quality Management practices given the rise of GenAI and LLMs.”

Responses & Consensus: All 15 panelists agreed with this statement (100% agreement), indicating unanimous consensus that the advent of GenAI/LLMs warrants a reevaluation of current DQM practices. Several experts emphasized that GenAI has *“exacerbated and accelerated the data quality problem”* – in other words, existing data issues are amplified under AI.

Common Feedback Themes:

- **Amplification of Existing Issues:** Many noted that GenAI/LLMs amplify longstanding data quality problems (e.g., bias, drift, privacy breaches) and thus demand renewed focus on DQM (five panelists). For instance, one panelist warned that *GenAI trained on biased data may amplify problems in ways that are hard to detect.*
- **Different Data & New Criteria:** Panelists highlighted that LLMs rely on unstructured, probabilistic data with different quality criteria than traditional structured databases (two panelists). GenAI introduces new data types and risks, so most current frameworks (built for structured data) are insufficient.
- **Lack of DQ Awareness:** It was observed that many organizations still lack basic DQM awareness. One expert described colleagues *“unsure of what the definition of data quality is”* and no clear ownership for defining DQ requirements. This lack of clarity, coupled with GenAI’s complexity, makes reexamination critical.
- **GenAI as a Tool for DQM:** A few panelists suggested GenAI/LLMs themselves could help improve or automate parts of DQM processes (one panelist). They advocated exploring GenAI as a tool within DQM frameworks (for instance, to aid data cleaning or monitoring).

Second Round - Question 1

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 2 – First Round Summary:

Original Statement: “GenAI and LLMs introduce complex data challenges in practice.”

Responses & Consensus: All 15 panelists agreed (100%) that GenAI/LLM adoption brings new, complex data quality challenges. There was a strong consensus that these technologies present novel issues beyond those handled by traditional DQ methods.

Common Feedback Themes:

- **New Types of DQ Problems:** Panelists listed specific challenges unique to GenAI/LLMs, including AI *“hallucinations,”* prompt sensitivity, context-window limitations, and stochastic/unpredictable outputs. These issues can degrade data quality in ways not seen with traditional IT systems.
- **“Garbage In, Garbage Out” at Scale:** Many experts echoed that feeding low-quality data into LLMs produces low-quality (and potentially harmful) outputs at much greater scale and speed. One noted that organizations rushing to implement GenAI without assessing data quality will get *“garbage out very fast and in massive volumes.”*
- **Governance and Oversight Gaps:** Panelists highlighted difficulties in governing AI systems

and measuring success. Traditional data governance struggles when *“the input data is unknown or not easily governed”* emphasized by panelists. Ensuring oversight of what data LLMs consume – and how to validate it – is a complex new frontier.

- Volume and Discovery: The enormous volume of data used to train and run LLMs makes it hard to discover data quality issues. Traditional DQ tools may not scale to identify errors or inconsistencies across such vast, diverse datasets.

- Confirmation of Existing Issues: Panelists noted that these challenges reinforce long-known truths: ultimately, *“AI succeeds or fails based on the quality of inputs provided”*, so data quality remains fundamental. GenAI’s emergence simply shines a brighter light on data issues that may have been ignored before.

Second Round - Question 2:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 3 – First Round Summary:

Original Statement: “Traditional Data Quality Management practices may fall short due to the adoption of GenAI and LLMs for companies who are not enhancing their DQ practices.”

Responses & Consensus: A strong majority agreed, 11 of 15 panelists (73.33%), supporting the notion that without updates, traditional DQM will be insufficient for GenAI. However, 2 panelists disagreed and 2 were neutral/uncertain, indicating some debate. Notably, one panelist explicitly *“disagreed”* and argued current practices can still work, while another could neither agree nor disagree at this stage.

Common Feedback Themes:

- Need for Evolution: Experts agreeing stressed that many traditional DQM frameworks focus on structured data and static rules, which cannot keep pace with GenAI’s dynamic, unstructured inputs. GenAI uses heterogeneous data (text, images, etc.), so *“unstructured data has a lot of variability... making it extremely difficult for traditional DQ methods to be successful”*.

- Frameworks vs. Agility: A recurring observation was that organizations often treat DQM frameworks rigidly (*“a framework, not a straitjacket”* as one put it). Panelists felt companies must be more agile and continuously improve DQ processes in response to GenAI. Stagnant “checkbox” approaches to DQ were already struggling even *“before GenAI came along.”*

- Not a Blanket Dismissal: Some experts cautioned that saying traditional DQM “falls short” should not imply abandoning existing best practices entirely. For example, one agreed there could be gaps, *“NOT to say that traditional DQM practices should be dismissed... assume many of them do apply and/or can be refined.”* This nuanced view suggests we must build on, rather than replace, traditional DQ fundamentals when integrating GenAI.

- Dissenting Views: The two dissenting panelists believed current DQ practices can still suffice if properly applied. One argued *“current practices will continue to work,”* even if GenAI tools can add efficiency. Another panelist outright disagreed and later clarified that both traditional and enhanced practices are needed together. Essentially, they felt the statement’s phrasing risked implying an either/or choice between old and new frameworks, which they rejected.

Second Round - Question 3:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 4 – First Round Summary:

Original Statement: “Robust Data Quality Management frameworks are critical for Artificial Intelligence performance.”

Responses & Consensus: The panel showed near-unanimous agreement on this point: all 15 panelists agreed (100%). Panelists strongly affirmed that having a solid DQM framework (and methodology) is crucial to reliable AI outcomes, although a few noted caveats in practice.

Common Feedback Themes:

- **Foundational Importance:** Many emphasized that “*AI models trained on poor quality data will likely produce unreliable outputs,*” so robust DQM is indeed critical to AI success. Especially in high-stakes domains (finance, healthcare), comprehensive DQ frameworks underpin AI reliability and trust.

- **Framework + Methodology:** Panelists not only agreed on frameworks’ importance, but some highlighted that a DQM framework must be paired with effective processes/methodologies. One noted the framework provides the “what and why,” but “*a methodology is equally important to provide the how and when.*” This suggests organizations need both a strong DQ model and an operational plan to implement it.

- **Preventing “Silver Bullet” Thinking:** Several comments warned against neglecting data fundamentals amid AI hype. Too often organizations chase AI as a “*silver bullet*” and ignore data quality, only to be reminded that “garbage in, garbage out” still applies. The consensus is that robust DQM must not be skipped if one expects AI to perform.

- **Real-World Use of Frameworks:** One panelist agreed in principle but remarked that while frameworks are conceptually critical, in practice they are “*rarely used... in my organization.*” This implies a gap between DQ theory and practice – even if everyone *knows* DQ is critical, many practitioners may not actively use formal frameworks day-to-day. This insight supports the need to bridge IQ theory and practice (a core objective of this research).

Second Round - Question 4:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 5 – First Round Summary:

Original Statement: “Effective Data Quality Management practice would tremendously help industries minimize risk and maximize value through strong and robust data governance and data quality management programs in light of GenAI and LLMs adoption.”

Responses & Consensus: All panelists agreed or partially agreed with this broad statement. 93.33% (14/15) agreed outright, while one panelist expressed “*partial*” agreement, noting reservations about current governance efficacy. Overall, consensus was that strong DQM/governance is indeed essential to reap GenAI benefits while mitigating its risks.

Common Feedback Themes:

- **DQM as Risk Mitigator:** Panelists concurred that high-quality, well-governed data ensures GenAI/LLM outputs are accurate, trustworthy, and compliant, “*thereby reducing risk while maximizing value.*” In other words, effective DQM acts as a safety net against the pitfalls of GenAI (like bad decisions from bad data).

- **Fundamental Principle:** Several noted this is true “*Generative AI or not*” – quality data is always critical for sound decisions. GenAI’s arrival simply raises the stakes. One participant remarked this statement is almost “*a given*,” implying it is intuitively true.
- **Defining “High Quality” for AI:** A few experts pointed out that in practice we need to define what “high-quality information” means for LLMs. Traditional data quality metrics may not directly translate to AI outputs. This suggests a future focus on establishing concrete DQ criteria/metrics for AI contexts (e.g., factual accuracy of outputs, absence of bias, etc.).
- **Organizational Understanding:** Panelists stressed the importance of a shared understanding of data quality within organizations. One wrote that it would help if everyone agreed on “*what the definition of data quality is, and what issues a sound DQM practice ameliorates*.” This hints that part of maximizing GenAI’s value is educational – ensuring stakeholders know why DQM matters for AI projects.
- **Governance Challenges:** One expert commented that “*as currently practiced, data governance is failing*” in many organizations. This sober note means that while effective DQM/governance would help, many current programs are not up to the task. It underscores the need to improve governance practices themselves to truly realize GenAI’s value safely.

Second Round - Question 5:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 6 – First Round Summary:

Original Statement: “Data Quality (good or poor) directly impacts GenAI and LLMs outputs such as ‘Accuracy, Reliability and Ethical integrity’.”

Responses & Consensus: 14 out of 15 (93.33%) agreed that the quality of data directly determines the quality of AI outputs. No one disagreed; one panelist did not select agree/disagree, instead asking for clarification on “*ethical integrity*.” The overwhelming consensus was that “garbage in, garbage out” applies good data yields better AI results, bad data yields poor results.

Common Feedback Themes:

- **Universal Principle:** Many noted this is true for *all* systems, not just AI – if anything, GenAI just makes the link more apparent. As one panelist succinctly put it, “*Ultimately AI succeeds or fails based on the quality of answers it provides, which in turn depends on the quality of inputs*.” Quality data is the fuel for quality AI output.
- **Reiterating GIGO:** Multiple experts literally echoed the phrase “*garbage in, garbage out*,” emphasizing that LLMs are no exception. If anything, GenAI can mask garbage with fluent language, but the underlying accuracy/reliability will still be poor if the data is poor.
- **Strong Programs Needed:** Panelists argued that organizations need very strong DQ programs to monitor and understand the data fed into AI models. Since GenAI can ingest massive, diverse data, only a robust DQ framework can ensure that “good” data (accurate, relevant, up to date) is being used to train or prompt the models.
- **Question Terminology:** One panelist did not fully understand the inclusion of “*ethical integrity*” as an output attribute. They commented that they do not usually consider “ethical integrity” an output quality per se, but rather a property of the process or system. This suggests the phrasing might mix concepts; accuracy and reliability are clear output metrics, but “ethical integrity” might need definition or could be interpreted as model behavior (e.g., not producing harmful content).

Second Round - Question 6:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 7 – First Round Summary:

Original Statement: “If DQM frameworks and practices are not examined and enhanced, mitigation of biases in GenAI/LLM outputs is among the challenges industries will be facing.”

Responses & Consensus: 13 panelists (86.67%) agreed that poor/unrevised DQM will make bias in AI output harder to tackle. One panelist disagreed and one was uncertain. The majority view is that stronger DQM is needed to address AI bias, though a couple of experts felt the logical connection in the statement was not clearly expressed.

Common Feedback Themes:

- **Weak DQM => Bias Risk:** The consensus was that without robust DQM, organizations will struggle to detect and mitigate biases in AI. Bias in training data or outputs can lead to “*unfair and unethical decisions that undermine public trust*,” and weak DQ practices leave these issues unchecked. In short, enhancing DQM is seen as part of the solution to AI bias.

- **Lack of Awareness:** Panelists noted many organizations do not even realize DQ frameworks exist, let alone consider bias in that context. This lack of DQM maturity exacerbates the bias problem - companies may deploy AI with biased data without any DQ oversight until something goes wrong.

- **Not Just Bias:** A few comments broadened the issue: it’s not only bias that’s a risk when DQM is neglected, but any number of output quality problems (e.g., misinformation, lack of context). One expert wrote “*the problem is not only about bias... it goes to the core of how we use language itself*,” implying that semantic nuances can introduce subtler biases or misinterpretations beyond obvious demographic bias. This underscores that semantic data quality (meaning, context) is deeply intertwined with bias mitigation.

Second Round - Question 7:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 8 – First Round Summary:

Original Statement: “If DQM frameworks and practices are not examined and enhanced, mitigation of hallucination in GenAI/LLM output is among the challenges industries will be facing.”

Responses & Consensus: 11 panelists (73.33%) agreed that poor DQM will make it hard to address AI “hallucinations” (fabricated outputs), while 3 panelists disagreed. One panelist was uncertain (7%). Agreement was lower here than prior questions, reflecting some divergence on whether data quality practices can directly mitigate model hallucination, which some see as more of a model issue than a data issue.

Common Feedback Themes:

- **Data Quality Can Influence Hallucination:** Those in agreement argued that low-quality or

insufficient training data contributes to hallucinations, so better data management can reduce this risk. One noted *“low quality data is one of the leading causes of hallucination in your models,”* implying that improving data quality (e.g., ensuring factual sources) should mitigate hallucinated answers. Another said models might present false info (hallucinate) if not grounded in high-quality data, which impacts trust.

- Limits of DQ’s Role: The dissenting panelists felt hallucination is primarily a model architecture/training issue, not purely a data quality issue. One disagreed in principle, stating *“hallucination and drift are specifically model issues, not data issues.”* Another simply said, *“hallucination is about the model, not about the data.”* These experts suggest that even if you feed an LLM perfect data, the model might still hallucinate due to how it generates responses. In their view, enhanced DQM alone won’t solve hallucinations; other measures (like better model training techniques or prompt designs) are needed.

- Partial Agreement – DQ Helps Indirectly: Some who agreed still provided nuance. For example, one panelist fully agreed hallucination is a risk and that more reliable data can *“partially mitigate”* it, but they were *“not sure how much a data quality framework can mitigate this risk”* because factors like prompting and training objectives also play a role. This acknowledges DQM is necessary but maybe not sufficient to tackle hallucinations.

Second Round - Question 8:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 9 – First Round Summary:

Original Statement: “If DQM frameworks and practices are not examined and enhanced, mitigation of misinformation in GenAI/LLM output is among the challenges industries will be facing.”

Responses & Consensus: The results for this question 13 of 15 (86.67%) panelists agreed, 1 panelist disagreed, and 1 panelist was uncertain. The panel mostly concurred that lacking strong DQM will make it hard to prevent or correct misinformation from AI. One dissent and one uncertainty again pointed to phrasing or nuance rather than rejecting the concept outright.

Common Feedback Themes:

- Acknowledgment of the Misinformation Problem: Many panelists agreed that AI systems can produce outputs that are *believable but absolutely WRONG*, and that improving data quality processes is part of the solution. Robust DQM helps ensure sources are verified and context is correct, reducing AI’s tendency to generate misinformation. One expert wrote that *“true accuracy... demands explicit alignment with authoritative semantic standards or records”*, reinforcing that authoritative data comparison (a DQ practice) is needed to catch misinformation.

- GenAI Heightens Scrutiny: A few noted that society is now far more likely to question the quality of inputs and outputs because of GenAI’s high visibility. Thus, if DQM isn’t improved, misinformation could proliferate and be quickly noticed, damaging trust. This underscores the urgency of having better DQ checks in place.

- Definitional Clarity: One panelist who agreed provided helpful definitions distinguishing *misinformation* (unintentional falsehood) vs *disinformation* (deliberately false), noting robust DQM can help address both. This suggests that enhanced DQM would include verifying facts and context to combat misinformation output, and maybe also tracking provenance to detect deliberate manipulation.

Second Round - Question 9:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 10 – First Round Summary:

Original Statement: “If DQM frameworks and practices are not examined and enhanced, there will be a lack of transparency of the GenAI and LLMs output.”

Responses & Consensus: 11 panelists (73.33%) agreed, 2 panelists disagreed, and 2 panelists were unsure. A couple of panelists weren’t certain what was meant by “transparency of output” or how DQM ties to it, leading to the mixed responses.

Common Feedback Themes:

- DQM and Transparency Links: Those who agreed interpreted it as meaning that improved DQM (especially tracking data lineage, documentation, etc.) aids transparency of AI outputs. Effective DQM lets organizations “*track data provenance, audit data usage, and understand model reasoning*,” thereby making AI decisions more transparent. In this sense, if we don’t enhance DQ practices (like metadata management, lineage tracking), AI outputs will remain a “black box” – i.e., lacking transparency.

- Transparency Beyond Data Quality: Panelists noted that transparency is also about the model and process, not just data. One panelist agreed but pointed out that even with good DQM, “*transparency will always be an issue since companies will be reluctant to share their GenAI input–procedure–output.*” Another expert agreed with the statement but added that the “*black-box nature*” of models themselves causes a lack of transparency beyond the data. In essence, poor data quality can further obscure output reasoning, but even great data practices won’t fully open the black box of complex models. This nuance suggests transparency is a multi-faceted challenge (data transparency, model interpretability, corporate openness, etc.).

- Disagreement Reasons: The two panelists who disagreed felt DQM has limited impact on transparency. One expert said they “*do not believe DQ management will impact the transparency of the LLM output unless this refers to DQ practices on the training data inputs.*” This panelist saw transparency as more related to explainable AI techniques or the model itself rather than data quality processes.

Second Round - Question 10:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 11 – First Round Summary:

Original Statement: “Inadvertent leakage of sensitive information is an emerging challenge industries will be facing due to the adoption of GenAI and LLMs, hence the need to examine traditional data quality frameworks and practices.”

Responses & Consensus: 11 panelists (73.33%) agreed, 2 panelists disagreed, and 2 panelists were neutral on this statement. Most panelists acknowledged the risk of sensitive data leakage with GenAI, but some questioned whether this is a “data quality” issue per se or more of a

security/governance issue. The consensus on the core risk was high but tying it to DQ frameworks drew mixed feelings.

Common Feedback Themes:

- **Sensitive Data Leakage is Real:** Almost everyone agreed that GenAI/LLM adoption introduces new privacy and confidentiality risks. Examples were noted of employees unintentionally pasting proprietary data into ChatGPT and thereby exposing it. The panel widely recognized this challenge.
- **Role of DQ Frameworks:** Where opinions diverged was on whether examining/enhancing DQ frameworks is the appropriate solution. Those agreeing felt that traditional DQ frameworks might not sufficiently address data sensitivity and should be updated. One wrote *“this is where frameworks really help”* – presumably in establishing guidelines for what data can be used in AI and how to handle sensitive fields. Others highlighted focusing on data sensitivity as part of data quality governance.
- **Security vs. Quality:** Both dissenting panelists argued that preventing leakage is more of a security and policy matter than a data quality matter. One said, *“information leakage would be best addressed by solid security,”* not necessarily by DQ processes. Another expert similarly stated this falls under “security... not directly quality of data,” except insofar as some DQ frameworks include security as a dimension. Essentially, they don’t disagree that leakage is a challenge, but they question linking it to DQ framework enhancements. They likely interpreted “data quality frameworks” in a partial sense that does not cover data security controls.

Second Round - Question 11:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 12 – First Round Summary:

Original Statement: “Inconsistent accuracy based on training context is an emerging issue due to adoption of GenAI and LLMs, hence the need to examine data quality practices and frameworks.”

Responses & Consensus: 13 panelists (86%) agreed that GenAI can exhibit inconsistent accuracy across different contexts/domains and that this issue merits reexamining DQ practices. None disagreed; however, 1 panelist gave a neutral response, and 1 panelist did not know how to answer. The high agreement suggests the phenomenon was understood by most, but the phrase “inconsistent accuracy” confused a couple of experts.

Common Feedback Themes:

- **Understanding the Issue:** Those who agreed interpreted “inconsistent accuracy based on training context” to mean that a model’s accuracy varies across different subject areas or scenarios depending on how well the training data covers those contexts. Panelists concurred this is indeed a problem: GenAI might be very accurate in domains it has lots of data for, and poor in others. *“The accuracy of GenAI outputs varies if data quality and context shift,”* one noted. In other words, a model can be inconsistently accurate – great on some inputs, terrible on others – due to uneven training data quality.
- **Context & Coverage as DQ Concerns:** Experts pointed out that this ties back to data quality practices like ensuring comprehensive and representative training data. If certain contexts are not covered (or the data in them is lower quality), accuracy in those areas will suffer. One panelist commented on “training context,” and suggested that they read it as domain coverage

and agreed that it's a DQ concern if coverage is patchy.

- Importance of Context Metadata: Some related this to the need for context and metadata management in DQ. If organizations do not manage what context their data covers, they will not know where the model might fail. *"Context is always an essential part of understanding data,"* one panelist remarked, supporting that DQ frameworks should explicitly consider context to ensure consistent accuracy.

Second Round - Question 12:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 13 – First Round Summary:

Original Statement: "With poor data quality management practices, industries will be facing the 'Digital Inbreeding Risk' – meaning that AI-trained data or AI-generated data of low quality can degrade over time, leading to challenges like bias, hallucination, and misinformation."

Responses & Consensus: This statement garnered near-unanimous agreement (14 of 15, 93.33%). Every panelist agreed except one who did not provide a clear agree/disagree but instead commented that the question *"needs rephrasing or further explanation."* Thus, the concept of "digital inbreeding" resonated strongly, though the wording was evidently confusing to at least one person.

Common Feedback Themes:

- Strong Recognition of the Risk: Panelists broadly agreed that if AI systems start training on data that includes outputs from other AI (especially if those outputs are low-quality), there is a compounding degradation effect – essentially a feedback loop of poor quality. One described that *"LLMs training on LLM-generated data tends to severely degrade the quality of the resulting model."* Another worried that *"if the questionable output of GenAI becomes input to train another, the degradation of results can happen quite quickly."* This captures the crux of "digital inbreeding": AI feeding on its own imperfect progeny and getting worse.

- Amplification of Errors: Panelists noted this leads to amplified biases and errors. Over generations, biases can compound, and misinformation can self-reinforce. One expert explicitly linked it to observed output concerns: *"I have already been concerned that it (GenAI output) looks good, sounds good, but it is not right... when that output is used as input to another model, it quickly degrades results."* This scenario exemplifies why strong DQM is needed to prevent tainted outputs from contaminating the training pool.

- Agreement on Terminology (once explained): Many panelists essentially rephrased the statement in their own words, indicating they grasped it. For example, *"the models increasingly learn from poor data which over time produce unreliable output"* or *"training a model on its own outputs leads to all kinds of risks."* This shows that after parsing; the panel overwhelmingly concurred with the idea.

Second Round - Question 13:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 14 – First Round Summary:

Original Statement: “Generative AI models (e.g., LLMs, GANs, Diffusion Models) rely on large, diverse datasets.”

Responses & Consensus: 11 panelists (73.33%) agreed, 2 panelists disagreed, and 2 panelists did not answer. Essentially everyone accepted this statement as a factual premise, aside from one dissent emphasizing that not *all* models require such scale or diversity. The high agreement suggests it was seen as a mostly self-evident truth (with a caveat for smaller models).

Common Feedback Themes:

- **General Agreement:** Most panelists simply agreed without additional comment, some noting “this is a fact.” It is broadly understood that large-scale models like GPT-4 are trained on enormous, diverse corpora. Panelists concurred that diversity and volume of data are part of what gives these models their power.
- **Nuance on Model Size:** One expert pointed out that while “*large general-purpose models...do require such large datasets,*” there exist “*smaller, more focused models*” that may not need as diverse a dataset. They added that even for smaller models, having larger pools of high-quality data helps accuracy. This nuance is valid: not every ML model is an LLM; some narrow-domain models train on limited data. However, the statement was phrased generally about “Generative AI models (e.g., LLMs)” which by definition implies large scale, so the disagreement likely stems from the panelist considering exceptions (like fine-tuned small models).
- **Relevance to DQ:** Although not many wrote feedback, one panelist mentioned this reliance on diverse data “*by definition...if they are to have general utility, the datasets must be diverse*”. Some also brought up that there are indeed approaches like LoRA that allow working with smaller datasets for specific LLMs. In terms of DQ management, the implication is that ensuring the quality and representativeness of these large, diverse datasets is paramount.

Second Round - Question 14:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 15 – First Round Summary:

Original Statement: “There is a growing need for improved data quality frameworks in industry to effectively support data quality management amid the adoption of GenAI and LLMs.”

Responses & Consensus: The panel showed virtually unanimous agreement (100%). All 15 panelists agreed (some with minor commentary). This strong consensus underscores that experts believe current industry DQ frameworks need upgrading to handle the challenges introduced by GenAI.

Common Feedback Themes:

- **Resounding Yes:** Panelists widely affirmed that industries need better DQ frameworks to cope with GenAI. The adoption of AI is multiplying opportunities for poor data to cause harm, thus creating urgency for stronger DQ structures. One noted, “*AI is providing many more opportunities for poor DQ to have negative impacts in multiple domains,*” hence improved frameworks are required. Others simply agreed without comment, implying the statement was self-evident.
- **Fit-for-Purpose Frameworks:** Some emphasized that new or improved frameworks must be “fit for purpose” for GenAI, which differs significantly from traditional structured data

environments. LLMs deal with unstructured data, so frameworks should incorporate semantic quality checks, contextual evaluations, etc., not just traditional dimensions.

- Volume, Speed, New Data Types: Panelists mentioned factors driving this need: the amount, speed, and types of data in GenAI are unprecedented. Traditional DQ frameworks were not built with continuously generated AI outputs or massive unstructured training corpora in mind, hence they require evolution. For example, frameworks may need to address how to validate model outputs (which is a new angle for DQ).

- Building on Existing Models: One expert advised that improving DQ frameworks for GenAI doesn't mean reinventing the wheel from scratch: *"Assume you will build on existing models, not throw them away"*. Many/most traditional DQ practices still apply; they can be refined or expanded to include GenAI considerations. This aligns with earlier sentiments that we should extend rather than discard established DQM principles.

- Panelist Reservations: No one disagreed, and one expert commented that they *do not know of any practitioner calling for them (improved frameworks)*. This implies that while the need exists in theory, the demand has not been strongly voiced in industry yet. It could be an emerging realization - one this study aims to validate and bring to light. It also suggests an action item: raising awareness among practitioners about the importance of updating DQ frameworks for AI.

Second Round - Question 15:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 16 – First Round Summary:

Original Statement: "Data professionals should advocate for urgent review of existing data quality frameworks in organizations adopting GenAI and LLMs to identify gaps and implement necessary improvements."

Responses & Consensus: 12 panelists (80%) agreed with this prescriptive statement, 1 panelist disagreed, and 2 panelists gave neutral/uncertain comments. The majority supported the call to action that data professionals take the lead in reviewing and updating DQ frameworks for AI. The few non-agreeing responses provided thoughtful nuances about who would conduct such reviews and the realism of organizations doing it independently.

Common Feedback Themes:

- Strong Advocacy Role: Those who agreed championed the idea that data professionals (DQ managers, data stewards, etc.) must proactively push updating DQ frameworks when their orgs embrace GenAI. They saw it as part of professional responsibility to ensure AI projects do not outpace data quality. One said it's not just about advocating – *"it's a necessity at this point"*, indicating how critical this review is.

- Gaps to Address: Panelists acknowledged that a thorough review would likely reveal needed additions such as provenance/lineage tracking, bias detection measures, semantic consistency checks, etc. (some of which were mentioned in feedback to other questions). In essence, existing frameworks might lack these GenAI-specific considerations, hence professionals should identify and fill those gaps.

- Leverage Existing Foundations: A theme emerged that while review and improvement are urgent, it should build on what exists. For example, one expert noted if core traditional dimensions (like Wang and Strong's top dimensions) are addressed, *"almost all requirements for GenAI will be resolved, except provenance/lineage/traceability."* This implies current

frameworks are a starting point and mostly adequate, with just a few critical enhancements needed for AI contexts (especially around traceability).

- Realism and Responsibility: The panelist who disagreed did so not because review isn't needed, but because they doubt organizations actually *have* DQ frameworks to review or will do it. *"I don't think organizations adopting AI employ such frameworks,"* they said, suggesting many companies lack a formal DQ framework altogether. They did mention they personally called for an independent review of foundations (indicating support for the idea in principle). Another panelist agreed with the need but argued it is *"unrealistic to think individual organizations will do this review"* - instead, industry leaders (associations, consultants, researchers) should do it and publish guidance. These points reflect that many companies might not have the expertise or will to self-audit their DQ frameworks for AI, so data professionals may need external support or frameworks.

- Neutral View: One panelist did not explicitly agree or disagree but commented that reviewing existing frameworks is part of figuring out solutions, though it's uncertain if any current framework can serve as basis for a new one. Another expert provided a nuanced take without agreeing or disagreeing. These highlight considerations on how such reviews should be conducted (collaboratively, industry-wide perhaps).

Second Round - Question 16:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 17 – First Round Summary:

Original Statement: "To ensure trustworthy GenAI and LLM outcomes and maintain data quality integrity, industries must adopt enhanced data quality frameworks rather than relying solely on traditional ones."

Responses & Consensus: A majority agreed (12 of 15, 80%) with this statement, 3 panelists disagreed. The split indicates that while most experts think enhanced frameworks are needed for trust in GenAI, a notable minority cautioned against the phrasing that might undervalue traditional frameworks. Indeed, their comments stressed a balanced approach.

Common Feedback Themes:

- General Agreement: Those in favor argued that modernized or enhanced DQ frameworks are essential to keep GenAI outputs trustworthy. Traditional frameworks alone, if used "as is," may not cover aspects like real-time monitoring, semantic quality, or AI-specific risks. One panelist wrote, *"Trust in GenAI outputs depends on modernized DQM frameworks integrating dynamic quality measures, metadata curation, ongoing audits,"* etc. This exemplifies the enhancements envisioned.

- Areas of Enhancement: Panelists suggested enhanced frameworks should include things like semantic analysis for unstructured data, explicit linking of DQM to model lifecycle stages, and other practices beyond what traditional data quality programs do. The general vibe was that clinging strictly to old DQ methods *without* evolution would be inadequate for AI – some new components are "necessary" (though not sufficient).

- Not Either/Or: The dissenting voices pushed back on the phrasing "rather than relying solely on traditional ones." They cautioned that traditional practices are still valuable and should not be thrown out. For instance, one expert said even enhanced frameworks will not *ensure* trustworthy outcomes by themselves; good data is necessary but *"not sufficient."* One expert provided a comment suggesting to *"be careful of assuming that none of the traditional practices apply..."*

assume many do apply and can be refined/modified, with additional practices for GenAI.” Two panelists noted that if an organization is not adopting GenAI, traditional methods suffice, which also show a desire to balance new with old. In summary, even among those who agreed, there is an implicit understanding that hybrid approaches are best - use enhanced frameworks but also leverage proven traditional DQ practices.

- Need to Adopt Enhanced Frameworks: Despite the above nuance, no one disputed that some enhancements are needed. The debate is more about messaging. “Must adopt enhanced frameworks rather than solely traditional” was slightly polarizing; refining the wording to “augment traditional frameworks with enhancements” might bring full consensus. Essentially all panelists accept the need for improvement (as seen in Q15 and Q16), so the key is to not imply an abandonment of the past.

Second Round - Question 17:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Prioritization of Semantic Data Quality Dimensions – (Questions 18–23)

In this section, the Delphi panel evaluated statements comparing semantic data quality dimensions (focused on meaning, context, etc.) versus syntactic dimensions (structure, format, etc.) in current practice. The goal was to validate observations that semantic DQ is often neglected and to gauge support for emphasizing semantic factors in the GenAI era. The panel strongly affirmed the importance of semantic DQ, though a few members provided caution that syntactic fundamentals remain important and that measurability of semantics is challenging.

Question 18 – First Round Summary:

Original Statement: “Based on this research’s preliminary analysis and your expertise, do you agree that semantic data quality dimensions (meaning and context) are often overlooked and not adequately measured in current practice?”

Responses & Consensus: An overwhelming 14 of 15 panelists (93.33%) agreed that semantic DQ dimensions are indeed often overlooked and under-measured in today’s data quality practice. This confirms our preliminary finding that meaning/context get short shrift in real-world DQM compared to more technical aspects.

Common Feedback Themes:

- Widespread Agreement with Premise: The majority of experts echoed that organizations focus on syntactic data quality (valid codes, formats, completeness) and pay far less attention to semantic aspects like coherence, context relevance, or interpretability. One stated plainly: *“Most organizations focus on resolving syntactic issues... while essential aspects such as coherence, meaning, and context are much less prioritized.”* Another noted their organization *“doesn’t know the difference between semantic and syntactic... semantic isn’t part of our data quality vocabulary.”* These comments exemplify the oversight of semantic DQ in practice.

- Challenges in Defining/Measuring: Panelists agreed it is happening, and some gave reasons why semantic dimensions are neglected. They are *“more challenging to define and measure,”* often requiring good metadata management (which many companies lack) and can be expensive or hard to quantify. In contrast, syntactic checks can be automated (e.g., format checks), so organizations default to those. Essentially, semantics fall by the wayside because they’re harder.

- **Metadata & Foundations:** It was pointed out that if basics (technical quality) aren't right, semantic quality is hard to achieve – *"if you don't have the technical aspects right, it's really hard to have the semantic and contextual pieces right"*, and those pieces depend on metadata which is "painfully lacking" in many orgs. This suggests a maturity model: some organizations have not even mastered syntactic DQ, so they never progress to semantic DQ. However, those that manage their metadata well tend to also do better on data meaning/context.

- **Dissenting View:** One expert seems to *"disagree that semantic dimensions of DQ are overlooked,"* noting that it is well-known that providing context in AI is essential. This disagreement suggests that not all corners of industry ignore semantics; some may be paying attention. But given the weight of other responses, those are likely exceptions.

- **Panelist Insights:** Others added that while overlooked in many places, the importance of semantic DQ is rising due to GenAI. Data curation for LLM success was cited. Also, one panelist cross-referenced our provided frequency counts table and said, *"I have to agree"* it shows semantic dims are not prioritized - *"they are not ALWAYS overlooked,"* they noted, but often. This nuance acknowledges there are cases or organizations where semantics are considered but generally supports the statement's truth.

Second Round - Question 18:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 19 – First Round Summary:

Original Statement: "Do you agree that syntactic data quality dimensions (structure and format) are prioritized over semantic dimensions in practice?"

Responses & Consensus: 13 panelists (86.67%) agreed that in practice, organizations give more attention to syntactic aspects of data quality than to semantic ones and 2 panelists disagreed. The strong agreement further validates the pattern that format/accuracy checks take precedence, while meaning/context gets less attention.

Common Feedback Themes:

- **Syntactic is Easier, So It's the Focus:** Many pointed out that syntactic DQ is *"easier to automate and measure, so it receives most attention in data quality programs"*, whereas semantic issues (misclassification, ambiguity, context) are *"harder to quantify and often left out of routine checks"*. This summarizes why organizations prioritize syntactic: it is low-hanging fruit. Panelists concur that because you can readily run scripts for validity, completeness, consistency, etc., those get done, and semantic consistency or relevance checks (which might require human judgment or complex algorithms) often don't.

- **Evidence in Practice:** One panelist remarked on exactly this: *"In practice, syntactic DQ dimensions are more often the focus because they are 'easier' to measure and manage"*. Another said *"this is the easier of the two (syntactic) but doesn't address correctness or meaningfulness"* - implying organizations know they are focusing on the easy part and potentially ignoring the actual meaning. The consensus is that efficiency drives emphasis - companies tackle what they can with existing tools (formats, schema checks) and shy away from the messy semantic territory.

- **Reinforcing Q18 Reasons:** Panelists referenced earlier points: lack of metadata, difficulty, cost. Experts emphasized that semantic DQ dimensions are put on the back burner due to operational challenges and lack of know-how, leaving syntactic as the mainstay of DQ efforts.

- **Disagreeing Opinions:** The two who disagreed provided interesting perspectives. One said,

“some dimensions fall under both semantic and syntactic sets,” hinting that the dichotomy is not always clear-cut – e.g., an issue like completeness has both semantic and syntactic facets. This could be interpreted as: in practice, professionals might be addressing some semantic elements without labeling them as such because of overlap. It is possible in these experts' experiences, certain organizations do prioritize semantics (for instance, where data meaning is critical, like medical data, some semantic controls exist). Still, 13 out of 15 agreeing is a decisive majority.

- Unified View: Combining Q18 and Q19 feedback, the panel clearly sees a priority gap favoring syntactic DQ. This sets the stage for treating semantic dimensions as essential and promoting them.

Second Round - Question 19:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 20 – First Round Summary:

Original Statement: “Do you consider semantic data quality dimensions (DQ dimensions centered on meaning and context) as essential measurable factors for defining data quality requirements and evaluating industry data quality performance and risks in this age of GenAI and LLMs adoption?”

Responses & Consensus: All panelists agreed (100%) that semantic DQ dimensions are essential factors for defining DQ requirements and assessing performance/risk in the GenAI era - with a slight caveat from one panelist about measurability. This unanimous endorsement underlines that the experts see semantic quality not as a luxury, but as a necessity now.

Common Feedback Themes:

- Semantic Dimensions = Vital in GenAI Context: The panel uniformly affirmed that meaning- and context-related dimensions (e.g., accuracy of meaning, relevance, completeness of context, etc.) are crucial metrics for data quality when dealing with AI. Comments included: “*Semantic DQ dimensions are vital for defining requirements and evaluating data risks, especially with GenAI... models are highly sensitive to subtle shifts in meaning and context.*” Also “*Semantic is equally if not more important than syntactic... because it's the business that drives the semantic aspects of data quality*” - highlighting that ultimately it's the business use (meaning and context) that matters most for data fitness.

- Acknowledging Measurability Challenge but Necessity: One panelist wrote they agree semantic dimensions are important, “*Disagree that they are measurable*” easily. They note “*DQ measurement is always tough. Data does not possess physical properties.*” - implying that while essential, semantic quality often requires subjective or complex evaluation. Despite this, they still marked “Agree” to the overall question. This view suggests yes, we must include semantic dimensions as requirements, even if measuring them is challenging, because excluding them would ignore key aspects of quality. Another panelist also said “*Yes, these are measurable; however, measuring them takes a different kind of work than syntactic dimensions*”, indicating it's possible but perhaps through qualitative assessments or specialized techniques.

- Examples of Semantic DQ Dimensions Emphasized: Panelists enumerated specific semantic factors: factual accuracy, relevance, consistency of meaning, interpretability, etc., as things that need to be considered now. One looked forward to “*more techniques and measures for semantic DQ dimensions*” becoming available, showing that the field is evolving to quantify semantics (through NLP, perhaps human-in-loop evaluations). Another stressed that simply

telling organizations to apply semantic measures is not enough - you have to guide them *“how to do so and be more detailed.”* This underscores that while everyone agrees on importance, there is a need for practical guidance on implementation.

- GenAI Makes Semantics Mandatory: The context of GenAI was repeatedly mentioned. Because LLMs work with natural language (which is loaded with context and nuance), ignoring semantic quality can directly lead to model failures (hallucinations, irrelevance, etc.). Panelists conveyed that industries must incorporate semantic quality checks now to ensure AI outputs are fit for use. The consensus here aligns with the research’s objective: bridging IQ theory and practice by injecting semantic IQ into everyday DQM.

Second Round - Question 20:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 21 – First Round Summary:

Original Statement: “Should data quality professionals promote prioritizing semantic data quality dimensions within DQM practices in this era of GenAI and LLMs adoption?”

Responses & Consensus: 10 of 15 panelists (66.67%) agreed that DQ professionals should indeed push for greater priority on semantic DQ dimensions now, 3 disagreed, and 2 were neutral/unsure. The majority favors championing semantic DQ, but a sizable minority cautioned that priorities should be situation-dependent and not at the expense of other dimensions.

Common Feedback Themes:

- General Support for Emphasizing Semantics: Most panelists think DQ practitioners need to lead a shift in focus. *“Yes, semantic data quality is especially important in LLMs,”* one wrote plainly. Others elaborated that ensuring accurate meaning and context will *“minimize the risk of harmful or misleading outcomes”* and is advocated by emerging best practices (mentioning tools like data catalogs, knowledge graphs to support semantic understanding). This indicates many experts believe it is part of a DQ professional’s role today to elevate the discussion around semantic quality (which historically has been undervalued).

- Not Ignoring Other Dimensions: Four dissenting panelists each argued against *prioritizing* semantics to the exclusion or downgrading of other dimensions. One expert said DQ practitioners should pick dimensions based on context/purpose - *“quality dimensions [should be] based on who will use the AI and for what purposes,”* rather than a blanket prioritization of semantics. Another expert said, *“all dimensions should be treated equally and as important.”* Another panelist echoed that both syntactic and semantic are needed, *“would not prioritize one over the other... semantic without syntactic doesn’t work.”* These responses warn against swinging the pendulum too far - you cannot abandon basic data integrity (syntactic) in favor of semantics; they are complementary. Their disagreement seems rooted in the semantics of the word “prioritizing.” They likely agree semantics needs more attention than currently, but they resist any implication that semantic dimensions outrank or replace others in all cases.

- Conditional Agreement: Even among some who agreed, there was nuance. One expert provided a comment: *“Not sure how to answer because it depends on the situation. Promoting semantic DQ generally is helpful, but not always at the expense of other DQ dimensions. Semantic dimensions should become part of the ‘menu’ of DQ dimensions available... organizations can determine which are most applicable to various situations.”* This aligns with the dissenters - context matters, and DQ professionals should champion semantics by adding them to the toolkit and highlighting them but not necessarily making everything all about

semantics all the time. One expert (who agreed) also noted *“prioritizing these dimensions is not an end in itself”*, implying they must be part of a solid framework and solve real problems, not just a buzzword.

- Overall Insight: The panel clearly wants greater inclusion of semantic DQ in practice, but many prefer a balanced approach. The question’s phrasing “promote prioritizing” might have been interpreted as “promote above all else.” Wording like “promote increased focus on semantic DQ dimensions” might capture the intent without implying that syntactic or other aspects should be neglected. After all, the previous answers show semantics have been under-prioritized historically, so “prioritizing” could be read as “bringing them to equal footing” rather than “making them top dog.”

Second Round - Question 21:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 22 – First Round Summary:

Original Statement: “Do you agree that measuring accuracy of meaning, relevance, and context requires comparison with an Authoritative Data Source (ADS) or official System of Record (SOR)?”

Responses & Consensus: 11 panelists (73.33%) agreed that to assess a data element’s meaning/relevance accuracy, you generally need an authoritative reference. 3 panelists disagreed and 1 panelist was neutral. The majority aligns with the idea that you need some ground truth (ADS/SOR) to judge semantic accuracy, but a notable minority pointed out other ways or complexities in evaluating meaning and relevance.

Common Feedback Themes:

- **Agreement – ADS/SOR as Benchmark:** Those in agreement argued that without a trusted reference point, subtle errors in meaning or context can go undetected. *“Without a trusted reference, subtle errors... can persist undetected and undermine AI results,”* one stated. Another said, *“Semantic accuracy will be challenging even with an ADS but extremely hard without.”* These comments support using authoritative data sources or knowledge bases as a yardstick for truth when evaluating if an AI output’s content is correct in context. In practice, this might mean if an LLM says “X is Y”, you check against an official database or source of truth to verify that fact (meaning).
- **Source of Truth & DQ:** Panelists highlighted that a big difficulty is often identifying *what* the authoritative source is for a given piece of information. One noted their organization struggles to determine the ADS and responsible owner for data, implying that establishing authoritative sources is itself a data governance challenge. Nonetheless, having that ADS (and ensuring it’s of high quality) is seen as key to measuring semantic correctness.
- **Human-in-Loop Alternatives:** One expert gave a perspective that meaning/relevance might be tied to the purposes and business processes, and if ADS/SOR are used, then those systems should include high-quality metadata to judge context reliability. They implied other methods like human judgment or business rules might also play a role, not strictly ADS comparison.
- **Disagreements - Other Ways to Measure or Not Always Possible:** The dissenting panelists provided two main counterpoints: **(a) Subjectivity of meaning:** One said *“what if there is no standard? I do not think one is even possible when it comes to ‘meaning’ - things can mean different things to different people.”* They argue meaning is sometimes inherently subjective or context-dependent, so an official source may not exist. **(b) Alternative approaches:** Another

disagreed even though they acknowledge ADS/SOR can help - *“I disagreed even though meaning, relevance, context could benefit from ADS comparison. However, they are often subjective, which means other measurements have to be employed.* This suggests using methods like human-in-the-loop evaluations, user feedback, or AI-based heuristics to judge semantic quality. Similarly, another panelist disagreed and noted *“other ways are also possible, e.g., human-in-the-loop or AI-in-the-loop approaches... ADS or SOR is possible but not necessarily required.”* They envision leveraging expert review or even another AI to cross-check, rather than strictly needing an official dataset.

- Takeaway: Most believe having an authoritative reference is the safest and most objective way to measure semantic accuracy. The disagreement highlights fair points stating that the ADS/SOR approach may not always be feasible (if no authority exists) and that we might need to develop other evaluative techniques.

Second Round - Question 22:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*

Question 23 – First Round Summary:

Original Statement: “Do you believe the practice of assuming validated data is inherently accurate without a proper comparison of data with an approved Authoritative Data Source should end, especially with the rise of GenAI and LLMs?”

Responses & Consensus: 10 of 15 panelists (66.67%) agreed that blindly trusting “validated” data without cross-checking against an ADS should stop, particularly in the GenAI context. 1 panelist disagreed, but notably 4 panelists were neutral or unsure. This indicates that while a majority sees unverified trust in data as risky, a considerable number found the question unclear or needed more context to answer.

Common Feedback Themes:

- General Agreement – Don’t Assume, Always Verify: Those who agreed voiced that just because data passed some validation rules or came from a supposedly validated source, one shouldn’t assume it is 100% accurate. This is increasingly true with AI in the mix, where stakes and scrutiny are higher. One wrote *“assuming ‘validated data’ is inherently accurate is dangerous!”* Another noted it is *“a slippery and dangerous slope”* to believe validated = accurate without independent verification. The consensus here is that contextual and semantic validation (e.g., comparing to an ADS) is needed, not just internal validation checks. Especially as GenAI can amplify any errors, not double-checking against a source of truth can propagate inaccuracies widely.

- Complexity and Cost Considerations: A few agreeing panelists acknowledged that while ideal, requiring ADS comparison for every case might be impractical. One said, *“while we should never assume validated data is accurate... without context it’s hard to determine since requiring an ADS in every case could make many tasks impossible due to cost.”* So, they agree in principle but note real-world constraints – you have to pick battles (some data might be “good enough” after standard validation if further checking is too costly). This suggests a nuanced approach: where possible, verify against ADS, but weigh the effort vs risk.

- Trusted ADS/SOR: Panelists commented that trust in data depends on validation criteria, and ADS comparison is one way, but even ADS must be trusted and rules appropriate. They conclude if people trust validations they do not understand, that’s bad – which aligns with the idea that blind trust is dangerous. Similarly, another panelist commented *“... it is increasingly*

risky to assume ‘validated’ data is accurate without confirming against a formally approved ADS,” effectively agreeing but phrasing it as an observation. Another expert gave an example about definitions of “validated” and “accurate” - illustrating that something could be technically valid (e.g., code in a list) but still not accurate contextually. This insight supports the statement: just because data passes a validity check doesn’t ensure it’s accurate in the real-world sense.

- Disagreement: One panelist disagreed outright and referenced back to Q16 nuance: essentially saying sometimes using validated data without checking an ADS is okay depending on context, and requiring an ADS always could be overkill. They advocate context-based decision - sometimes “validated” data is sufficient if it meets known criteria. This is a moderate dissent focusing on practicality rather than principle.

Second Round - Question 23:

- (OPTIONAL) - *Please provide any additional feedback or comments on this question/statement:*